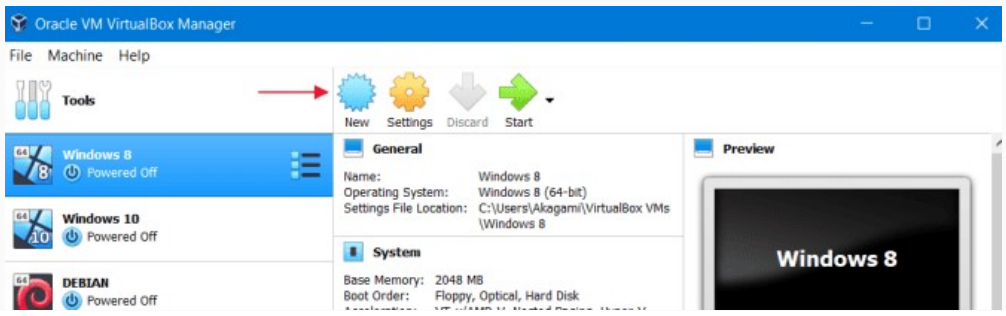
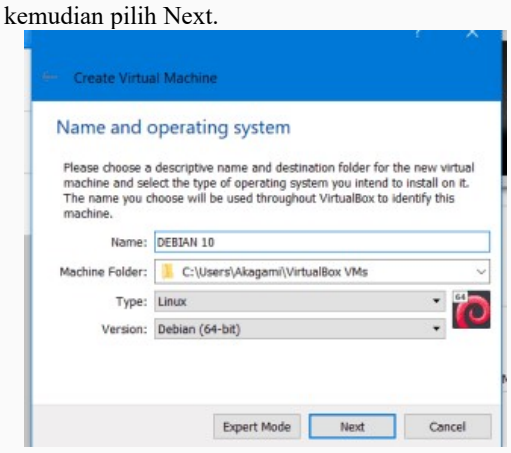
A. Penginstalan Debian via Virtual Box

Di dalam Debian, tersusun kernel Linux, sistem operasi GNU, dan juga manajer paket APT maupun DPKG. Salah satu alasan mengapa Debian menjadi salah satu sistem operasi yang banyak digunakan adalah karena OS ini bukan hanya mendistribusikan software bagi penggunanya, melainkan di dalamnya juga terdapat puluhan ribu perangkat lunak yang tersimpan di dalam repositorinya. Berikut ini tutorial cara install Debian 9 di VirtualBox lengkap untuk pemula:

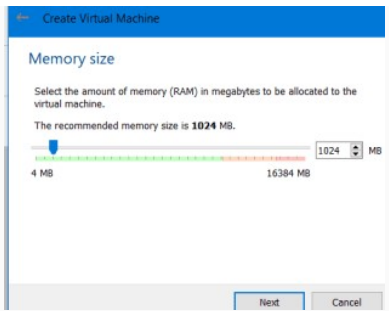
1. Untuk membuat mesin virtual baru di VirtualBox, kita klik tombol New.



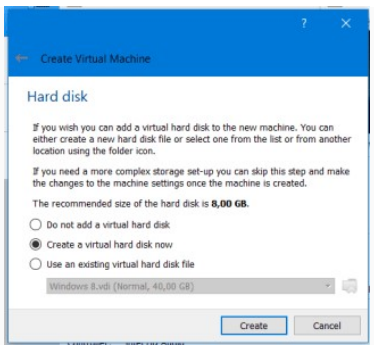
2. Beri nama pada mesin virtual anda kemudian pilih Next.



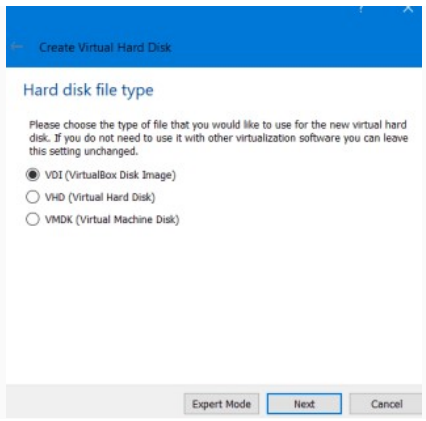
3. Untuk ukuran memori (RAM) yang digunakan pada mesin virtual anda, saya sarankan *1024MB* namun jika spesifikasi laptop / komputer anda pas-pasan, *512MB* juga tidak apa-apa. Selanjutnya pilih Next.



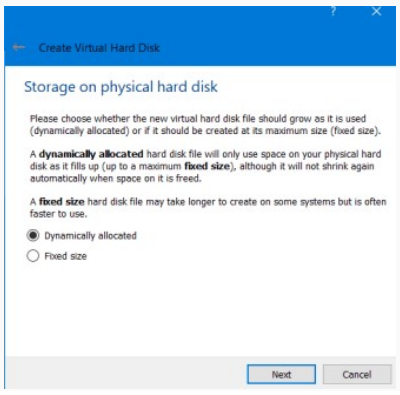
4. Setelah itu kita membuat harddisk virtual untuk menyimpan data dan file mesin virtual kita. Pilih Create a virtual hard disk now kemudian pilih Create.



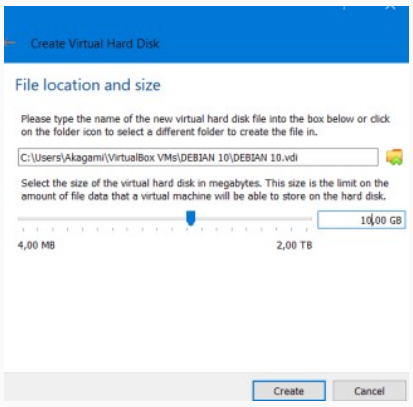
5. Setelah itu pilih VDI (VirtualBox Disk Image).



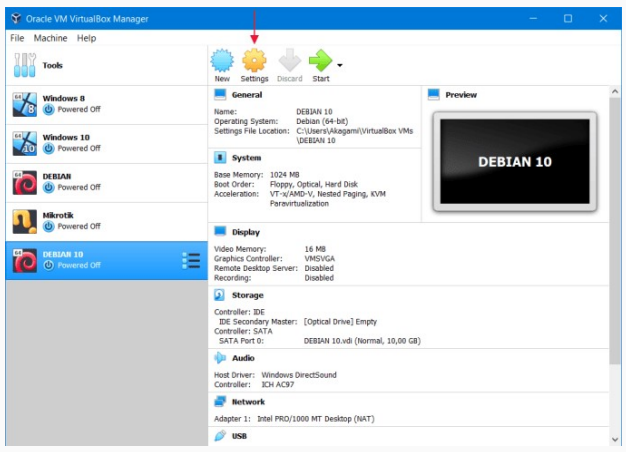
6. Selanjutnya pilih dynamically allocated kemudian pilih Next.



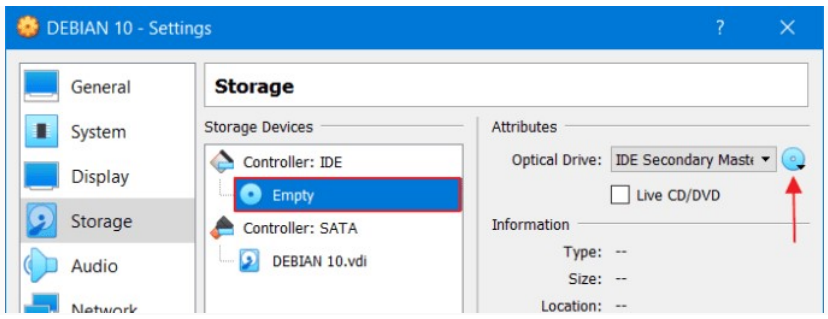
7. Untuk ukuran harddisk virtual yang digunakan, 10GB sudah lebih dari cukup. Setelah itu pilih Create.



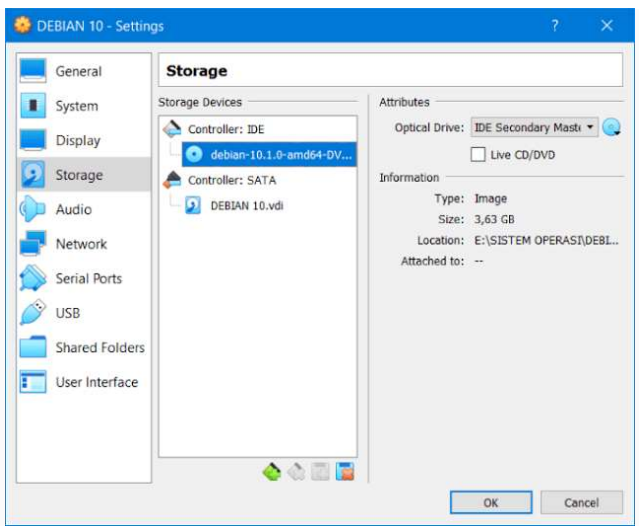
8. Setelah selesai membuat mesin virtual, selanjutnya kita perlu melakukan sedikit pengaturan. Pilih mesin virtual yang telah kita buat tadi (DEBIAN 9) kemudian pilih Settings untuk masuk ke menu pengaturan.



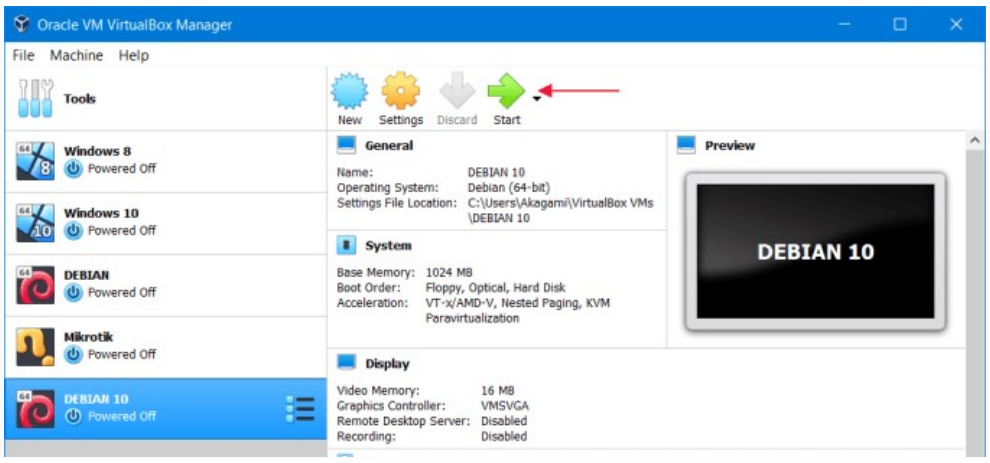
9. Pada menu Settings, kita pilih Storage. Kemudian pada bagian Controller: IDE, pilih Empty kemudian klik ikon CD seperti yang ditunjuk tanda panah. Pilih file ISO Debian 9 / 10 yang sebelumnya sudah anda download.



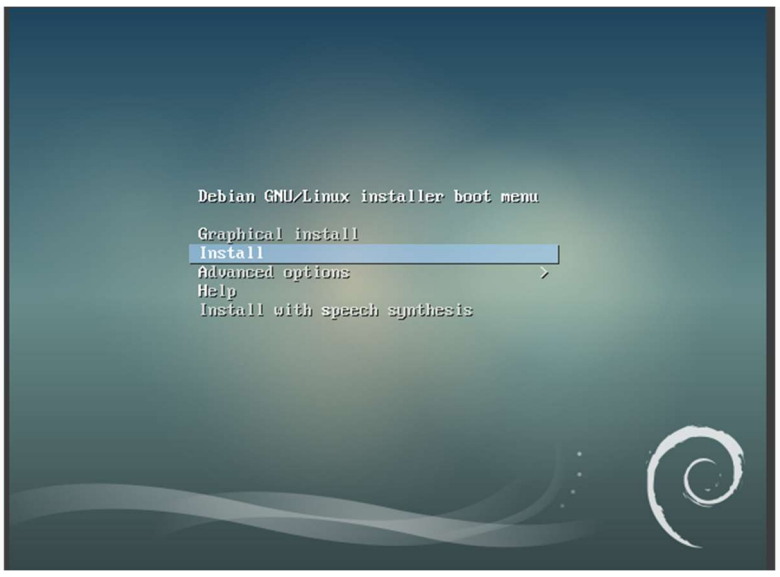
10. Jika file ISO Debian berhasil ditambahkan, tampilannya nanti kurang lebih seperti ini. Selanjutnya pilih *OK*.



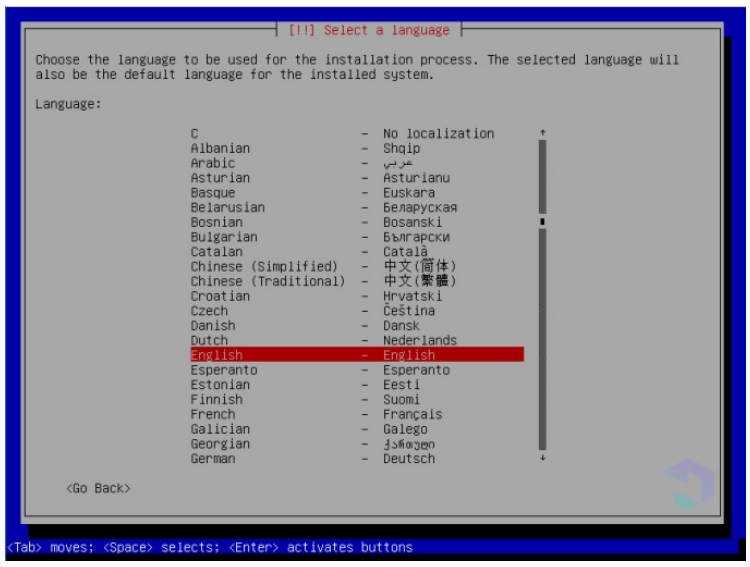
11. Pilih Start untuk memulai proses instalasi.



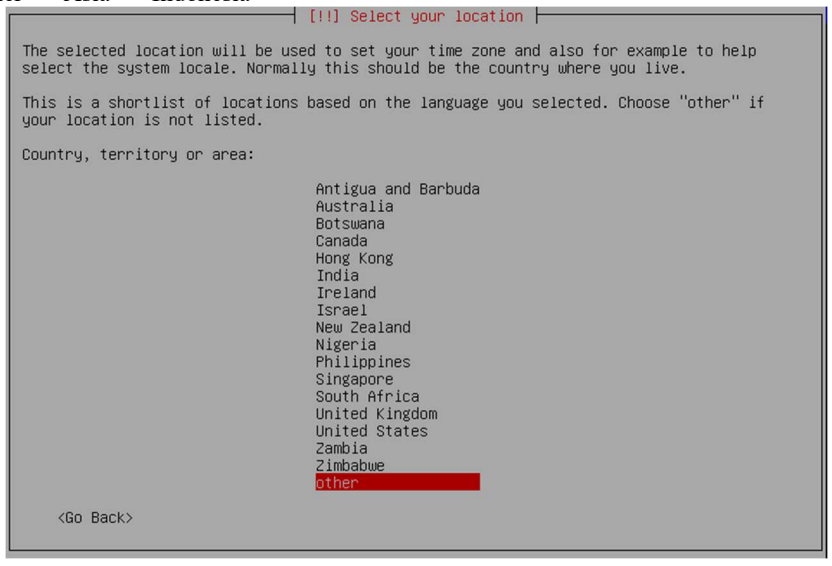
12. Berikut ini tampilan awal instalasi Debian 9. Pilih Install.

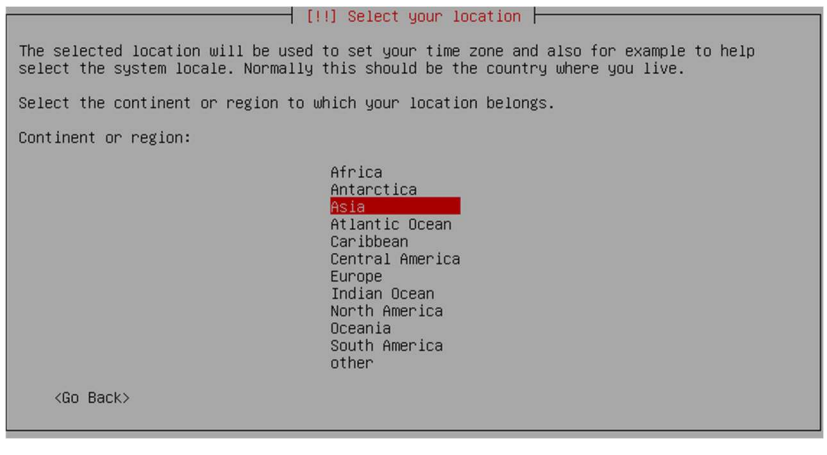


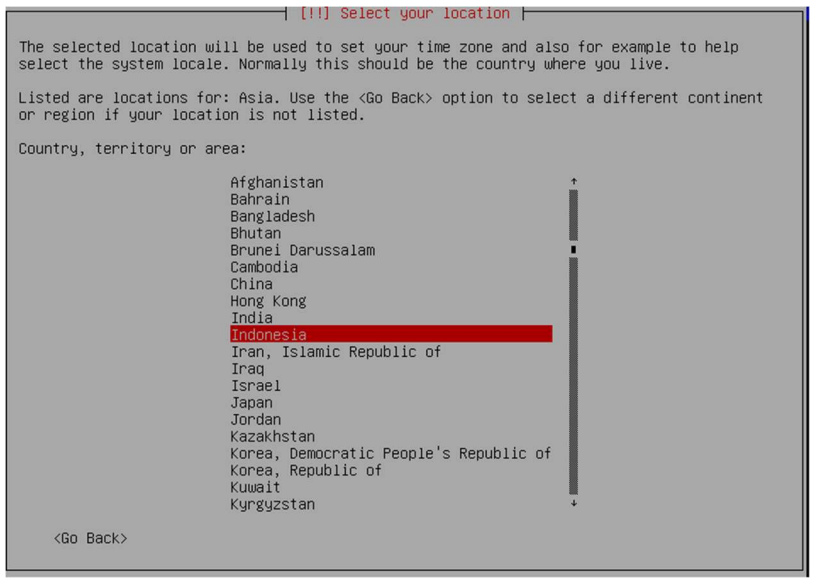
13. Select a Luanguage = English



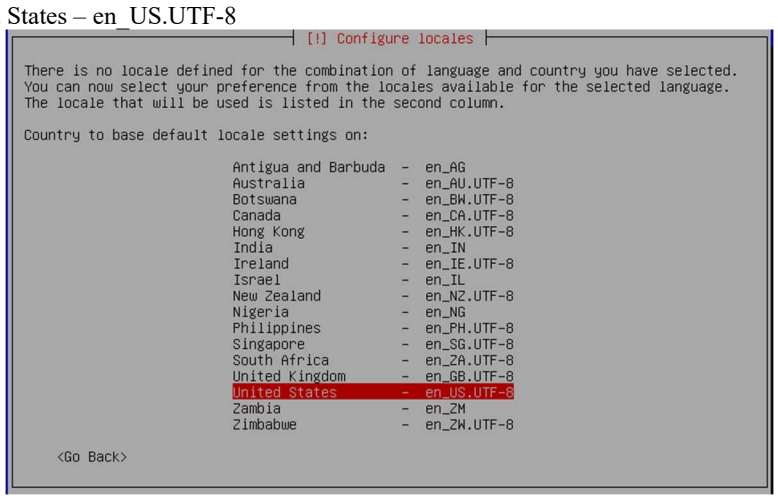
14. Selec your location = other => Asia => Indonesia







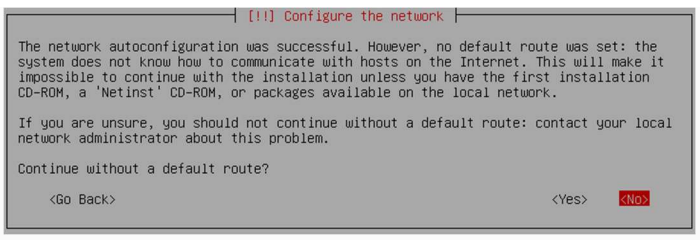
15. Configure locales = Uneted States – en_US.UTF-8

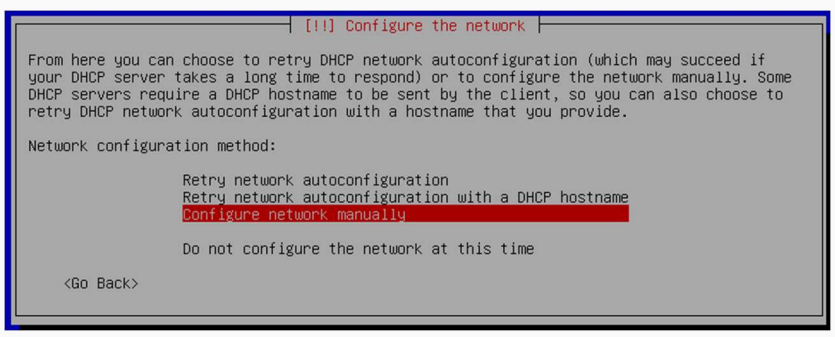


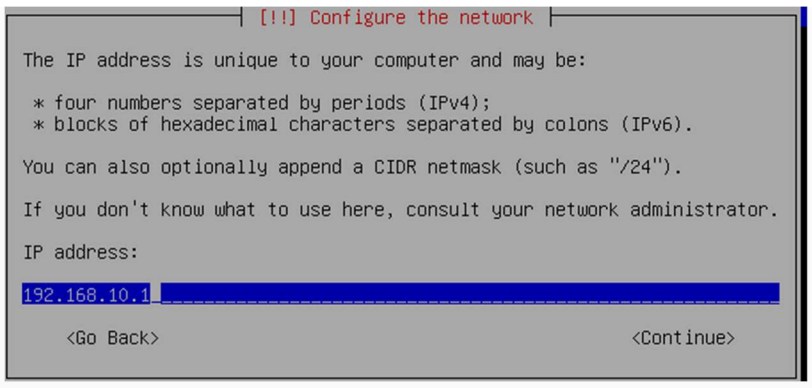
16. Configure the Keyboard = American English

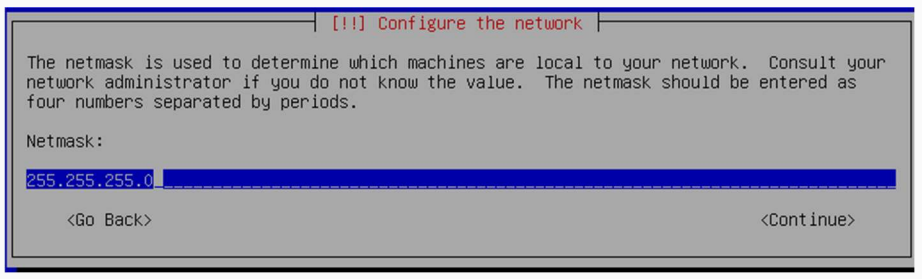


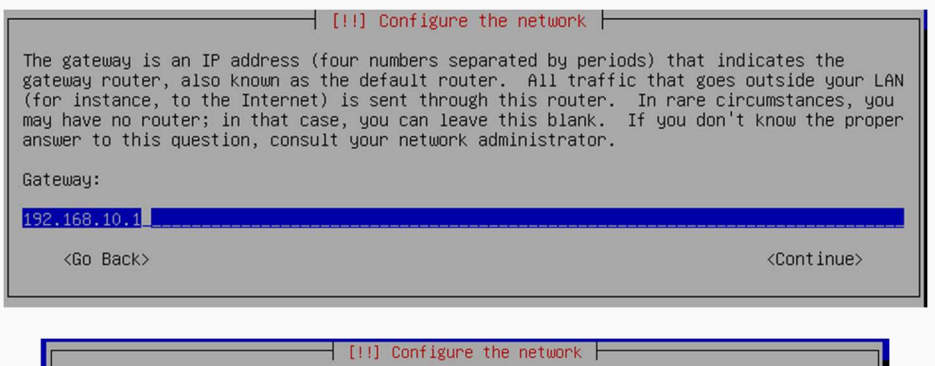
17. Configure network = No => Configure network manually => IP address: 192.168.10.1 => Netmask: 255.255.255.0 => Gateway: 192.168.10.1=> Name server addresses: 192.168.10.1=>Hostname: smkn3tator => Domain name: smkn3tator.net

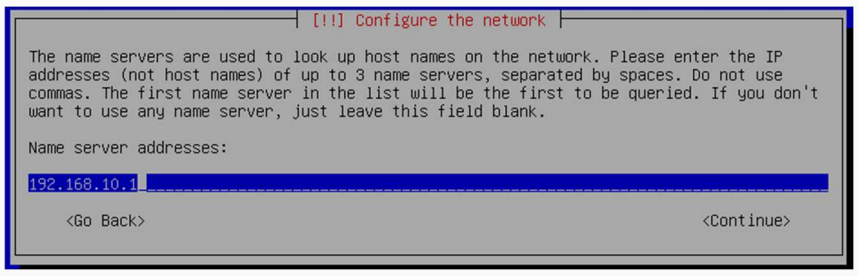


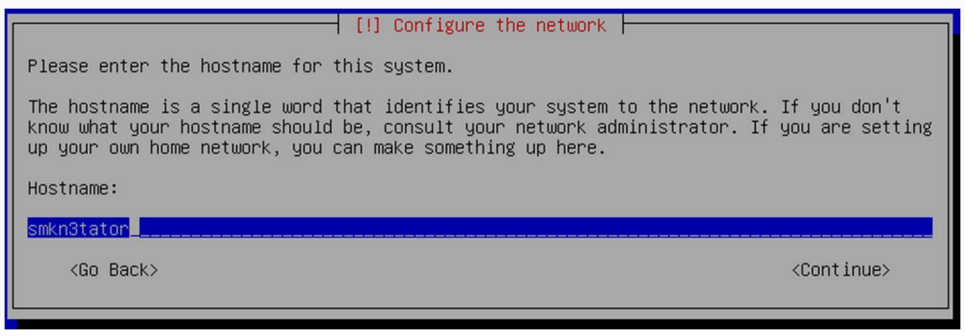


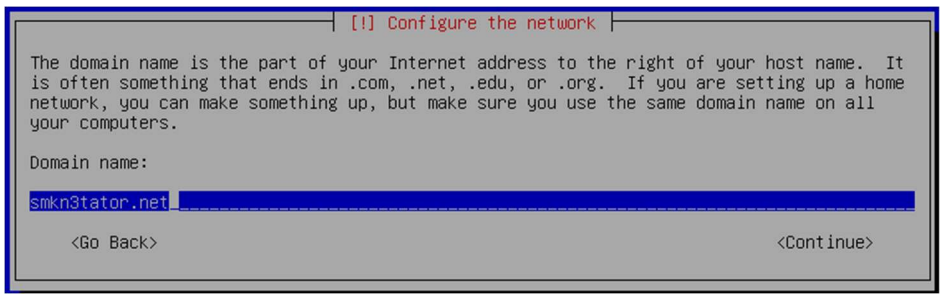




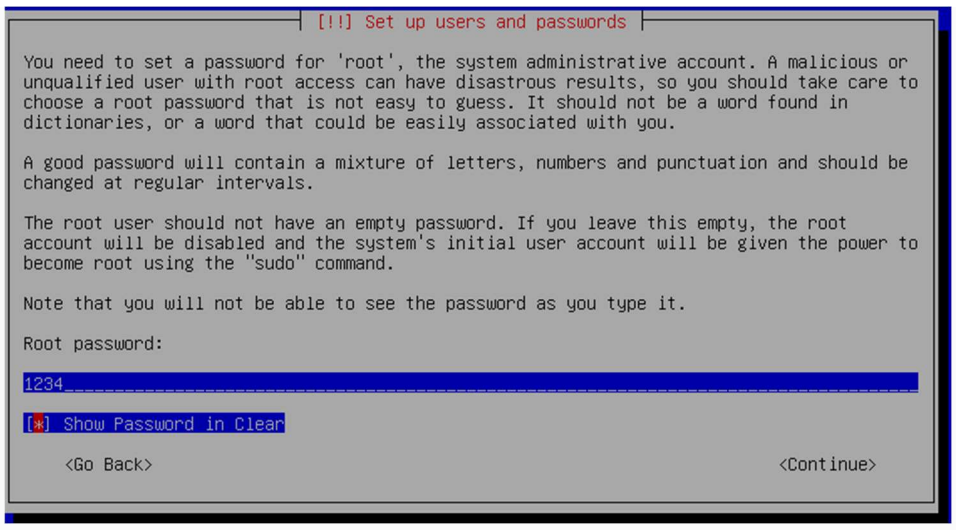


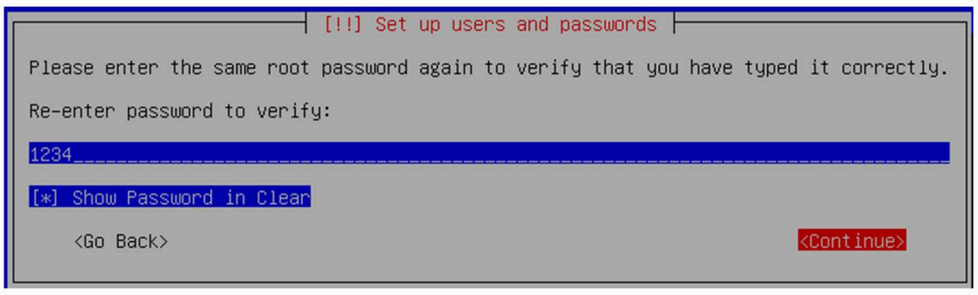


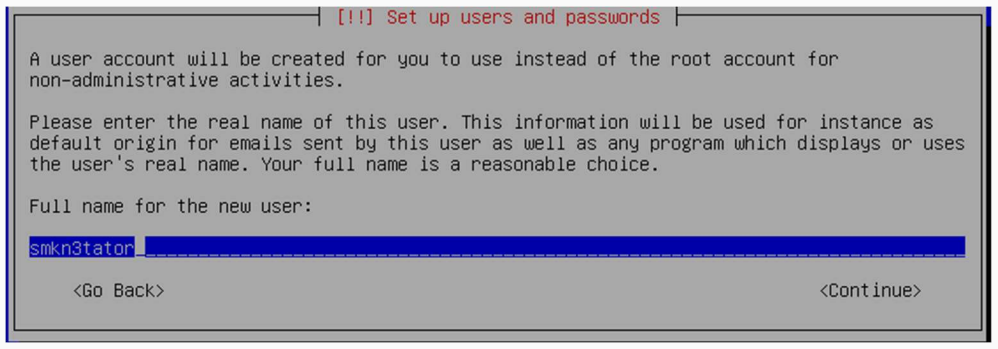


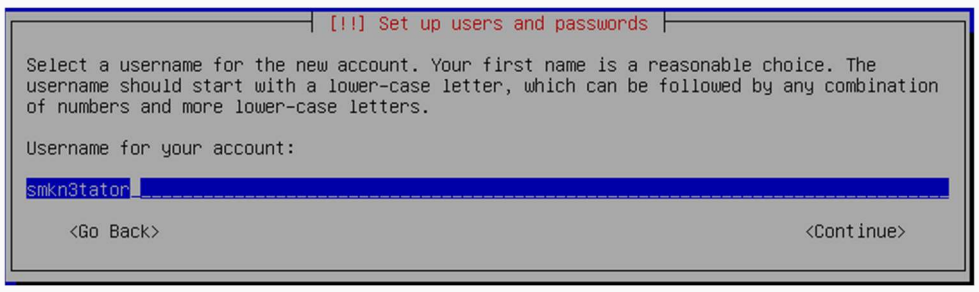


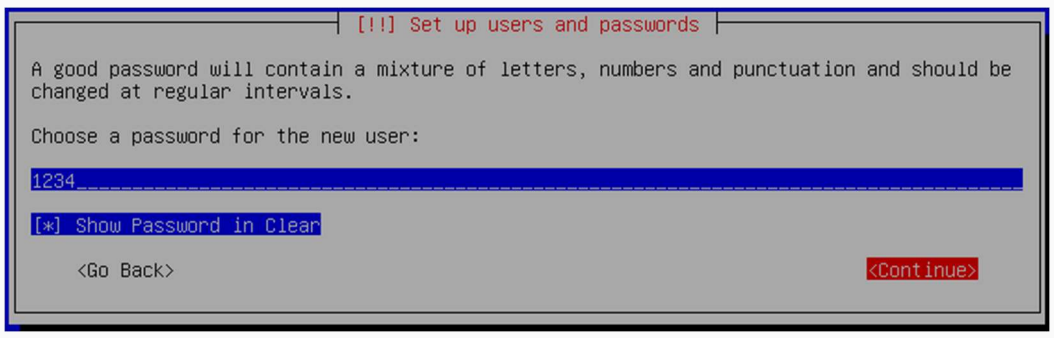
18. Set up users and passwords => Root password : 1234 => Set up users and passwords: 1234 => Full namefor the new user: smkn3tator => Username for your account: smkn3tator => Choose apassword for the new user: 1234 => Re-enter password to verify: 1234

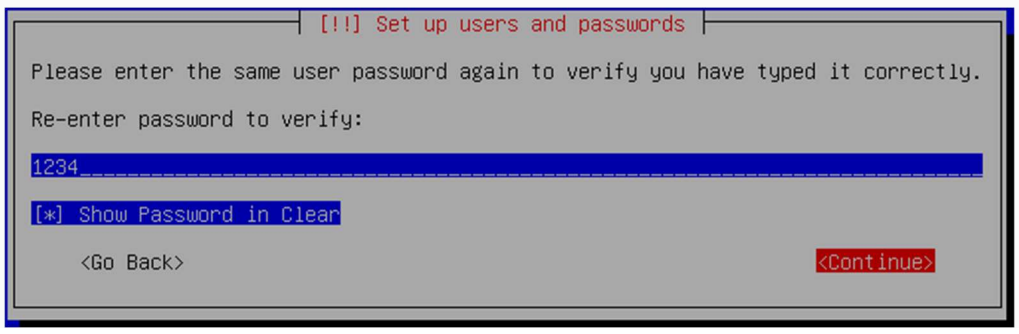

[!!] Set up users and passwords
You need to set a password for 'root', the system administrative account. A malicious or unqualified user with root access can have disastrous results, so you should take care to choose a root password that is not easy to guess. It should not be a word found in dictionaries, or a word that could be easily associated with you.
A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals.
The root user should not have an empty password. If you leave this empty, the root account will be disabled and the system's initial user account will be given the power to become root using the "sudo" command.
Note that you will not be able to see the password as you type it.
Root password:
1234
[*] Show Password in Clear
<Go Back>
<Continue>


[!!] Set up users and passwords
Please enter the same root password again to verify that you have typed it correctly.
Re-enter password to verify:
1234
[*] Show Password in Clear
<Go Back>
<Continue>

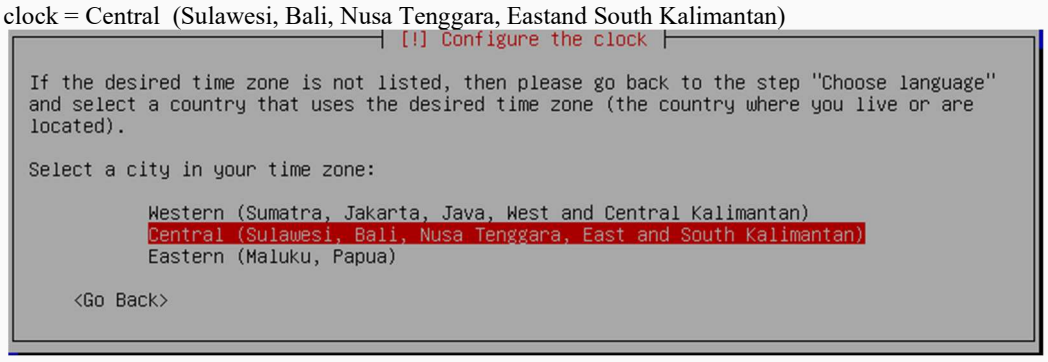

[!!] Set up users and passwords
A user account will be created for you to use instead of the root account for non-administrative activities.
Please enter the real name of this user. This information will be used for instance as default origin for emails sent by this user as well as any program which displays or uses the user's real name. Your full name is a reasonable choice.
Full name for the new user:
smkn3tator
<Go Back>
<Continue>


[!!] Set up users and passwords
Select a username for the new account. Your first name is a reasonable choice. The username should start with a lower-case letter, which can be followed by any combination of numbers and more lower-case letters.
Username for your account:
smkn3tator
<Go Back>
<Continue>

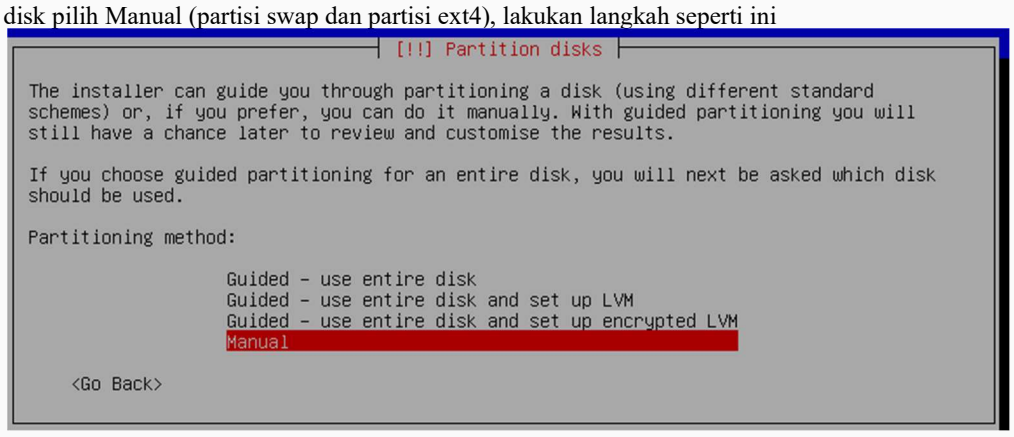

[!!] Set up users and passwords
A good password will contain a mixture of letters, numbers and punctuation and should be changed at regular intervals.
Choose a password for the new user:
1234
[*] Show Password in Clear
<Go Back>
<Continue>

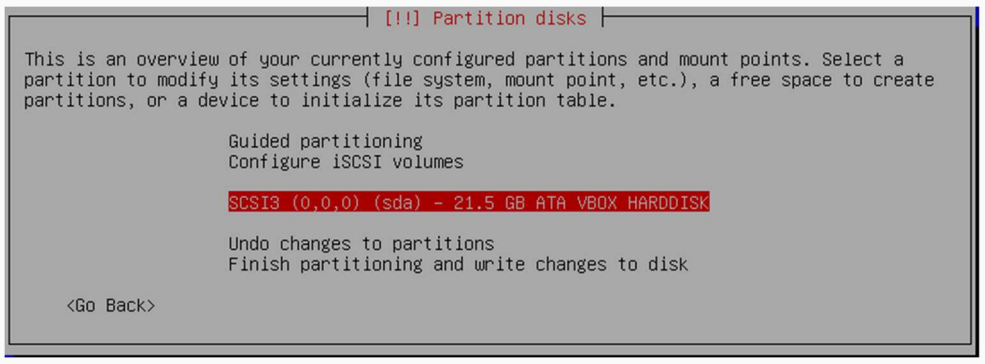

[!!] Set up users and passwords
Please enter the same user password again to verify you have typed it correctly.
Re-enter password to verify:
1234
[*] Show Password in Clear
<Go Back>
<Continue>

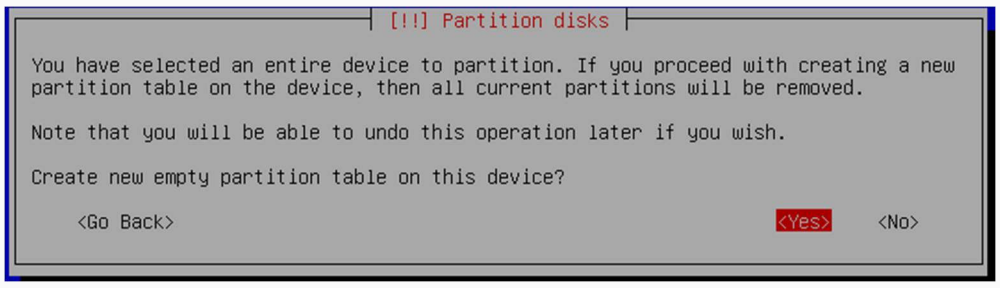
19. Configure the clock = Central (Sulawesi, Bali, Nusa Tenggara, Eastand South Kalimantan)



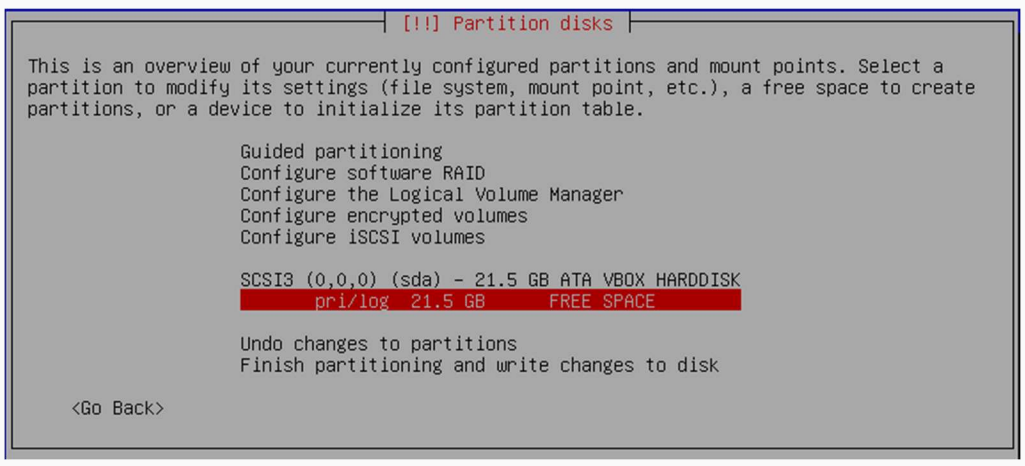
20. Untuk Partition disk pilih Manual (partisi swap dan partisi ext4), lakukan langkah seperti ini



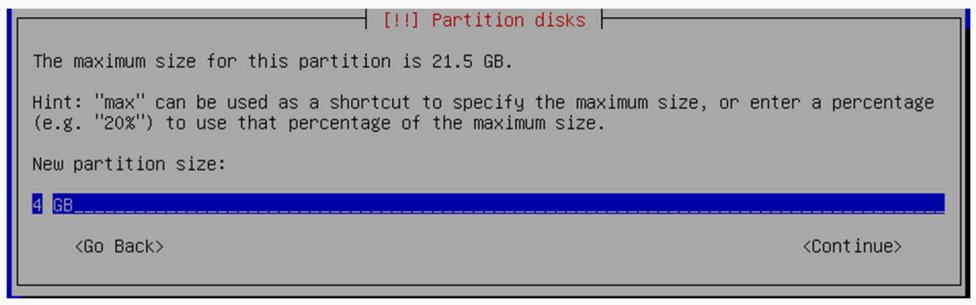




PARTISI SWAP

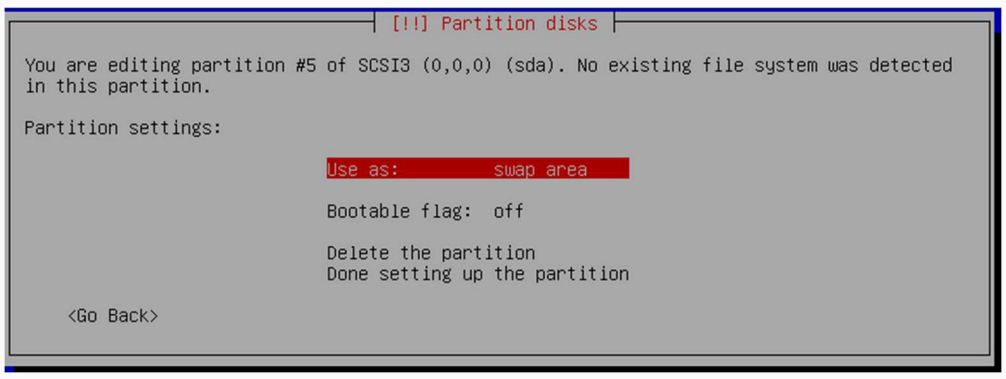


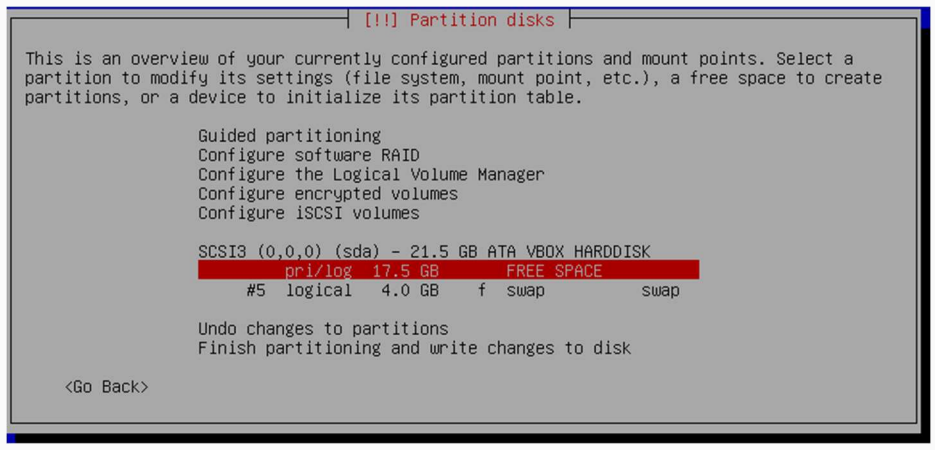

[!!] Partition disks
How to use this free space:
Create a new partition
Automatically partition the free space
Show Cylinder/Head/Sector information
<Go Back>


[!!] Partition disks
The maximum size for this partition is 21.5 GB.
Hint: "max" can be used as a shortcut to specify the maximum size, or enter a percentage (e.g. "20%") to use that percentage of the maximum size.
New partition size:
4 GB
<Go Back>
<Continue>


[!!] Partition disks
Type for the new partition:
Primary
Logical
<Go Back>

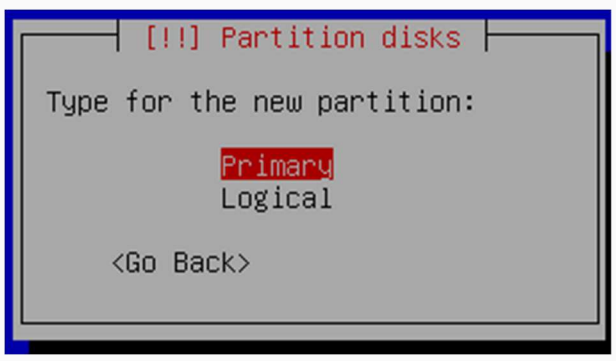

[!!] Partition disks
Please choose whether you want the new partition to be created at the beginning or at the end of the available space.
Location for the new partition:
Beginning
End
<Go Back>


[!!] Partition disks
You are editing partition #5 of SCSI3 (0,0,0) (sda). No existing file system was detected in this partition.
Partition settings:
Use as: swap area
Bootable flag: off
Delete the partition
Done setting up the partition
<Go Back>

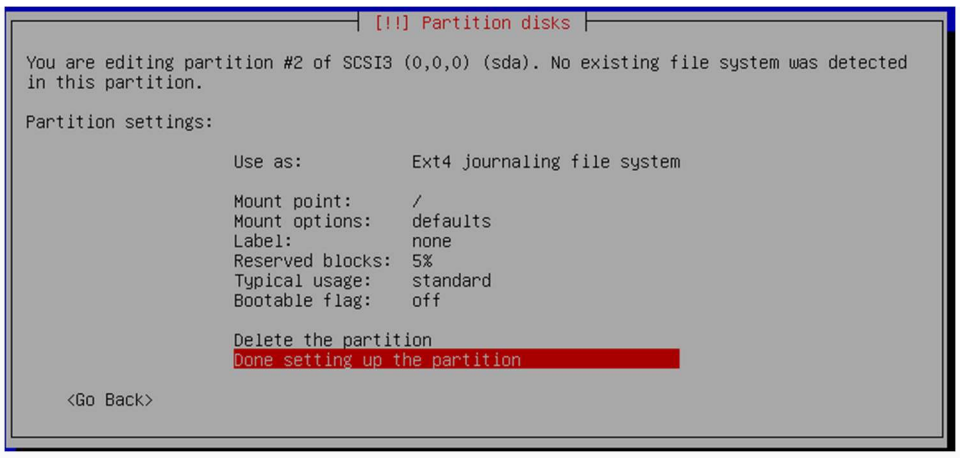

[!!] Partition disks
This is an overview of your currently configured partitions and mount points. Select a partition to modify its settings (file system, mount point, etc.), a free space to create partitions, or a device to initialize its partition table.
Guided partitioning
Configure software RAID
Configure the Logical Volume Manager
Configure encrypted volumes
Configure iSCSI volumes
SCSI3 (0,0,0) (sda) - 21.5 GB ATA VBOX HARDDISK
pri/log 17.5 GB FREE SPACE
#5 logical 4.0 GB f swap swap
Undo changes to partitions
Finish partitioning and write changes to disk
<Go Back>


[!!] Partition disks
How to use this free space:
Create a new partition
Automatically partition the free space
Show Cylinder/Head/Sector information
<Go Back>

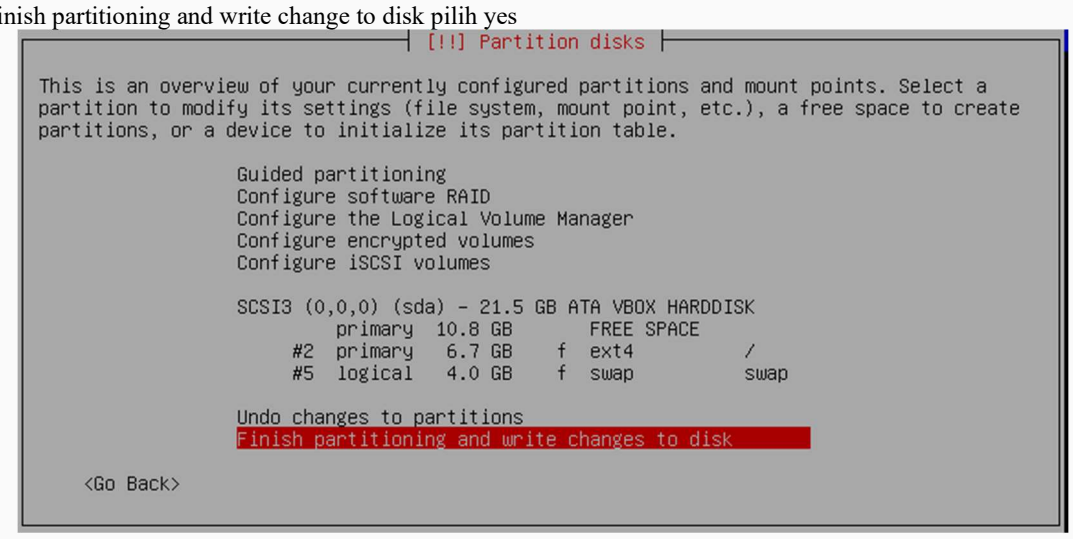

[!!] Partition disks
The maximum size for this partition is 17.5 GB.
Hint: "max" can be used as a shortcut to specify the maximum size, or enter a percentage (e.g. "20%") to use that percentage of the maximum size.
New partition size:
6.7 GB
<Go Back>
<Continue>

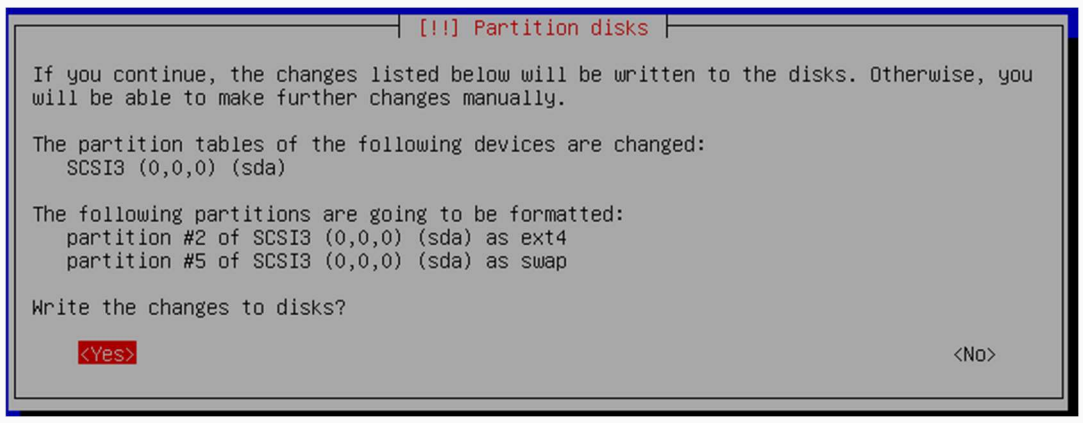

[!!] Partition disks
Type for the new partition:
Primary
Logical
<Go Back>


[!!] Partition disks
Please choose whether you want the new partition to be created at the beginning or at the end of the available space.
Location for the new partition:
Beginning
End
<Go Back>

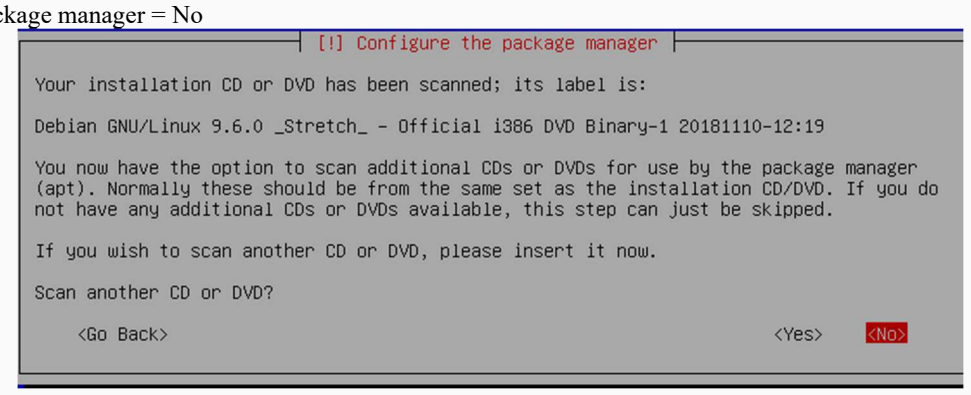


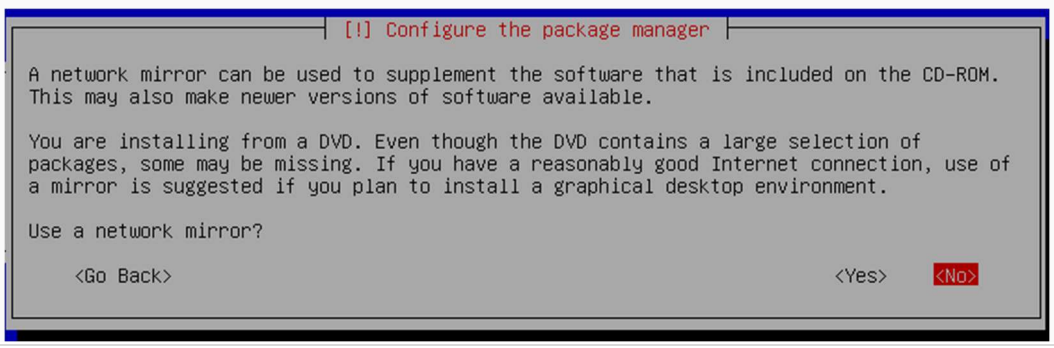
21. Klik menu finish partitioning and write change to disk pilih yes

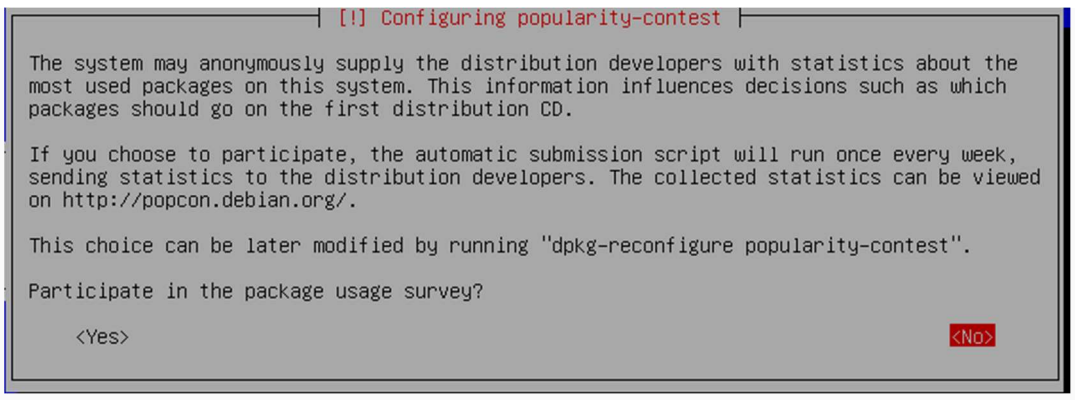




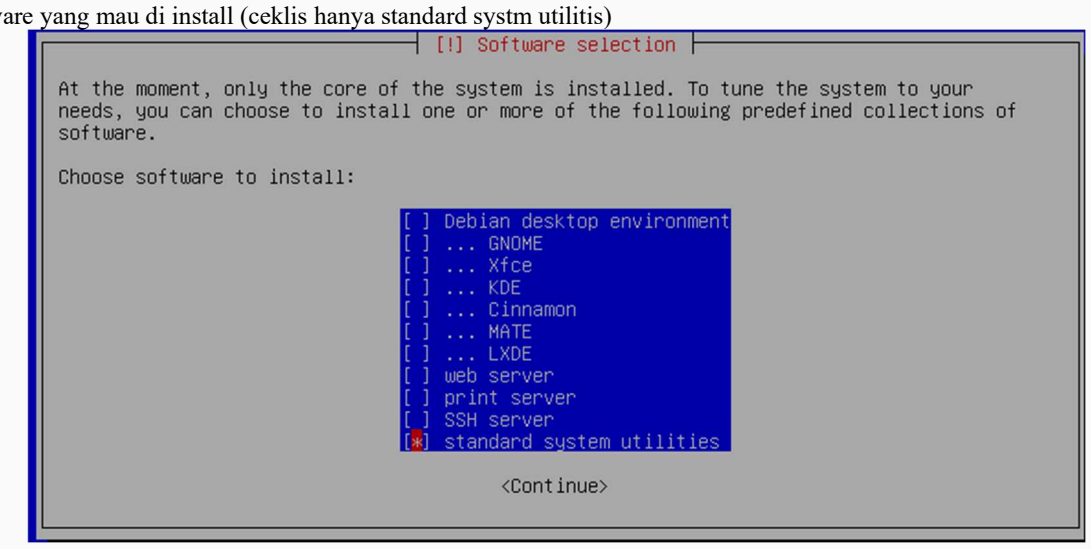
22. Configure the package manager = No



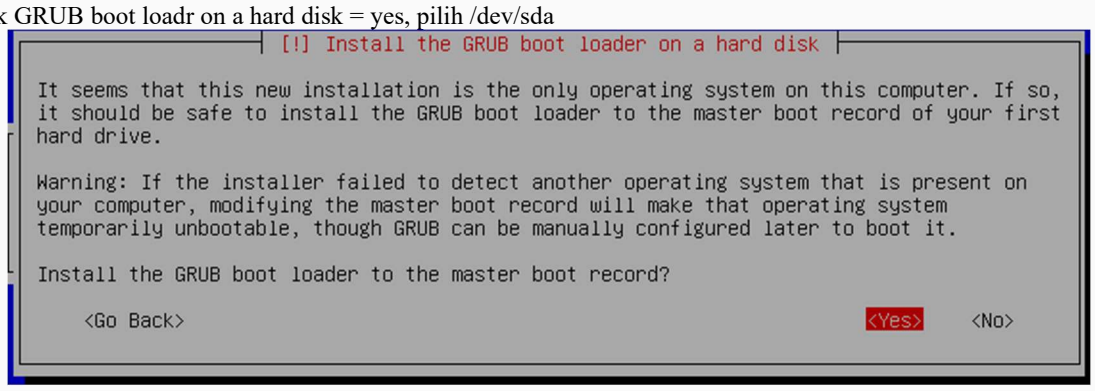


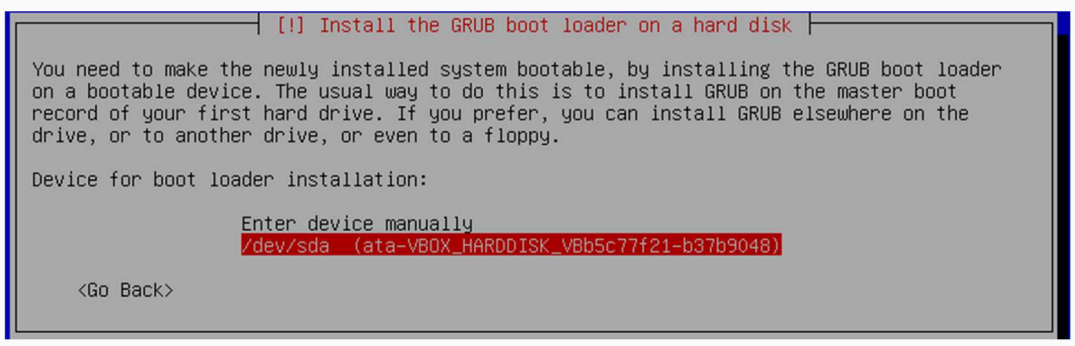


23. Pilih Software yang mau di install (ceklis hanya standard systm utilitis)



24. Install untuk GRUB boot loadr on a hard disk = yes, pilih /dev/sda





25. Klik Finish installasi dan pilih continue, lalu enter.. install sudah selesai dan sekarang kalian bisa login menggunakan user yang sudah di buat.



26. Berikut tampilan jika sudah berhasil di install dan bisa untuk di coba login.



B. Perintah dasar Debian linux

pada sebuah OS baik windows maupun linux pasti memiliki perintah-perintah dasar yang berfungsi untuk membantu para administrator melakukan tugasnya. Berikut perintah dasar dari linux debian:

1. **hostname**, berfungsi untuk merubah nama host pada pc atau perangkat jaringan lainnya. Host ini adalah sebuah identitas atau tanda pengenal untuk membedakan antara device satu dengan yang lainnya.

```
root@debian:~# hostname
debian
```

2. **man**, berfungsi untuk melihat kegunaan dari perintah (melihat buku manual dari sebuah program). Contohnya seperti perintah "*man apt*" akan menampilkan manual penggunaan dari program apt

```
root@debian:~# man apt
APT(8)                                  APT                                  APT(8)

NAME
       apt - command-line interface

SYNOPSIS
       apt [-h] [-o=config_string] [-c=config_file] [-t=target_release]
           [-a=architecture] {list | search | show | update |
           install pkg [{=pkg_version_number | /target_release}]...  |
           remove pkg...  | upgrade | full-upgrade | edit-sources |
           {-v | --version} | {-h | --help}}

DESCRIPTION
       apt provides a high-level commandline interface for the package
       management system. It is intended as an end user interface and enables
       some options better suited for interactive usage by default compared to
       more specialized APT tools like apt-get(8) and apt-cache(8).

       Much like apt itself, its manpage is intended as an end user interface
       and as such only mentions the most used commands and options partly to
       not duplicate information in multiple places and partly to avoid
       overwhelming readers with a cornucopia of options and details.

       update (apt-get(8))
 Manual page apt(8) line 1 (press h for help or q to quit)
```

3. **help**, berfungsi sebagai bantuin untuk menampilkan perintah-perintah apa saja yang bisa dilakukan sehingga ketika kita lupa kita bisa menggunakan perintah ini.

```
root@debian:~# help_
command [-pVv] command [arg ...]        readonly [-aAf] [name[=value] ...] o>
compgen [-abcdefgjksuv] [-o option] [>  return [n]
complete [-abcdefgjksuv] [-pr] [-DE] >  select NAME [in WORDS ... ;] do COMM>
compopt [-o|+o option] [-DE] [name ..>  set [-abefhkmnptuvxBCHP] [-o option->
continue [n]                            shift [n]
coproc [NAME] command [redirections]    shopt [-pqsu] [-o] [optname ...]
declare [-aAfFgilnrtux] [-p] [name[=v>  source filename [arguments]
dirs [-clpv] [+N] [-N]                  suspend [-f]
disown [-h] [-ar] [jobspec ... | pid >  test [expr]
echo [-neE] [arg ...]                   time [-p] pipeline
enable [-a] [-dnps] [-f filename] [na>  times
eval [arg ...]                          trap [-lp] [[arg] signal_spec ...]
exec [-cl] [-a name] [command [argume>  true
exit [n]                                type [-afptP] name [name ...]
export [-fn] [name[=value] ...] or ex>  typeset [-aAffgilnrtux] [-p] name[=v>
false                                   ulimit [-SHabcdefiklmnpqrstuvxPT] [l>
fc [-e ename] [-lnr] [first] [last] o>  umask [-p] [-S] [mode]
fg [job_spec]                           unalias [-a] name [name ...]
for NAME [in WORDS ... ] ; do COMMAND>  unset [-f] [-v] [-n] [name ...]
for (( exp1; exp2; exp3 )); do COMMAN>  until COMMANDS; do COMMANDS; done
function name { COMMANDS ; } or name >  variables - Names and meanings of so>
getopts optstring name [arg]            wait [-n] [id ...]
hash [-lr] [-p pathname] [-dt] [name >  while COMMANDS; do COMMANDS; done
help [-dms] [pattern ...]               { COMMANDS ; }
```

4. **exit**, berfungsi untuk keluar dari super user ke user biasa.

```
root@debian:~# exit
exit
anu@debian:~$ _
```

5. **su** (sudo), berfungsi untuk masuk ke root atau super user.

```
anu@debian:~$ su
Password:
root@debian:/home/anu#
```

6. **mkdir**, berfungsi untuk membuat folder baru. ls, berfungsi untuk Melihat daftar file & folder yang ada pada direktori saat itu.

```
root@debian:~# mkdir latihan
root@debian:~# ls
latihan
root@debian:~#
```

7. **cd**, berfungsi untuk masuk ke dalam direktori atau folder. Misal, saya ingin masuk ke folder latihan, maka perintahnya "cd latihan".

```
root@debian:~# mkdir latihan
root@debian:~# ls
latihan
root@debian:~# cd latihan/
root@debian:~/latihan# _
```

8. **rmdir**, berfungsi untuk menghapus folder atau direktori. Misal, disini saya ingin menghapus folder atau direktori latihan, maka perintahnya "rmdir latihan". Setelah sudah coba tes dengan melakukan perintah ls dan masuk ke dalam direktori tersebut.

```
root@debian:~# rmdir latihan
root@debian:~# ls
root@debian:~# cd latihan
bash: cd: latihan: No such file or directory
root@debian:~# _
```

9. **touch**, berfungsi untuk membuat file dengan ekstensi yang anda butuhkan. contoh saya ingin membuat file document/text, maka perintahnya ""touch anu.ext.

```
root@debian:~# touch anu.txt
root@debian:~# ls
anu.txt
root@debian:~#
```

10. **pwd**, berfungsi untuk melihat direktori kerja yang pada saat itu aktif.

```
root@debian:~# pwd
/root
root@debian:~#
```

11. **cal**, berfungsi untuk melihat kalender. Perintah ini hanya menampilkan bulan yang sedang berlangsung saat ini.

```
root@debian:~# cal
     August 2022
Su Mo Tu We Th Fr Sa
    1  2  3  4  5  6
 7  8  9 10 11 12 13
14 15 16 17 18 19 20
21 22 23 24 25 26 27
28 29 30 31
```

12. **date**, untuk menampilkan detail waktu yaitu berupa hari, tanggal, bulan, tahun dan waktu saat ini.

```
root@debian:~# date
Tue Aug  9 06:03:35 EDT 2022
root@debian:~# _
```

13. **nano**, berfungsi sebagai text editor yang tidak perlu membuka jendela baru. Dengan perintah ini kita bisa memasukkan isi text pada file ekstensi .txt atau file document lainnya.

```
root@debian:~# nano anu.txt
```

```
 GNU nano 2.7.4                  File: anu.txt

PERINGATAN!!! JANGAN LUPA SENYUM!!!!
```

14. **cat**, berfungsi untuk menampilkan atau melihat isi text file.

```
root@debian:~# cat anu.txt
PERINGATAN!!! JANGAN LUPA SENYUM!!!!
root@debian:~#
```

15. **cp**, berfungsi untuk mengcopy file.

```
root@debian:~# cp anu.txt latihan/
root@debian:~# cd latihan/
root@debian:~/latihan# ls
anu.txt
root@debian:~/latihan# _
```

16. **cp -r**, berfungsi untuk mengcopy folder atau direktori.

```
root@debian:~# cp -r latihan/ testing
root@debian:~# ls
anu.txt  latihan  testing
root@debian:~#
```

17. **cd** /, berfungsi untuk pindah ke direktori root. Direktori root ditandai dengan tanda (/), sedangkan tanda (~) adalah direktori home.

```
root@debian:~# cd /
root@debian:/# _
```

18. **rm**, berfungsi untuk menghapus file.

```
root@debian:~/testing# rm anu.txt
```

19. **mv**, berfungsi untuk memindahkan file atau folder.

```
root@ikbal:/testing# ls
ikbal.txt  latihan
root@ikbal:/testing# mv ikbal.txt /latihan
root@ikbal:/testing# ls
latihan
root@ikbal:/testing# cd /latihan
root@ikbal:/latihan# ls
ikbal.txt
root@ikbal:/latihan# _
```

20. **find -name**, berfungsi untuk mencari file atau folder terletak dimana atau ada dimana saja.

```
root@ikbal:/# find -name ikbal.txt
./latihan/ikbal.txt
./testing/ikbal.txt
root@ikbal:/# _
```

21. **echo**, berfungsi untuk menampilkan satu baris teks. Bisa juga untuk menuliskan sebuah teks kedalam file, contohnya seperti berikut "echo "peringatan" >> ikbal.txt". Perintah tersebut akan menuliskan “Teks” ke dalam file “niagahoster.txt”, jika file tersebut belum ada maka otomatis akan dibuat.

```
root@ikbal:/# echo "PERINGATAN">>ikbal.txt
root@ikbal:/# cat ikbal.txt
PERINGATAN
root@ikbal:/# _
```

22. **who**, berfungsi untuk menampilkan user yang pada saat ini dipakai.

```
root@ikbal:/# who
ikbal    tty1         2019-04-20 11:03
```

23. **passwd**, berfungsi untuk membuat password baru.

```
root@ikbal:/# passwd
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
root@ikbal:/# _
```

24. **history**, berfungsi untuk melihat perintah apa saja yang sudah dilakukan.

```
root@ikbal:/# history_
```

25. **ps**, berfungsi untuk menampilkan snapshot process yang sedang berjalan.

```
root@ikbal:/# ps
  PID TTY          TIME CMD
  832 tty1     00:00:00 login
  847 tty1     00:00:00 bash
  853 tty1     00:00:00 ps
```

26. **fdisk -l**, berfungsi untuk menampilkan list partisi pada perangkat.

```
root@ikbal:/# fdisk -l
Disk /dev/sda: 200 GiB, 214748364800 bytes, 419430400 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x25f964c4

Device     Boot     Start       End   Sectors  Size Id Type
/dev/sda1  *         2048 195311615 195309568 93.1G 83 Linux
/dev/sda2       195313662 396482559 201168898 95.9G  5 Extended
/dev/sda5       195313664 390623231 195309568 93.1G 83 Linux
/dev/sda6       390625280 394520767   3903488  1.9G 82 Linux swap / Solaris
/dev/sda7       394530816 396482559   1951744  953M 83 Linux
```

27. **df**, berfungsi untuk menampilkan penggunaan ruang disk pada filesystem.

```
root@ikbal:/# df
Filesystem     1K-blocks   Used Available Use% Mounted on
udev              499032      0    499032   0% /dev
tmpfs             102036   2944     99092   3% /run
/dev/sda1       95595940 805484  89891336   1% /
tmpfs             510164      0    510164   0% /dev/shm
tmpfs               5120      0      5120   0% /run/lock
tmpfs             510164      0    510164   0% /sys/fs/cgroup
/dev/sda7         944120  36716    842228   5% /boot
/dev/sda5       95595940  61484  90635336   1% /home
tmpfs             102032      0    102032   0% /run/user/0
```

28. **reboot**, berfungsi untuk melakukan restart atau reboot pada pc.

```
root@ikbal:/# reboot_
```

29. **poweroff**, berfungsi untuk mematikan pc.

```
root@ikbal:/# poweroff_
```

30. **top**, berfungsi untuk melihat semua proses yang sedang berjalan, diurutkan dari proses yang paling besar. Fungsinya hampir sama seperti system monitor.

```
root@ikbal:/# top_
top - 13:10:12 up  2:22,  1 user,  load average: 0.05, 0.02, 0.00
Tasks:  70 total,   1 running,  69 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.2 sy,  0.0 ni, 99.7 id,  0.1 wa,  0.0 hi,  0.0 si,  0.0 st
KiB Mem :  1020332 total,   826716 free,    36908 used,   156708 buff/cache
KiB Swap:  1951740 total,  1951740 free,        0 used.   832064 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU %MEM     TIME+ COMMAND
    1 root      20   0   56900   6540   5152 S   0.0  0.6   0:00.97 systemd
    2 root      20   0       0      0      0 S   0.0  0.0   0:00.00 kthreadd
    3 root      20   0       0      0      0 S   0.0  0.0   0:00.04 ksoftirqd/0
    5 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 kworker/0:0H
    7 root      20   0       0      0      0 S   0.0  0.0   0:00.22 rcu_sched
    8 root      20   0       0      0      0 S   0.0  0.0   0:00.00 rcu_bh
    9 root      rt   0       0      0      0 S   0.0  0.0   0:00.00 migration/0
   10 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 lru-add-dra+
   11 root      rt   0       0      0      0 S   0.0  0.0   0:00.03 watchdog/0
   12 root      20   0       0      0      0 S   0.0  0.0   0:00.00 cpuhp/0
   13 root      20   0       0      0      0 S   0.0  0.0   0:00.00 kdevtmpfs
   14 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 netns
   15 root      20   0       0      0      0 S   0.0  0.0   0:00.00 khungtaskd
   16 root      20   0       0      0      0 S   0.0  0.0   0:00.00 oom_reaper
   17 root       0 -20       0      0      0 S   0.0  0.0   0:00.00 writeback
   18 root      20   0       0      0      0 S   0.0  0.0   0:00.00 kcompactd0
   19 root      25   5       0      0      0 S   0.0  0.0   0:00.00 ksmd
   21 root      39  19       0      0      0 S   0.0  0.0   0:00.00 khugepaged
```

31. **uname -a**, berfungsi untuk menampilkan versi kernel yang dipakai, tanggal instalasi, dan jenis arsitektur sistem operasi.

```
root@ikbal:/# uname -a
Linux ikbal 4.9.0-8-amd64 #1 SMP Debian 4.9.130-2 (2018-10-27) x86_64 GNU/Linux
```

32. **clear**, berfungsi untuk membersihkan jendela terminal. Jadi isi jendela terminal akan kosong, namun jika di scroll keatas maka perintah yang sebelumnya dijalankan masih bisa terlihat.

33. **chmod**, berfungsi untuk mengganti hak akses pada sebuah file. Contohnya jika ingin mengganti hak akses ikbal.txt menjadi 124 maka menggunakan perintah "chmod 124 ikbal.txt".
ket : x = 1, w = 2, r = 4.

```
root@ikbal:/latihan# chmod 124 ikbal.txt
root@ikbal:/latihan# ls -l
total 0
---x-w-r-- 1 root root 0 Apr 20 11:38 ikbal.txt
```

34. **chown**, berfungsi untuk mengganti pemilik dan group dari sebuah file. Contohnya disini saya ingin mengubah kepemilikan ikbal.txt menjadi “ikbal root” bisa menggunakan perintah "chown ikbal:root ikbal.txt. Kata “ikbal” di depan merujuk pada user sedangkan “root” di belakang merujuk pada nama group.

```
root@ikbal:/latihan# chown ikbal:root ikbal.txt
root@ikbal:/latihan# ls -l
total 0
---x-w-r-- 1 ikbal root 0 Apr 20 11:38 ikbal.txt
```

35. **dir**, berfungsi untuk untuk melihat isi directory.

```
root@ikbal:/# dir
bin   home          latihan     media  root  sys      var
boot  ikbal.txt     lib         mnt    run   testing  vmlinuz
dev   initrd.img    lib64       opt    sbin  tmp      vmlinuz.old
etc   initrd.img.old lost+found  proc   srv   usr
```

36. **ls -lh**, berfungsi untuk menampilkan isi direktori secara lengkap, mulai dari hak akses, owner, group dan tanggal file atau direktori tersebut dibuat.

```
root@ikbal:/# dir
bin   home          latihan     media  root  sys      var
boot  ikbal.txt     lib         mnt    run   testing  vmlinuz
dev   initrd.img    lib64       opt    sbin  tmp      vmlinuz.old
etc   initrd.img.old lost+found  proc   srv   usr
```

37. **free -h**, berfungsi untuk melihat free memory.

```
root@ikbal:/# free -h
               total        used        free      shared  buff/cache   available
Mem:           996M         35M        807M        3.1M        153M        812M
Swap:          1.9G          0B        1.9G
```

38. **nano /etc/network/interfaces**, berfungsi untuk memasukkan network atau ip address secara static.

```
root@ikbal:/# nano /etc/network/interfaces_

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto enp0s3
iface enp0s3 inet static
address 172.16.0.220/16
gateway 172.16.0.1_
```

39. **systemctl restart networking**, berfungsi untuk melakukan restart system sehingga tidak perlu restart secara gui pada pc.

```
root@ikbal:/# systemctl restart networking
root@ikbal:/# reboot_
```

40. **nano /etc/resolv.conf**, berfungsi untuk memasukkan alamat dns sehingga bisa diakses.

```
root@ikbal:/# nano /etc/resolv.conf_
 GNU nano 2.7.4                File: /etc/resolv.conf

nameserver 8.8.8.8
```

41. **apropos**, berfungsi untuk melihat bantuan manual berdasarkan topik.

```
root@ikbal:/# apropos nano
editor (1)           - Nano's ANOther editor, an enhanced free Pico clone
nano (1)             - Nano's ANOther editor, an enhanced free Pico clone
nanorc (5)           - GNU nano's configuration file
pico (1)             - Nano's ANOther editor, an enhanced free Pico clone
rnano (1)            - a restricted nano
traceroute-nanog (1) - print the route packets trace to network host
root@ikbal:/# apropos mkdir
mkdir (1)            - make directories
```

42. **ping**, berfungsi untuk melakukan pengetesan koneksi antara pc dengan network lainnya atau alamat yang dituju.

```
root@ikbal:/# ping google.com
PING google.com (74.125.200.102) 56(84) bytes of data.
64 bytes from sa-in-f102.1e100.net (74.125.200.102): icmp_seq=1 ttl=43 time=18.3
 ms
64 bytes from sa-in-f102.1e100.net (74.125.200.102): icmp_seq=2 ttl=43 time=17.0
 ms
64 bytes from sa-in-f102.1e100.net (74.125.200.102): icmp_seq=3 ttl=43 time=17.5
 ms
64 bytes from sa-in-f102.1e100.net (74.125.200.102): icmp_seq=4 ttl=43 time=32.8
 ms
64 bytes from sa-in-f102.1e100.net (74.125.200.102): icmp_seq=5 ttl=43 time=18.0
 ms
64 bytes from sa-in-f102.1e100.net (74.125.200.102): icmp_seq=6 ttl=43 time=17.9
 ms
```

43. **uptime**, berfungsi untuk melihat jumlah waktu pemakaian komputer oleh seseorang, terhitung proses reboot terakhir.

```
root@ikbal:/# uptime
 13:43:44 up 8 min,  2 users,  load average: 0.00, 0.02, 0.00
```

44. **adduser**, berfungsi untuk menambahkan user baru pc. Dalam satu pc kita bisa membuat lebih dari satu user.

```
root@ikbal:/# adduser baihaqi
Adding user `baihaqi' ...
Adding new group `baihaqi' (1001) ...
Adding new user `baihaqi' (1001) with group `baihaqi' ...
Creating home directory `/home/baihaqi' ...
Copying files from `/etc/skel' ...
Enter new UNIX password:
Retype new UNIX password:
passwd: password updated successfully
Changing the user information for baihaqi
Enter the new value, or press ENTER for the default
        Full Name []:
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
```

45. **addgroup**, berfungsi untuk membuat sebuah group dalam satu jaringan.

```
root@ikbal:/# addgroup tkj
Adding group `tkj' (GID 1002) ...
Done.
```

46. **dmesg**, berfungsi untuk melihat hardware yang sedang beraktifitas.

```
root@ikbal:/# dmesg_
```

47. **apt-get**, berfungsi untuk memperoleh paket/software dari repository linux secara online. Untuk menggunakan perintah ini pc anda harus terkoneksi internet.

```
root@ikbal:/# apt-get
apt 1.4.8 (amd64)
Usage: apt-get [options] command
       apt-get [options] install|remove pkg1 [pkg2 ...]
       apt-get [options] source pkg1 [pkg2 ...]

apt-get is a command line interface for retrieval of packages
and information about them from authenticated sources and
for installation, upgrade and removal of packages together
with their dependencies.

Most used commands:
```

48. **ps axu**, berfungsi untuk melihat seluruh proses yang dijalankan.

```
root@ikbal:/# ps axu
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.6  57092  6820 ?        Ss   13:34   0:00 /sbin/init
root           2  0.0  0.0      0     0 ?        S    13:34   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        S    13:34   0:00 [ksoftirqd/0]
root           5  0.0  0.0      0     0 ?        S<   13:34   0:00 [kworker/0:0H]
root           7  0.0  0.0      0     0 ?        S    13:34   0:00 [rcu_sched]
root           8  0.0  0.0      0     0 ?        S    13:34   0:00 [rcu_bh]
root           9  0.0  0.0      0     0 ?        S    13:34   0:00 [migration/0]
root          10  0.0  0.0      0     0 ?        S<   13:34   0:00 [lru-add-drain]
root          11  0.0  0.0      0     0 ?        S    13:34   0:00 [watchdog/0]
root          12  0.0  0.0      0     0 ?        S    13:34   0:00 [cpuhp/0]
```

49. **service --status-all**, berfungsi untuk melihat service / program yang sedang bekerja.

```
root@ikbal:/# service --status-all
 [ + ]  apache-htcacheclean
 [ + ]  apache2
 [ - ]  console-setup.sh
 [ + ]  cron
 [ + ]  dbus
 [ - ]  hwclock.sh
 [ - ]  keyboard-setup.sh
 [ + ]  kmod
 [ + ]  networking
 [ + ]  procps
 [ + ]  rsyslog
 [ + ]  ssh
 [ + ]  udev
```

50. **cal -y**, berfungsi untuk menampilkan kalender selama setahun penuh.

```
root@ikbal:/# cal -y
                                2019
      January               February                March
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
       1  2  3  4  5                  1  2                  1  2
 6  7  8  9 10 11 12   3  4  5  6  7  8  9   3  4  5  6  7  8  9
13 14 15 16 17 18 19  10 11 12 13 14 15 16  10 11 12 13 14 15 16
20 21 22 23 24 25 26  17 18 19 20 21 22 23  17 18 19 20 21 22 23
27 28 29 30 31        24 25 26 27 28        24 25 26 27 28 29 30
                                            31

      April                 May                   June
Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa  Su Mo Tu We Th Fr Sa
    1  2  3  4  5  6            1  2  3  4                  1
```

51. **lspci**, berfungsi untuk melihat perangkat pci yang sedang terkoneksi ke komputer.

```
root@ikbal:/# lspci
00:00.0 Host bridge: Intel Corporation 440FX - 82441FX PMC [Natoma] (rev 02)
00:01.0 ISA bridge: Intel Corporation 82371SB PIIX3 ISA [Natoma/Triton II]
00:01.1 IDE interface: Intel Corporation 82371AB/EB/MB PIIX4 IDE (rev 01)
00:02.0 VGA compatible controller: InnoTek Systemberatung GmbH VirtualBox Graphi
cs Adapter
00:03.0 Ethernet controller: Intel Corporation 82540EM Gigabit Ethernet Controll
er (rev 02)
00:04.0 System peripheral: InnoTek Systemberatung GmbH VirtualBox Guest Service
00:05.0 Multimedia audio controller: Intel Corporation 82801AA AC'97 Audio Contr
oller (rev 01)
00:06.0 USB controller: Apple Inc. KeyLargo/Intrepid USB
00:07.0 Bridge: Intel Corporation 82371AB/EB/MB PIIX4 ACPI (rev 08)
00:0d.0 SATA controller: Intel Corporation 82801HM/HEM (ICH8M/ICH8M-E) SATA Cont
roller [AHCI mode] (rev 02)
```

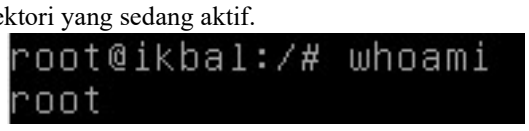
52. **deluser**, berfungsi untuk menghapus user pada pc.

```
root@ikbal:/# deluser baihaqi
Removing user `baihaqi' ...
Warning: group `baihaqi' has no more members.
Done.
```

53. **lsusb**, berfungsi untuk menampilkan informasi usb.

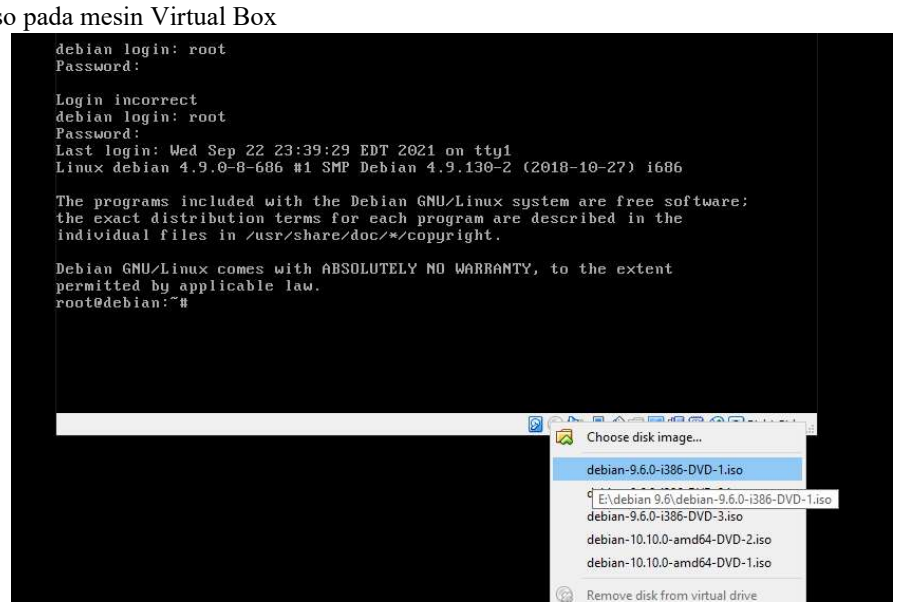


54. **whoami**, berfungsi untuk melihat direktori yang sedang aktif.



C. Konfigurasi IP Address

1. Masukkan DVD 1.iso pada mesin Virtual Box



2. Lakukan penginstallan paket - paket pada debian Dengan perintah **#apt install net-tools dnsutils**

```
permitted by applicable law.
root@debian:~# apt install net-tools dnsutils_
```

Tekan tombol Y untuk melanjutkan

3. Ketikkan perintah **#nano /etc/network/interfaces** untuk melakukan konfigurasi IP Address

```
root@debian:~# nano /etc/network/interfaces
```

4. Lakukan pengeditan pada direktori **/etc/network/interfaces**

```
  GNU nano 2.7.4            File: /etc/network/interfaces              Modifi

# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback

auto enp0s3
iface enp0s3 inet static
        address 192.168.10.1/24
```

Simpan dengan menekan tombol **CTRL+X → Y → Enter**

5. Konfigurasi file sysctl.conf untuk mengaktifkan ip forward.
#nano /etc/sysctl.conf
Hapus tanda pagar (#) pada line **net.ipv4.ip_forward=1**

```
# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=1
```

6. Restart konfigurasi ip address dengan perintah **#/etc/init.d/networking restart**

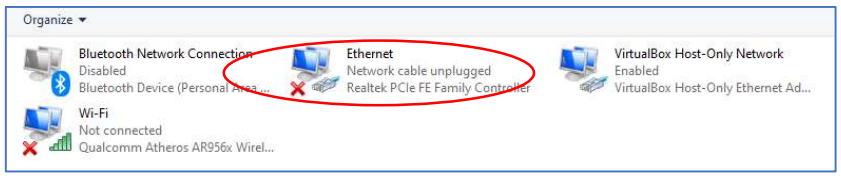
```
root@debian:~# /etc/init.d/networking restart
[ ok ] Restarting networking (via systemctl): networking.service.
root@debian:~# _
```

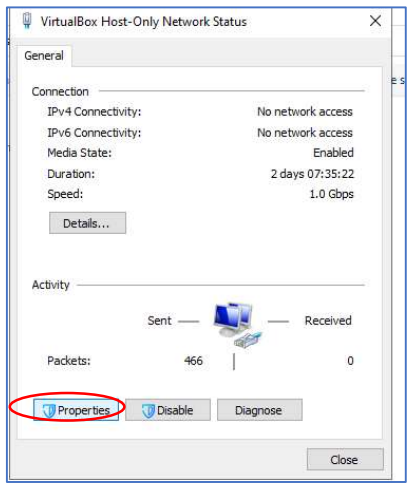
7. Lakukan pengujian dengan perintah **#ip a**

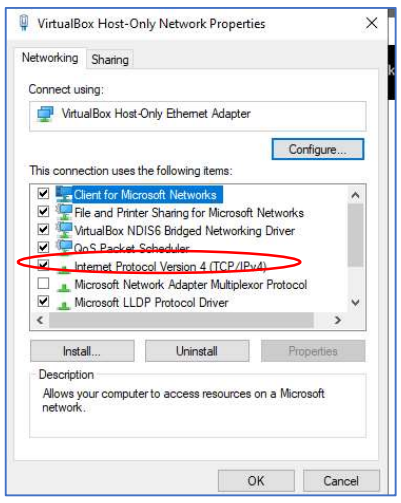
```
root@debian:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN gro
t qlen 1
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast
group default qlen 1000
    link/ether 08:00:27:3d:c6:d4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.1/24 brd 192.168.10.255 scope global enp0s3
       valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe3d:c6d4/64 scope link
       valid_lft forever preferred_lft forever
```

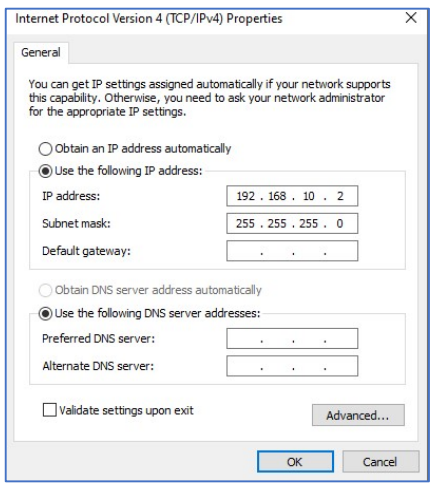
Catatan: jika ip addres tidak muncul pada linux, maka lakukan restart pada linux dengan perintah ***reboot***

8. Konfigurasi ip address pada TCP/IP windows (client)

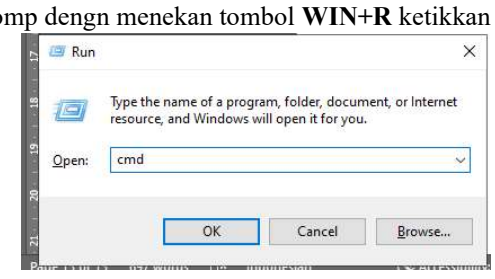








9. Lakukan pengujian pada command promp dengn menekan tombol **WIN+R** ketikkan **cmd** + **OK**



10. Ketikkan perintah ping 192.168.10.1

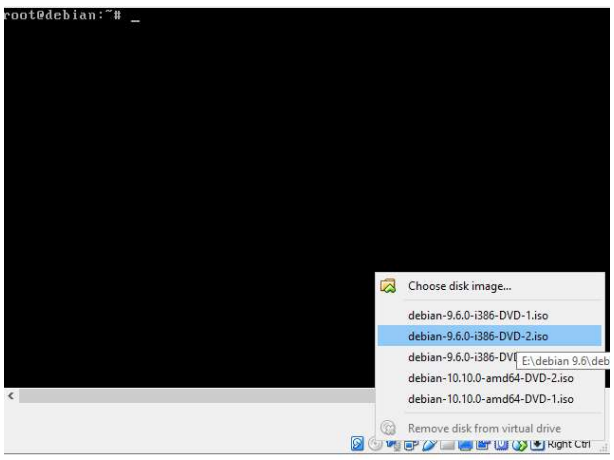
```
C:\Users\ANU>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=1ms TTL=64
Reply from 192.168.10.1: bytes=32 time=1ms TTL=64
Reply from 192.168.10.1: bytes=32 time=1ms TTL=64
Reply from 192.168.10.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

D. Konfigurasi DHCP Server

1. Masukkan DVD 2. Iso Debian



2. Ketikkan perintah **#apt-cdrom add**

```
root@debian:~# apt-cdroom add
-bash: apt-cdroom: command not found
root@debian:~# apt-cdrom add
Using CD-ROM mount point /media/cdrom/
Unmounting CD-ROM...
Waiting for disc...
Please insert a Disc in the drive and press [Enter]
Mounting CD-ROM...
Identifying... [f5062ef217e5ef53f5b1670f43e2af0a-2]
Scanning disc for index files...
Found 2 package indexes, 0 source indexes, 2 translation indexes and 0
s
Found label 'Debian GNU/Linux 9.6.0 _Stretch_ - Official i386 DVD Binar
110-12:19'
This disc is called:
'Debian GNU/Linux 9.6.0 _Stretch_ - Official i386 DVD Binary-2 20181110
Reading Package Indexes... Done
Reading Translation Indexes... Done
Writing new source list
Source list entries for this disc are:
deb cdrom:[Debian GNU/Linux 9.6.0 _Stretch_ - Official i386 DVD Binary-
0-12:19]/ stretch contrib main
Unmounting CD-ROM...
Repeat this process for the rest of the CDs in your set.
```

3. Lakukan penginstalan paket DHCP Server dengan perintah **#apt install isc-dhcp-server**

```
root@debian:~# apt install isc-dhcp-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libirs-export141 libisccc-export140 libisccfg-export140 policycoreuti
  selinux-utils
Suggested packages:
  policykit-1 isc-dhcp-server-ldap
The following NEW packages will be installed:
  isc-dhcp-server libirs-export141 libisccc-export140 libisccfg-export1
  policycoreutils selinux-utils
0 upgraded, 6 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/1,777 kB of archives.
After this operation, 6,779 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Tekan ombol **Y** untuk melanjutkan penginstallan

```
● isc-dhcp-server.service - LSB: DHCP server
   Loaded: loaded (/etc/init.d/isc-dhcp-server; generated; vendor prese
d)
   Active: failed (Result: exit-code) since Sat 2021-10-23 07:47:04 EDT
o
     Docs: man:systemd-sysv-generator(8)
  Process: 1148 ExecStart=/etc/init.d/isc-dhcp-server start (code=exite
=1/FAILURE)

Oct 23 07:47:02 debian dhcpd[1159]: No subnet declaration for enp0s3 (1
Oct 23 07:47:02 debian dhcpd[1159]: ** Ignoring requests on enp0s3.  If
Oct 23 07:47:02 debian dhcpd[1159]:    you want, please write a subnet
Oct 23 07:47:02 debian dhcpd[1159]:    in your dhcpd.conf file for the
Oct 23 07:47:04 debian isc-dhcp-server[1148]: Starting ISC DHCPv4 serve
Oct 23 07:47:04 debian isc-dhcp-server[1148]:  failed!
Oct 23 07:47:04 debian systemd[1]: isc-dhcp-server.service: Control pro
Oct 23 07:47:04 debian systemd[1]: Failed to start LSB: DHCP server.
Oct 23 07:47:04 debian systemd[1]: isc-dhcp-server.service: Unit entere
Oct 23 07:47:04 debian systemd[1]: isc-dhcp-server.service: Failed with
Hint: Some lines were ellipsized, use -l to show in full.
Processing triggers for libc-bin (2.24-11+deb9u3) ...
Processing triggers for systemd (232-25+deb9u6) ...
root@debian:~# _
```

4. Lakukan konfigurasi pada directory **/etc/dhcp/dhcpd.conf** dengan perintah **#nano /etc/dhcp/dhcpd.conf**
   Lakukan pengeditan

```
# A slightly different configuration for an internal subnet.
#subnet 10.5.5.0 netmask 255.255.255.224 {
#  range 10.5.5.26 10.5.5.30;
#  option domain-name-servers ns1.internal.example.org;
#  option domain-name "internal.example.org";
#  option routers 10.5.5.1;
#  option broadcast-address 10.5.5.31;
#  default-lease-time 600;
#  max-lease-time 7200;
#}
```

Menjadi

```
# A slightly different configuration for an internal subnet.
subnet 192.168.10.0 netmask 255.255.255.0 {
  range 192.168.10.25 192.168.10.60;
  option domain-name-servers 192.168.10.1;
  option domain-name "smkn3tator.net";
  option routers 192.168.10.1;
  option broadcast-address 192.168.10.254;
  default-lease-time 600;
  max-lease-time 7200;
}
```

5. Lakukan pengeditan pada **/etc/default/isc-dhcp-server** dengan perintah **#nano /etc/default/isc-dhcp-server**

```
 GNU nano 2.7.4          File: /etc/default/isc-dhcp-server

# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID in
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4=""
INTERFACESv6=""
```

Menjadi

```
 GNU nano 2.7.4          File: /etc/default/isc-dhcp-server          M

# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID in
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s3"
INTERFACESv6=""
```

6. Lakukan restart pada **isc-dhcp-server** dengan perintah **# /etc/init.d/isc-dhcp-server restart**

```
root@debian:~# /etc/init.d/isc-dhcp-server restart
[ ok ] Restarting isc-dhcp-server (via systemctl): isc-dhcp-server.serv
root@debian:~#
```

7. Lakukan pengujian pada windows dengan menghapus IP Address pada TCP/IP kemudian buka Common promp, ketikkan perintah **ipconfig /release** untuk menghapus IP Address

```
C:\Users\ANU>ipconfig /release

Windows IP Configuration

No operation can be performed on Ethernet while it has its media disconn
No operation can be performed on Local Area Connection* 1 while it has i
No operation can be performed on Wi-Fi while it has its media disconnect

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter VirtualBox Host-Only Network:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::89e6:9145:2936:6365%10
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

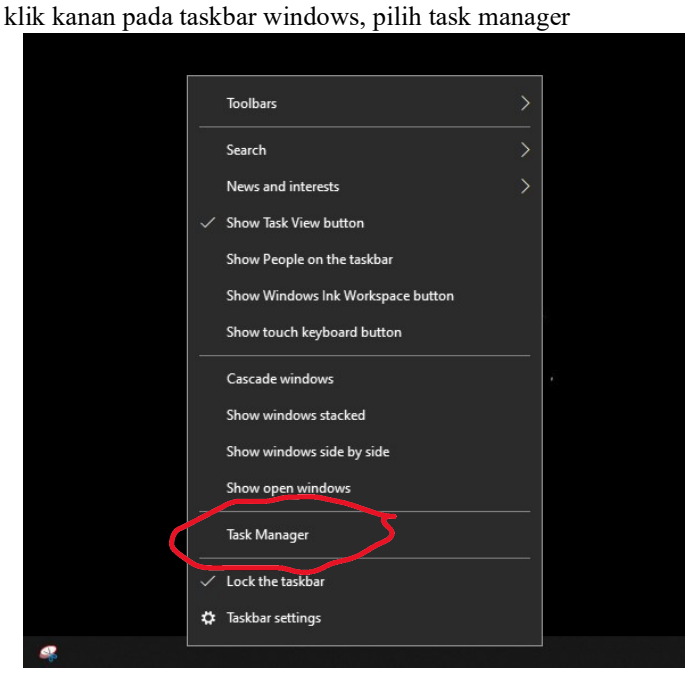
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

8. Ketikkan perintah **ipconfig /renew** untuk meminta IP Address ke server

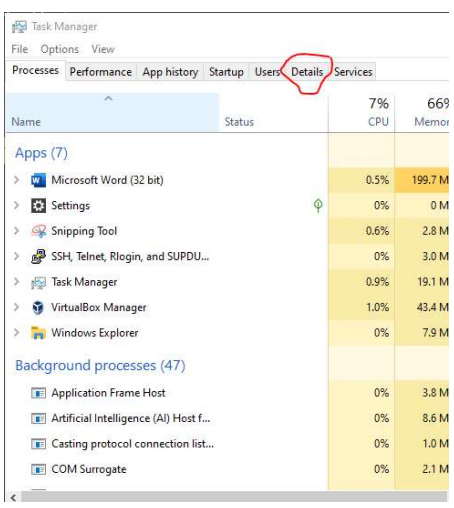
```
Ethernet adapter VirtualBox Host-Only Network:

   Connection-specific DNS Suffix  . : smkn3.net
   Link-local IPv6 Address . . . . . : fe80::256b:90f9:5e17:7636%23
   IPv4 Address. . . . . . . . . . . : 192.168.100.26
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.100.1
```

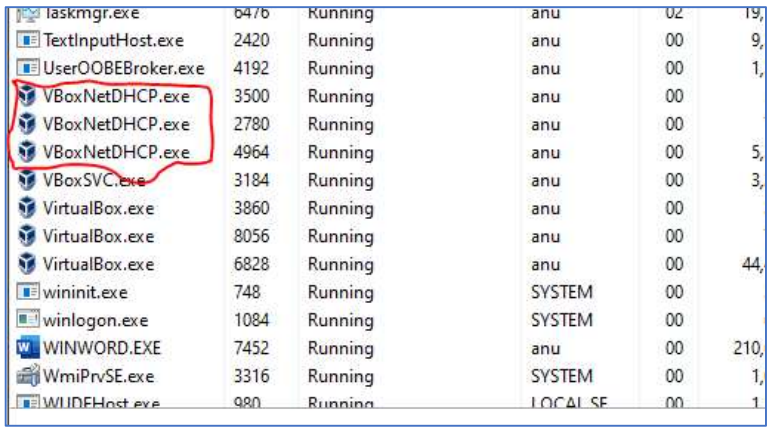
9. Jika pada windows 10 tidak mendapat IP maka lakukanlah Langkah-langkah seperti berikut:
   a. Buka task maneger dengan klik kanan pada taskbar windows, pilih task manager



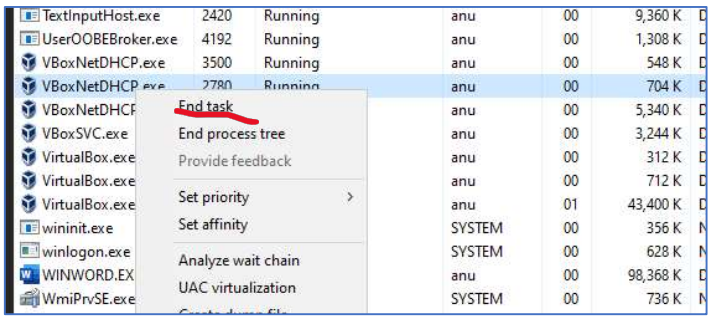
   b. Pilih Details



c. Scroll kebawah, cari **VboxNetDHCP.EXE**



d. Pilih salah satu **VboxNetDHCP.EXE** klik kanan, pilih **End task**, kemudian pilih **end proses**



e. Ulangi Langkah ke 7 untuk mendapatkan ip address

E. Konfigurasi FTP Server

1. Lakukan penginstalan paket **proftpd** dengan perintah **#apt install proftpd-basic**

```
root@debian:~# apt install proftpd-basic
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libmemcached11 libmemcachedutil2 libwrap0 tcpd
Suggested packages:
  openbsd-inetd | inet-superserver proftpd-doc proftpd-mod-ldap
  proftpd-mod-mysql proftpd-mod-odbc proftpd-mod-pgsql proftpd-mod-sqli
  proftpd-mod-geoip
The following NEW packages will be installed:
  libmemcached11 libmemcachedutil2 libwrap0 proftpd-basic tcpd
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/2,736 kB of archives.
After this operation, 5,278 kB of additional disk space will be used.
Do you want to continue? [Y/n] _
```

Tekan tombol **Y** untuk Melanjutkan penginstalan, masukkan DVD 1.iso dan DVD 2.iso pada mesin virtual Box

```
root@debian:~# apt install proftpd-basic
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libmemcached11 libmemcachedutil2 libwrap0 tcpd
Suggested packages:
  openbsd-inetd | inet-superserver proftpd-doc proftpd-mod-ldap
  proftpd-mod-mysql proftpd-mod-odbc proftpd-mod-pgsql proftpd-mod-sqli
  proftpd-mod-geoip
The following NEW packages will be installed:
  libmemcached11 libmemcachedutil2 libwrap0 proftpd-basic tcpd
0 upgraded, 5 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/2,736 kB of archives.
After this operation, 5,278 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Media change: please insert the disc labeled
 'Debian GNU/Linux 9.6.0 _Stretch_ - Official i386 DVD Binary-1 2018111
in the drive '/media/cdrom/' and press [Enter]
```

2. Lakukan pengeditan pada **/etc/proftpd/proftpd.conf** dengan perintah **#nano /etc/proftpd/proftpd.conf**

```
# Set off to disable IPv6 support which is annoying on IPv4 only boxes.
UseIPv6                         on
# If set on you can experience a longer connection delay in many cases.
IdentLookups                    off

ServerName                      "Debian"
# Set to inetd only if you would run proftpd by inetd/xinetd.
# Read README.Debian for more information on proper configuration.
ServerType                              standalone
DeferWelcome                    off
```

Menjadi

```
# Set off to disable IPv6 support which is annoying on IPv4 only boxes.
UseIPv6                         on
# If set on you can experience a longer connection delay in many cases.
IdentLookups                    off

ServerName                      "ftp.smkn3tator.net"
# Set to inetd only if you would run proftpd by inetd/xinetd.
# Read README.Debian for more information on proper configuration.
ServerType                              standalone
DeferWelcome                    off
```

Pada baris paling bawah tambahkan scrib sebagai berikut:

```
# Include other custom configuration files
Include /etc/proftpd/conf.d/

<Anonymous /home/smk/tjkt/>
user smk
</Anonymous>
```

3. Buat direktori/folder baru di home untuk mengakses file yang akan di upload dan download dengan perintah

```
root@smkn3tator:~# cd /home/
root@smkn3tator:/home# mkdir smk
root@smkn3tator:/home# _
```

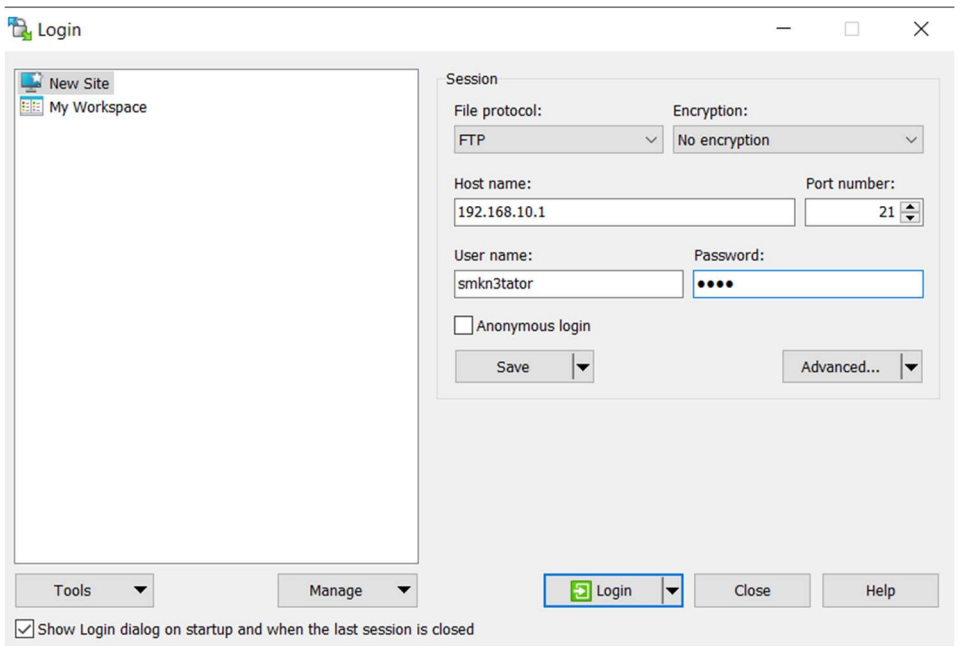
4. Berikan akses baru untuk mengakses direktori upload

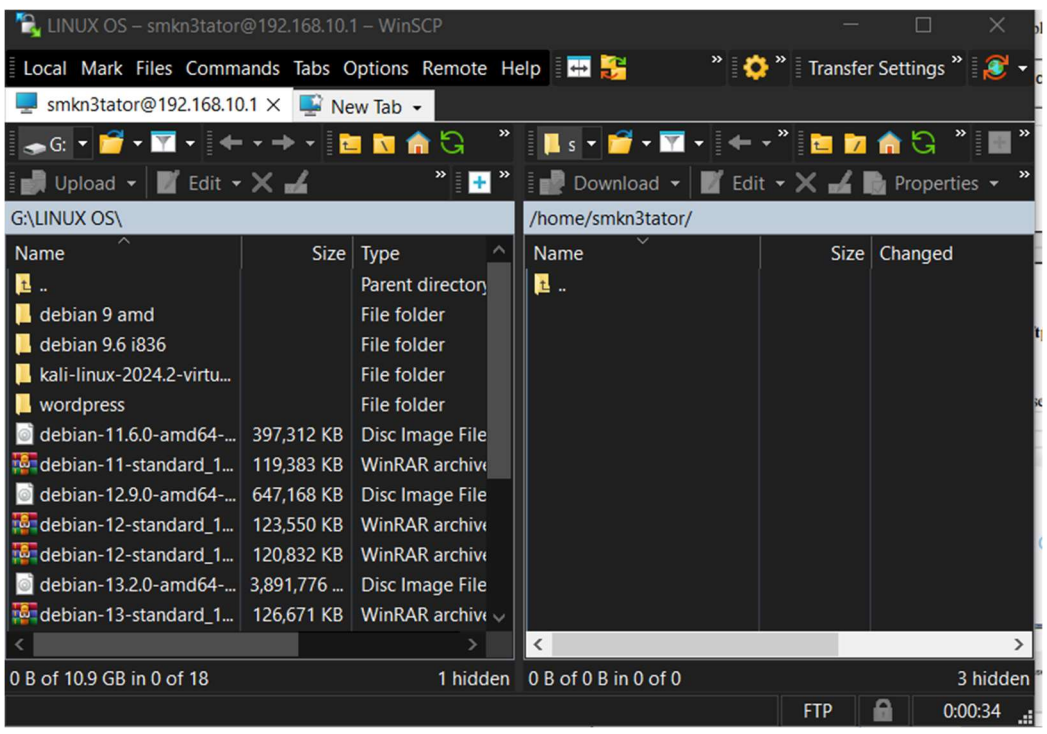
```
root@smkn3tator:/home# chmod 777 smk
```

5. Restart **proftpd** dengan perintah **/etc/init.d/proftpd restart**

6. Lakukan pengujian dengan membuka winsctp

   File protocol pilih FTP
   Host name: 192.168.10.1
   Port number: 21
   Username: smkn3tator
   Password: 1234





F. Konfigurasi SSH Server

1. Lakukan penginstalan SSH Server dengan Perintah **#apt install openssh-server** Tekan tombol Y untuk melanjutkan penginstalan

```
root@debian:/home# apt install openssh-server
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  openssh-sftp-server
Suggested packages:
  molly-guard monkeysphere rssh ssh-askpass ufw
The following NEW packages will be installed:
  openssh-server openssh-sftp-server
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/419 kB of archives.
After this operation, 1,189 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

2. Lakukan pengeditan pada directory **/etc/ssh/sshd_config** dengan perintah **#nano /etc/ssh/sshd_config**

```
  GNU nano 2.7.4                File: /etc/ssh/sshd_config

#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile     .ssh/authorized_keys .ssh/authorized_keys2
```

Tambahkan kalimat PermitRootLogin yes

Menjadi

```
  GNU nano 2.7.4                File: /etc/ssh/sshd_config

#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
PermitRootLogin Yes_
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

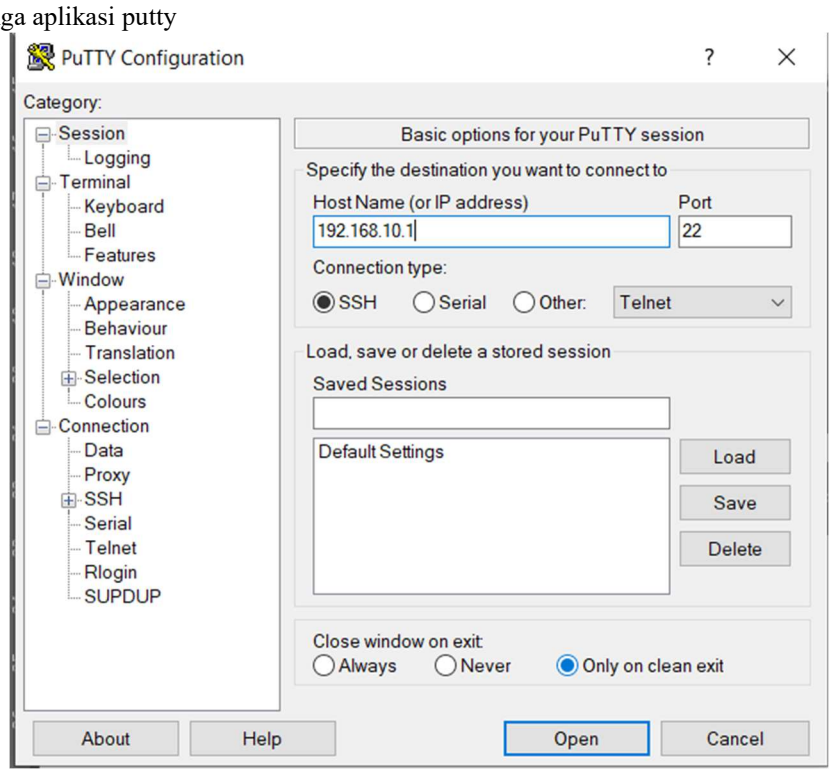
#PubkeyAuthentication yes

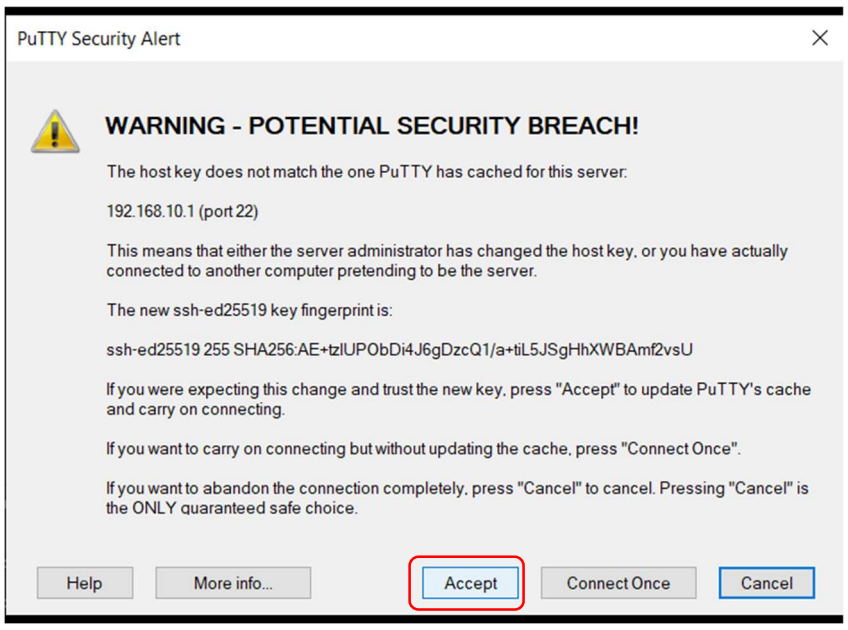
# Expect .ssh/authorized_keys2 to be disregarded by default in future
```

3. Restart ssh dengan perintah **/etc/init.d/ssh restart**

```
root@debian:~# /etc/init.d/ssh restart
[ ok ] Restarting ssh (via systemctl): ssh.service.
root@debian:~#
```

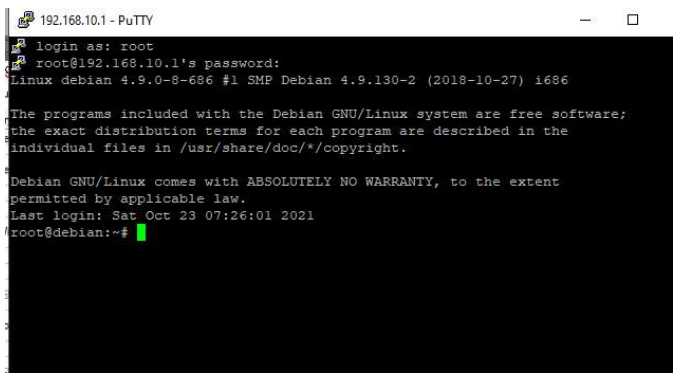
4. Lakukan pengujian denga aplikasi putty



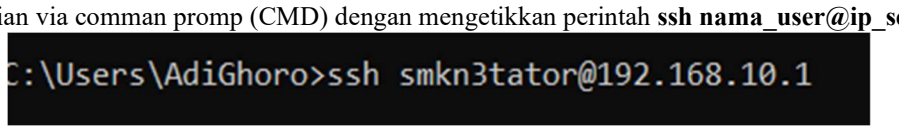


Klik Accept

Login mengunakan akun user dan password user



5. Lakukan juga pengujian via comman promp (CMD) dengan mengetikkan perintah **ssh nama_user@ip_server**



```
Microsoft Windows [Version 10.0.19045.5371]
(c) Microsoft Corporation. All rights reserved.

C:\Users\AdiGhoro>ssh anu@192.168.10.1
anu@192.168.10.1's password:
Linux debian 4.9.0-8-686-pae #1 SMP Debian 4.9.130-2 (2018-10-27) i686

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Jan 29 07:34:16 2025 from 192.168.100.2
anu@debian:~$
```

G. Konfigurasi DNS Server

1. Tambahkan iso2 **apt-cdrom add**
2. Lakukan penginstallan BIND9 dengan perintah **# apt install bind9**

```
oot@debian:~# apt install bind9
eading package lists... Done
uilding dependency tree
eading state information... Done
he following additional packages will be installed:
 bind9utils libirs141
uggested packages:
 bind9-doc resolvconf ufw
he following NEW packages will be installed:
 bind9 bind9utils libirs141
 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
eed to get 0 B/1,145 kB of archives.
fter this operation, 3,587 kB of additional disk space will be used.
o you want to continue? [Y/n]
```

3. Masuk ke directory **bind** dengan perintah **# cd /etc/bind**

```
root@smkn3tator:~# cd /etc/bind
root@smkn3tator:/etc/bind#
```

4. Copy file **db.local** ke **db.smkn3** serta file **db.127** ke **db.192**

```
cp db.local db.smkn3
cp db.127 db.192
```

5. Lakukan pengeditan pada **named.conf.default-zones** dengan perintah **nano named.conf.default-zones**

```
 GNU nano 2.7.4                File: named.conf.d

// prime the server with knowledge of the root s
zone "." {
        type hint;
        file "/etc/bind/db.root";
};

// be authoritative for the localhost forward an
// broadcast zones as per RFC 1912

zone "localhost" {
        type master;
        file "/etc/bind/db.local";
};

zone "127.in-addr.arpa" {
        type master;
        file "/etc/bind/db.127";
};

zone "0.in-addr.arpa" {
```

```
// be authoritative for the localhost
// broadcast zones as per RFC 1912

zone "smkn3.net" {
        type master;
        file "/etc/bind/db.smkn3";
};

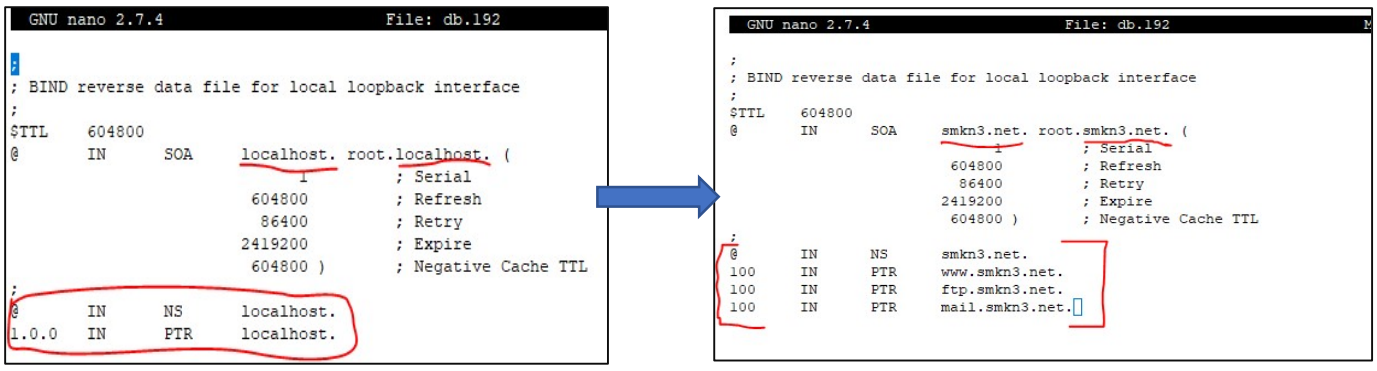
zone "1.100.168.192.in-addr.arpa" {
        type master;
        file "/etc/bind/db.192";
};
```

6. Lakukan pengeditan pada **db.smkn3** dengan perintah **# nano db.smkn3**

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     localhost. root.localhost. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      localhost.
@       IN      A       127.0.0.1
@       IN      AAAA    ::1
```

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     smkn3.net. root.smkn3.net. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      smkn3.net.
@       IN      A       192.168.100.1
www     IN      A       192.168.100.1
ftp     IN      A       192.168.100.1
mail    IN      A       192.168.100.1
```

7. Lakukan pengeditan pada **db.192** dengan perintah **# nano db.192**



8. Tambahkan teks pada file **resolv.conf** dengan perintah **#nano /etc/resolv.conf**

```
search smkn3.net
domain smkn3.net
nameserver 192.168.100.1
```

9. Restart **bind9** dengan perintah **# /etc/init.d/bind9 restart**

```
root@debian:/etc/bind# /etc/init.d/bind9 restart
[ ok ] Restarting bind9 (via systemctl): bind9.service.
```

10. Lakukan pengujian dengan perintah **#nslookup smkn3.net**

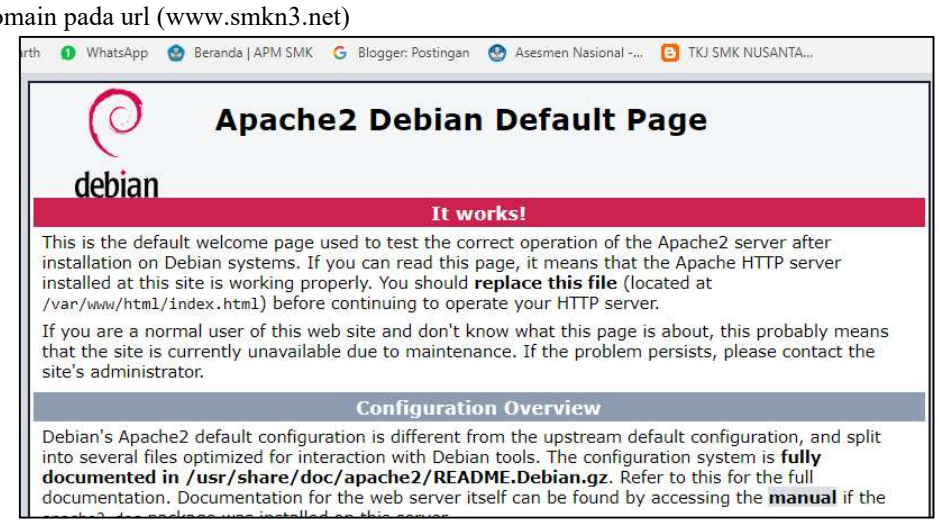
```
root@debian:/etc/bind# nslookup smkn3.net
Server:         192.168.100.1
Address:        192.168.100.1#53

Name:   smkn3.net
Address: 192.168.100.1
```

H. Konfigurasi WEB Server

1. Lakukan penginstallan apache2
   # apt-get install apache2

2. Lakukan pengujian dengan membuka web browser
   Ketikkan nama domain pada url (www.smkn3.net)



3. Buat beberapa scrib berikut "*index.html*, register.html dan *style.css*" pada client dengan aplikasi notepad, simpan di sebuah folder dengan nama "website".
   Index.html

```html
<html>
    <head>
        <title>Login</title>
        <link rel="stylesheet" href="style.css">
    </head>

    <body>
        <div class="container">
          <h1>login</h1>
            <form>
                <label>Username</label><br>
                <input type="text"><br>
                <label>Password</label><br>
                <input type="password"><br>
                <button>Log in</button>
                <p> Belum punya akun?
                  <a href="register.html">Register di sini</a>
                </p>
            </form>
        </div>
    </body>
</html>
```

register.html

```html
<html>
    <head>
        <title>Halaman Login</title>
        <link rel="stylesheet" href="style.css">
    </head>

    <body>
        <div class="container">
          <h1>Register</h1>
            <form>
                <label>Username</label>
                <br>
                <input type="text">
                <br>
                <label>Email</label>
                <br>
                <input type="text">
                <br>
                <label>Password</label>
                <br>
                <input type="password">
                <br>
                <button>Register</button>
                <p> Sudah punya akun?
                  <a href="index.html">Login di sini</a>
                </p>
            </form>
        </div>
    </body>
</html>
```

style.css

```css
*{
    margin: 0;
    padding: 0;
    outline: 0;
    font-family: 'Open Sans', sans-serif;
}
body{
    height: 100vh;
    background-image: url(img.jpg);
    background-size: cover;
    background-position: center;
    background-repeat: no-repeat;
}
p {
  color: white;
  font-family: 'Open Sans', sans-serif;
  padding-top: 10px;
}

h1 {
  text-align: center;
  padding-left: 100px;
  padding-bottom: 20px;
}

a {
  color: white;
  font-family: 'Open Sans', sans-serif;
}
.container{
    position: absolute;
    left: 50%;
    top: 50%;
    transform: translate(-50%,-50%);
    padding: 20px 25px;
    width: 300px;

    background-color: rgba(0,0,0,.7);
    box-shadow: 0 0 10px rgba(255,255,255,.3);
}
.container h1{
    text-align: left;
    color: #fafafa;
    margin-bottom: 30px;
    text-transform: uppercase;
    border-bottom: 4px solid #752BEA;
}
.container label{
    text-align: left;
    color: #90caf9;
}
.container form input{
    width: calc(100% - 20px);
    padding: 8px 10px;
    margin-bottom: 15px;
    border: none;
    background-color: transparent;
    border-bottom: 2px solid #752BEA;
    color: #fff;
    font-size: 20px;
}
.container form button{
    width: 100%;
    height: 40px;
    padding: 5px 0;
    border: none;
    background-color:#752BEA;
    font-size: 18px;
    color: #fafafa;
    border-radius: 20px;
}
```

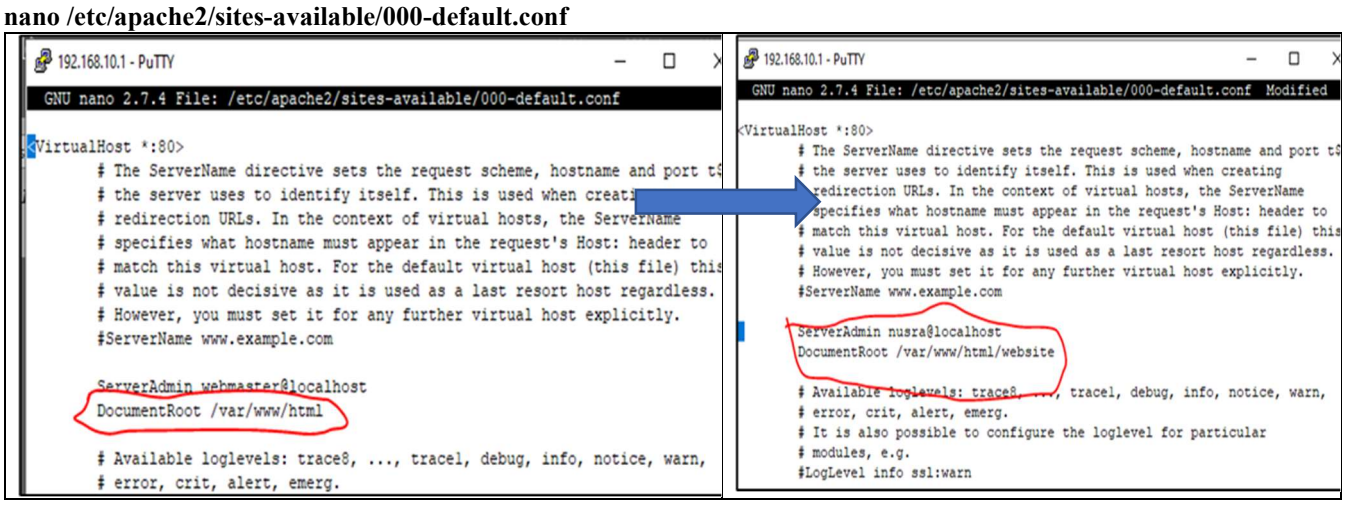
4. Tambahkan direktori baru dengan nama “website” pada direktory “/var/www/html” dengan perintah : **mkdir /var/www/html/website**

5. Berikan akses terhadap direktory “/var/www/html/website” dengan perintah :

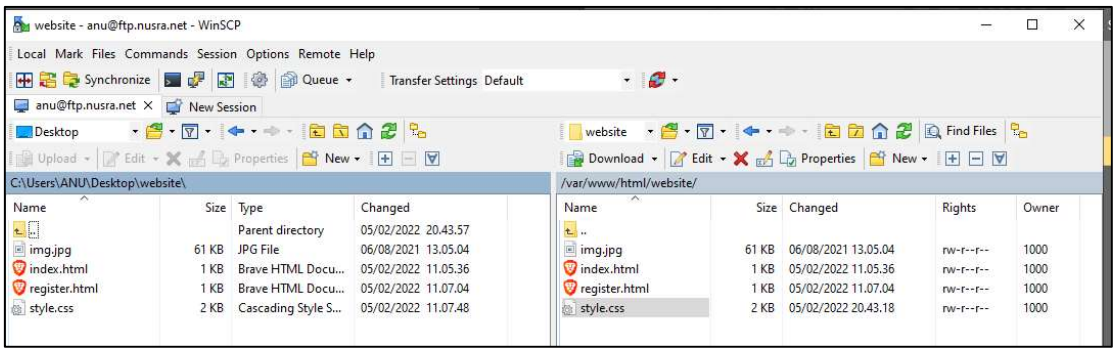
   chmod 777 /var/www/html/website

6. lakukan pengeditan pada */etc/apache2/sites-available/000-default.conf*, guna website kita dapat di akses langsung dari web browser dengan perintah :

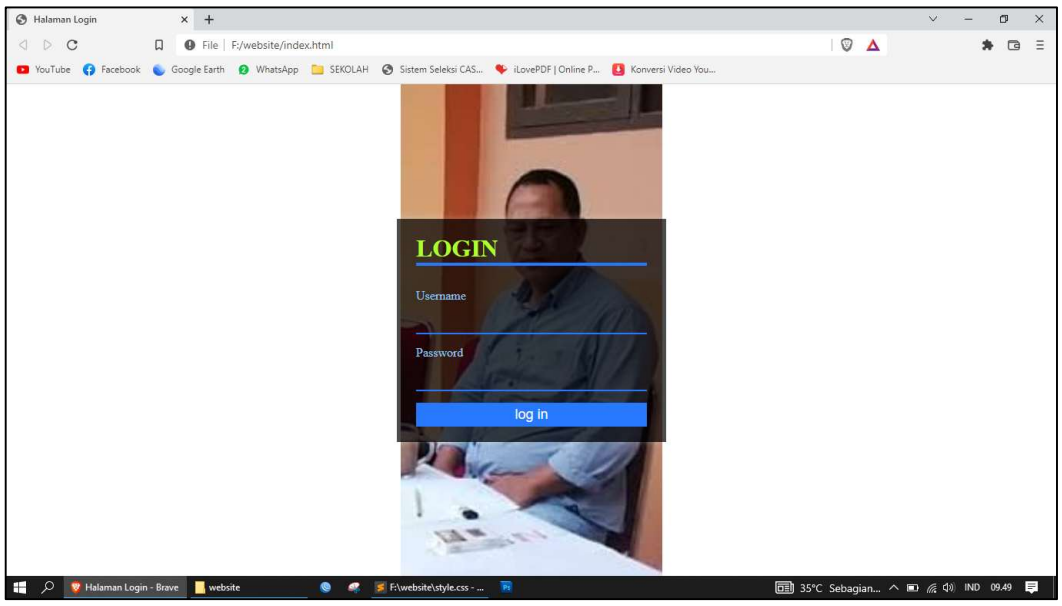
   nano /etc/apache2/sites-available/000-default.conf



7. upload file (hasil desain web) yang telah di buat ke direktori /var/www/html/website menggunakan aplikasi WinSCP

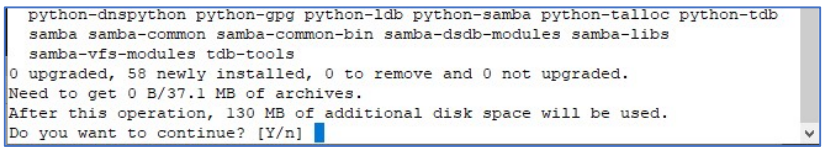


8. Restart apache2 pada server setelah selesai melakukan pengeditan dengan perintah **/etc/init.d/apache2 restart**
9. Lakukan pengujian dengan membuka web browser Ketikkan nama domain pada url: www.smkn3.net

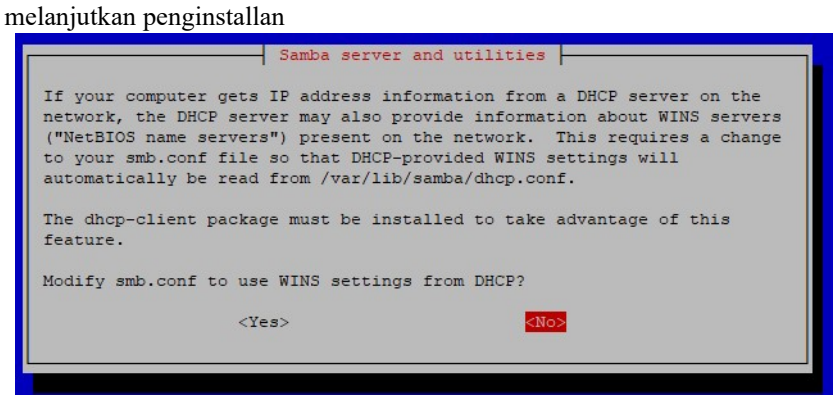


I. Konfigurasi samba server

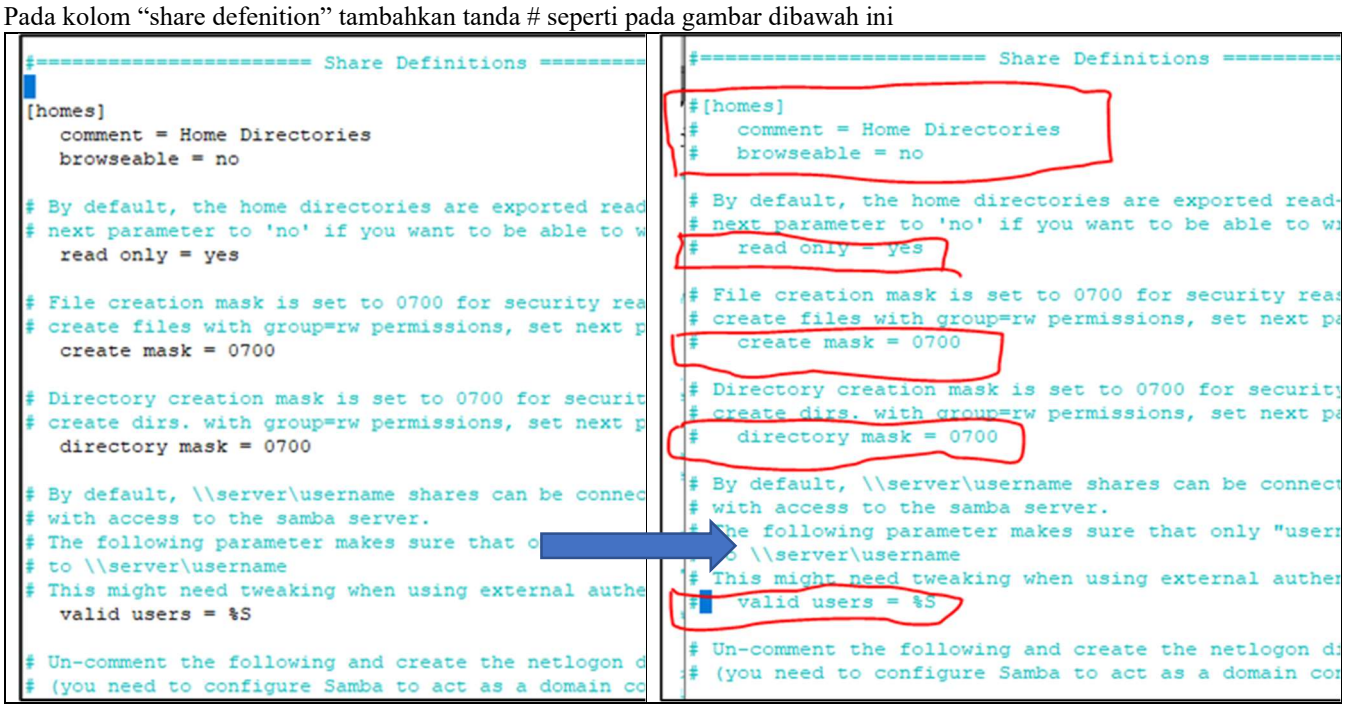
1. Lakukan penginstallan paket Samba dengan perintah # **apt install samba**



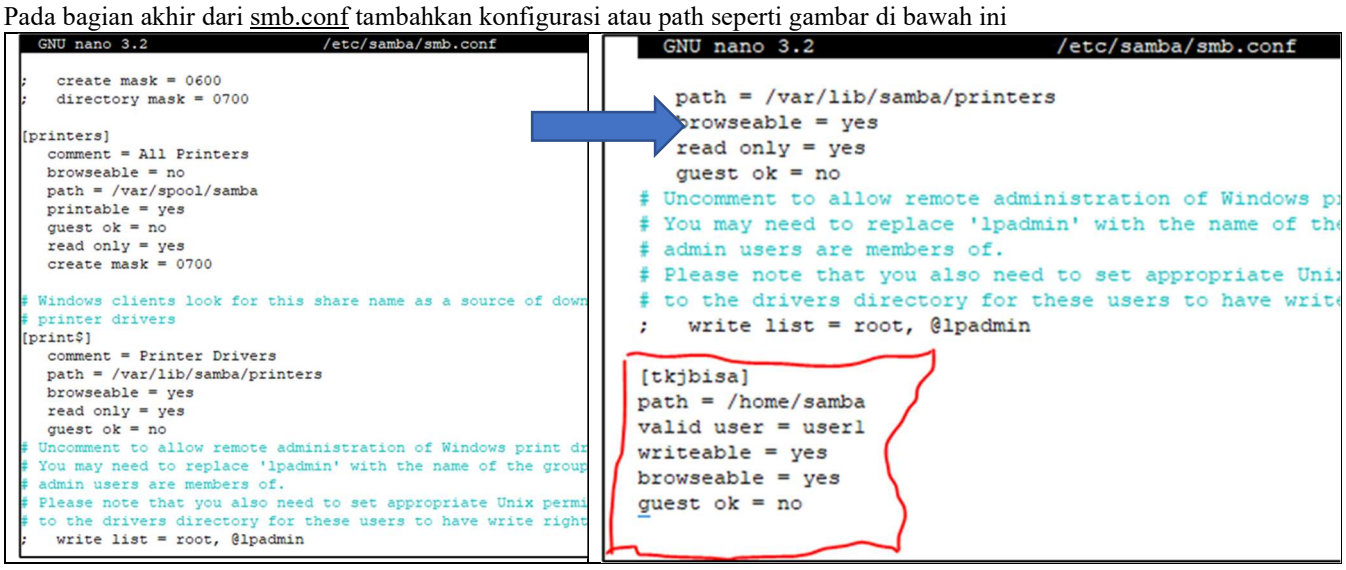
2. Tekan tombol **Y** untuk melanjutkan penginstallan



3. Pilih Yes jika sudah melakukan konfigurasi DHCP. Pilih NO jika belum sudah melakukan konfigurasi DHCP
4. Lakukan konfigurasi atau pengeditan dengan perintah # **nano /etc/samba/smb.conf**
5. Pada kolom “share defenition” tambahkan tanda # seperti pada gambar dibawah ini



6. Pada bagian akhir dari smb.conf tambahkan konfigurasi atau path seperti gambar di bawah ini



7. Lakukan penyimpanan pada konfigurasi di atas
8. Buatlah direktori **samba** pada direktori **home**

```
root@debian:~# mkdir /home/samba
```

9. Buat user dan password baru untuk mengakses direktory samba

```
root@debian:~# useradd -d /home/samba/ user1
root@debian:~# smbpasswd -a user1
New SMB password:
Retype new SMB password:
Added user user1.
```

10. Lakukan pengujian dengan perintah seperti pada gambar

```
root@debian:~# testparm
rlimit_max: increasing rlimit_max (1024) to minimum Windo
Registered MSG_REQ_POOL_USAGE
Registered MSG_REQ_DMALLOC_MARK and LOG_CHANGED
Load smb config files from /etc/samba/smb.conf
```

11. Buatlah sebuah file pada direktori **/home/samba** guna dapat di akses oleh client dengan perintah

```
nano /home/samba/filesamba.txt
```

12. Isikan teks seperti gambar dibawah ini setelah itu simpan

```
ini File SAMBA
```

13. Lakukan restart pada samba

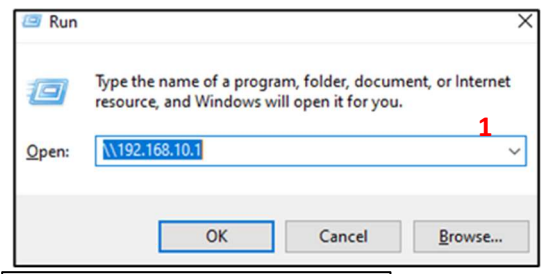
```
root@debian:~# /etc/init.d/smbd restart
[ ok ] Restarting smbd (via systemctl): smbd.service.
```

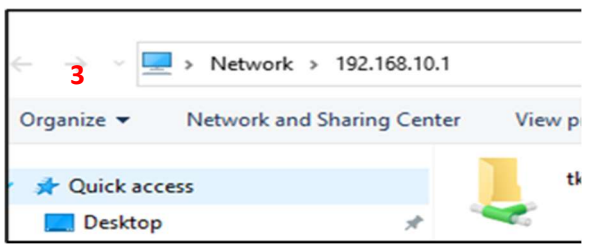
14. Supaya dapat di akses oleh linux client lakukan konfigurasi seperti berikut
    a. Masukkan perintah # **apt install smbclient** untuk penginstallan

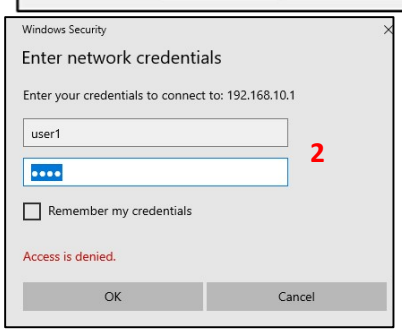
```
root@debian:~# apt install smbclient
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libarchive13 libsmbclient
Suggested packages:
  lrzip cifs-utils heimdal-clients
The following NEW packages will be installed:
  libarchive13 libsmbclient smbclient
0 upgraded, 3 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/935 kB of archives.
After this operation, 2,970 kB of additional disk space will be used.
Do you want to continue? [Y/n]
```

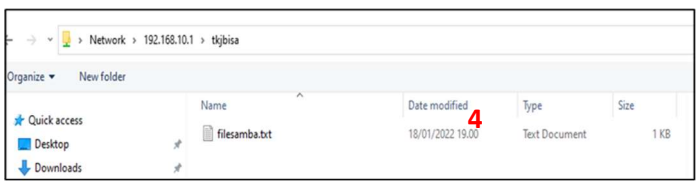
    b. Tekan tombol **Y** untuk melanjutkan penginstallan

15. Lakukan pengujian pada client windows dekan menekan tombol win+R , setelah itu masukkan alamat ip server









J. Konfigurasi Database Server

1. Lakukan penginstallan paket dengan perintah **# apt install php php-mysql**
2. Tekan tombol Y untuk melanjutkan penginstallan

```
After this operation, 21.5 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
```

3. Masukkan dvd 1 debian

```
Processing triggers for libc-bin (2.24-11+deb9u3) ...
Processing triggers for systemd (232-25+deb9u6) ...
Media change: please insert the disc labeled
 'Debian GNU/Linux 9.6.0 _Stretch_ - Official i386 DVD Binary-2 20181110-12:19'
in the drive '/media/cdrom/' and press [Enter]
```

4. Masukkan dvd 2 debian
5. Masukkan konfigurasi **<?php phpinfo (); ?>** pada **nano /var/www/html/info.php** untuk mengaktifkan layanan php

6. Lakukan pengujian pada client (web browser) apakah layanan php sudah aktif dengan mengetikkan pada kolom url yaitu *smkn3.net/info.php*

PHP Version 7.0.30-0+deb9u1

System	Linux debian 4.9.0-8-686 #1 SMP Debian 4.9.130-2 (2018-10-27) i686
Build Date	Jun 14 2018 13:50:25
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/7.0/apache2
Loaded Configuration File	/etc/php/7.0/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/7.0/apache2/conf.d
Additional .ini files parsed	/etc/php/7.0/apache2/conf.d/10-mysqlnd.ini, /etc/php/7.0/apache2/conf.d/10-opcache.ini, /etc/php/7.0/apache2/conf.d/10-pdo.ini, /etc/php/7.0/apache2/conf.d/20-calendar.ini, /etc/php/7.0/apache2/conf.d/20-ctype.ini, /etc/php/7.0/apache2/conf.d/20-exif.ini, /etc/php/7.0/apache2/conf.d/20-fileinfo.ini, /etc/php/7.0/apache2/conf.d/20-ftp.ini, /etc/php/7.0/apache2/conf.d/20-gettext.ini, /etc/php/7.0/apache2/conf.d/20-iconv.ini, /etc/php/7.0/apache2/conf.d/20-json.ini, /etc/php/7.0/apache2/conf.d/20-mysqli.ini, /etc/php/7.0/apache2/conf.d/20-pdo_mysql.ini, /etc/php/7.0/apache2/conf.d/20-phar.ini, /etc/php/7.0/apache2/conf.d/20-posix.ini, /etc/php/7.0/apache2/conf.d/20-readline.ini, /etc/php/7.0/apache2/conf.d/20-shmop.ini, /etc/php/7.0/apache2/conf.d/20-sockets.ini, /etc/php/7.0/apache2/conf.d/20-sysvmsg.ini, /etc/php/7.0/apache2/conf.d/20-sysvsem.ini, /etc/php/7.0/apache2/conf.d/20-sysvshm.ini, /etc/php/7.0/apache2/conf.d/20-tokenizer.ini
PHP API	20151012

7. Lakukan penginstallan maria db dengan perintah **apt install mariadb-server**

8. Tekan tombol Y untuk melanjutkan pengistallan

```
  galera-3 gawk libaio1 libcgi-fast-perl libcgi-pm-perl libdbd-mysql-perl
  libdbi-perl libencode-locale-perl libfcgi-perl libhtml-parser-perl
  libhtml-tagset-perl libhtml-template-perl libhttp-date-perl
  libhttp-message-perl libio-html-perl libjemalloc1 liblwp-mediatypes-perl
  libmariadbclient18 libmpfr4 libreadline5 libsigsegv2 libterm-readkey-perl
  libtimedate-perl liburi-perl mariadb-client-10.1 mariadb-client-core-10.1
  mariadb-common mariadb-server mariadb-server-10.1 mariadb-server-core-10.1
  mysql-common rsync socat
0 upgraded, 33 newly installed, 0 to remove and 0 not upgraded.
Need to get 0 B/26.8 MB of archives.
After this operation, 183 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

9. Masukkan DVD 1 dan DVD 2 debian pada saat pengistallan berlangsung kemudian tekan enter

```
?rocessing triggers for libc-bin (2.24-11+deb9u3) ...
?rocessing triggers for systemd (232-25+deb9u6) ...
edia change: please insert the disc labeled
 'Debian GNU/Linux 9.6.0 _Stretch_ - Official i386 DVD Binary-2 20181110-12:19'
n the drive '/media/cdrom/' and press [Enter]
```

10. Buat database berbasis console dengan perintah **mysql -u root -p**

    Dan masukkan password root

```
root@debian:~# mysql -u root -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 2
Server version: 10.1.26-MariaDB-0+deb9u1 Debian 9.1

Copyright (c) 2000, 2017, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input state

MariaDB [(none)]>
```

11. Lakukan pengecekan data bases pada mariadb dengan perintah

```
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
+--------------------+
3 rows in set (0.00 sec)

MariaDB [(none)]>
```

12. Buat databases dengan nama “TKJ” dengan perintah

```
MariaDB [(none)]> create database tkj;
Query OK, 1 row affected (0.00 sec)
```

13. Lakukan pengecekan apakah database TKJ telah ada di maria db

```
MariaDB [(none)]> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| tkj                |
+--------------------+
4 rows in set (0.00 sec)
```

14. Buat user untuk dapat mengakses databases tkj, dengan perintah

```
MariaDB [(none)]> create user 'anul'@'localhost' identified by 'anul23';
Query OK, 0 rows affected (0.00 sec)
```

15. Ijinkan user yg telah dibuat untuk mengakses database tkj, dengan perintah

```
MariaDB [(none)]> grant all privileges on *.* to 'anul'@'localhost';
Query OK, 0 rows affected (0.00 sec)

MariaDB [(none)]> flush privileges;
Query OK, 0 rows affected (0.00 sec)

MariaDB [(none)]>
```

16. Ketikkan perintah **quit** untuk keluar dari aplikasi maria db

```
MariaDB [(none)]> quit
Bye
root@debian:~#
```

17. Supaya database dapat di akses lewat web browser pada client, maka lakukanlah penginstalan phpmyadmin pada server. Dengan perintah **apt install phpmyadmin**

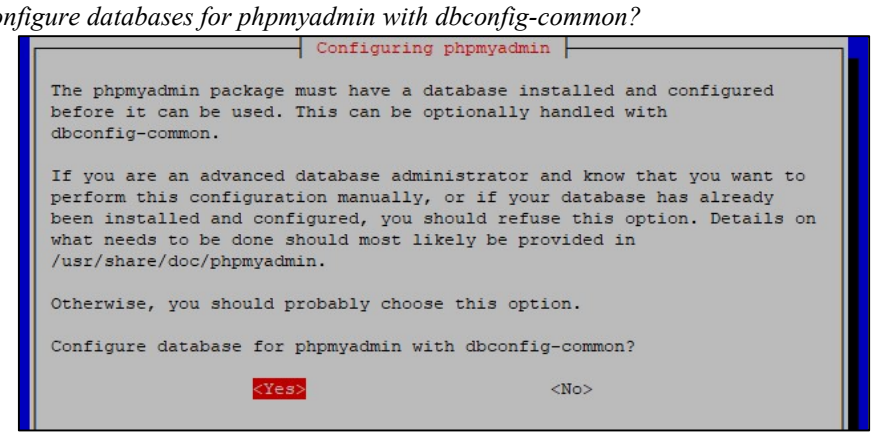
18. Tekan tombol Y untuk melanjutkan penginstallan serta masukkan DVD 1 debian

```
After this operation, 60.7 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Media change: please insert the disc labeled
 'Debian GNU/Linux 9.6.0 _Stretch_ - Official i386 DVD Binary-1 20181110-12:19'
in the drive '/media/cdrom/' and press [Enter]
```

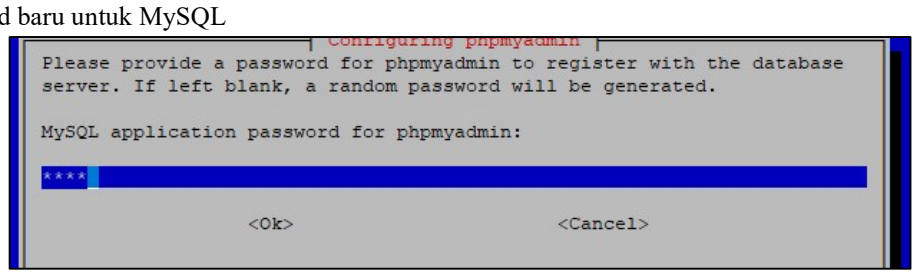
Masukkan DVD 2

```
Media change: please insert the disc labeled
 'Debian GNU/Linux 9.6.0 _Stretch_ - Official i386 DVD Binary-2 20181110-12
in the drive '/media/cdrom/' and press [Enter]
```

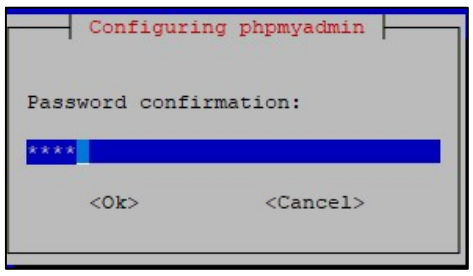
19. Pilih "Yes" untuk *configure databases for phpmyadmin with dbconfig-common?*



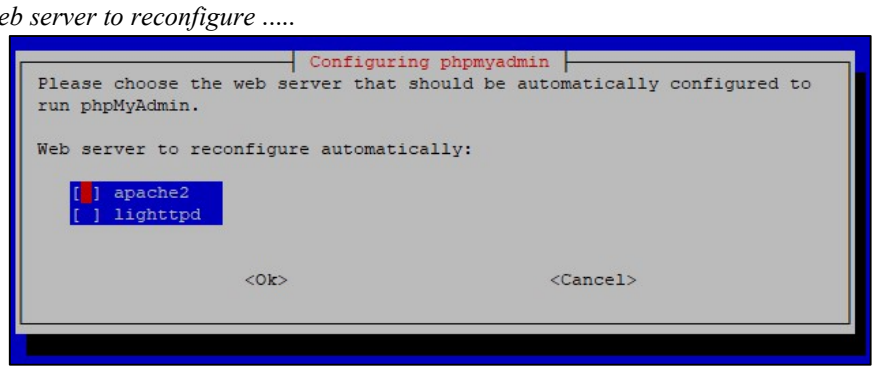
20. Masukkan password baru untuk MySQL



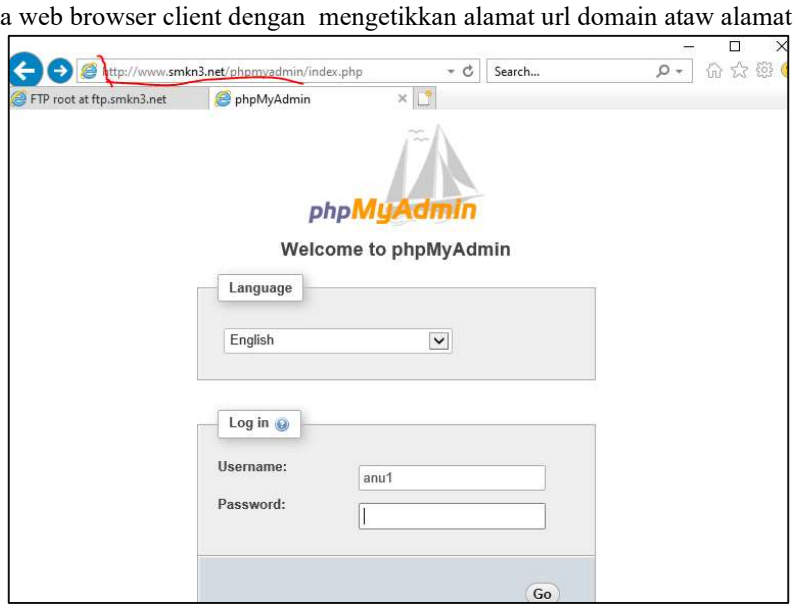
21. Ulangi pasword tersebut



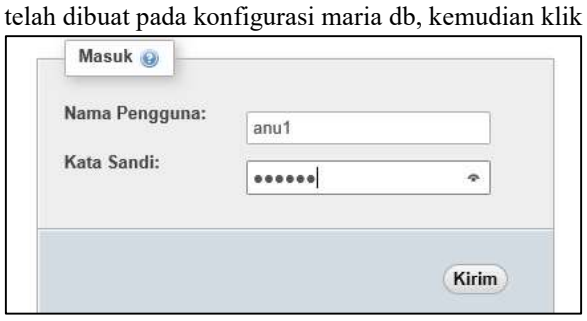
22. Pilih *apache* untuk *web server to reconfigure .....*



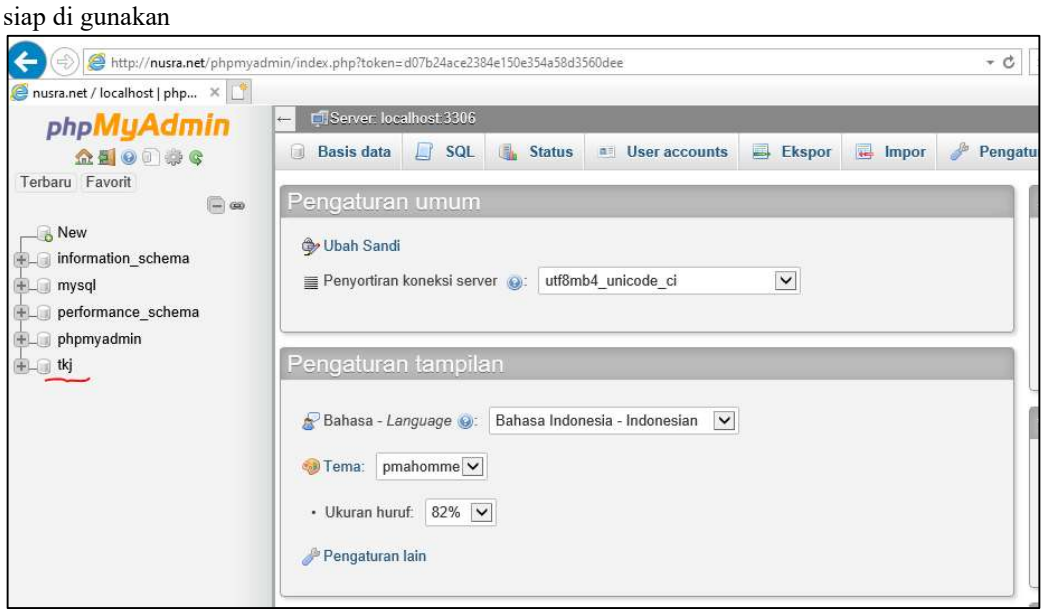
23. Lakukan pengecekan pada web browser client dengan mengetikkan alamat url domain ataw alamat ip server



24. Masukkan user dan password yang telah dibuat pada konfigurasi maria db, kemudian klik tombol kirim



25. Phpmyadmin siap di gunakan



26. Untuk memberikan akses root atau superuser dapat mengakses database php my admin, maka lakukan penyetingan sebagai berikut:
    a. Masukkan perintah

    mysql_secure_installation

    b. Masukkan password root

    ```
    Enter current password for root (enter for none):
    ```

    c. Pilih "**N"** jika tidak ingin menggati password root

    ```
    Change the root password? [Y/n]
    ```

    d. Pilih "**N"** jika tidak ingin menghapus anonymous user

    ```
    Remove anonymous users? [Y/n]
    ```

    e. Pilih "**Y"** pada

    ```
    Disallow root login remotely? [Y/n]
    ```

    f. Pilih "**Y"** pada

    ```
    Remove test database and access to it? [Y/n]
    ```

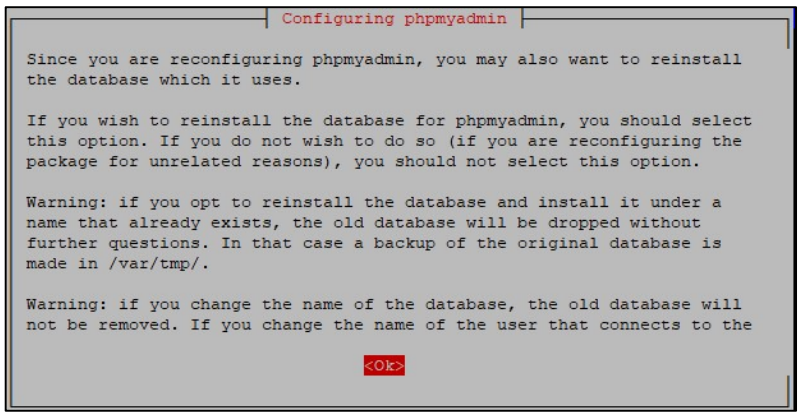
    g. Pilih "**Y"** pada

    ```
    Reload privilege tables now? [Y/n]
    ```

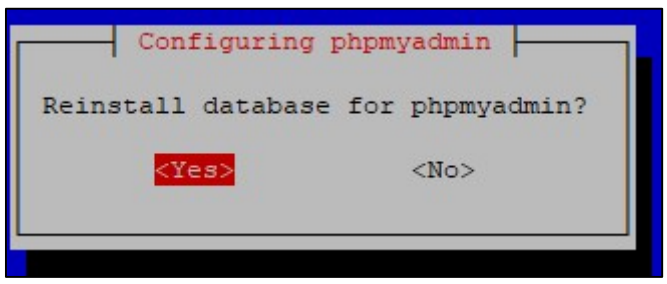
    h. Selanjutnya Lakukan kofigurasi pada *dpkg configure phpmyadmin*, dengan perintah

    dpkg-reconfigure phpmyadmin

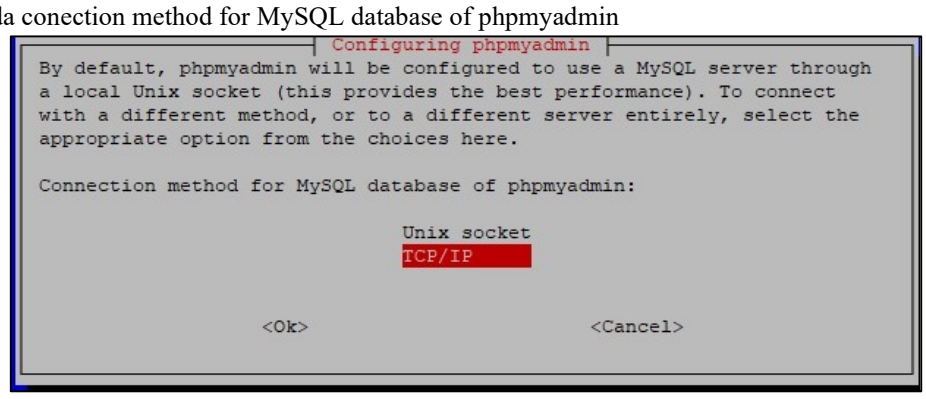
    i. Pilih ok jika muncul tampilan seperti dibawah ini



j. Pilih yes jika ingin reinstall database



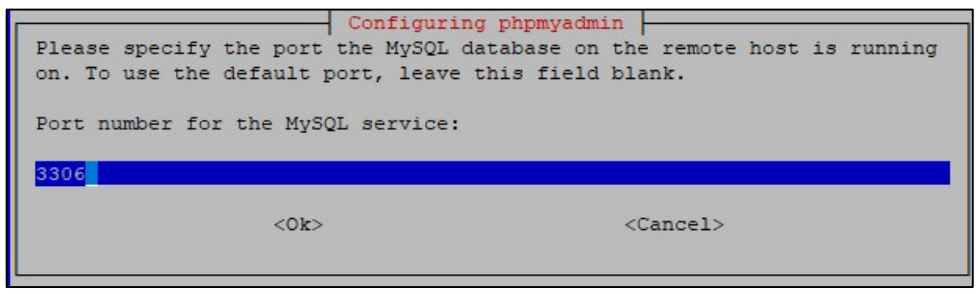
k. Pilih TCP/IP pada conection method for MySQL database of phpmyadmin



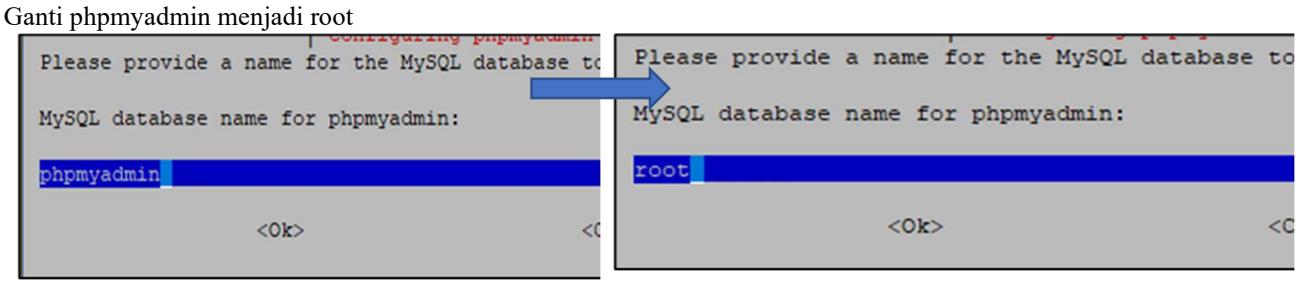
l. Pilih localhost



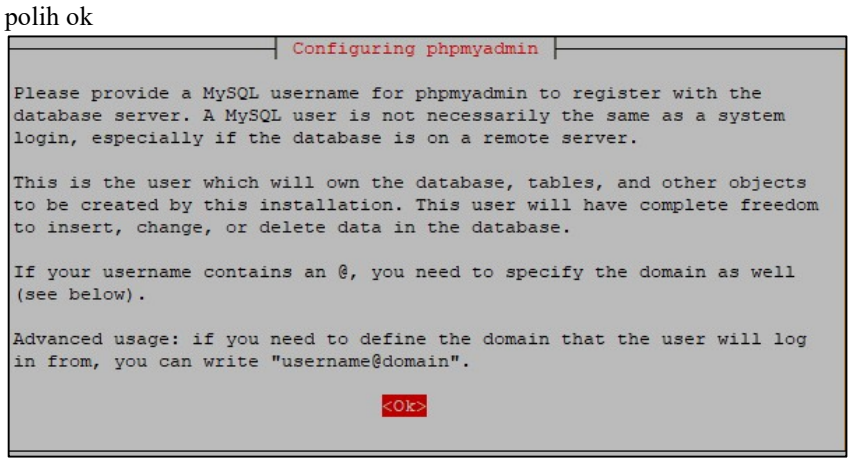
m. Untuk port number isikan 3306



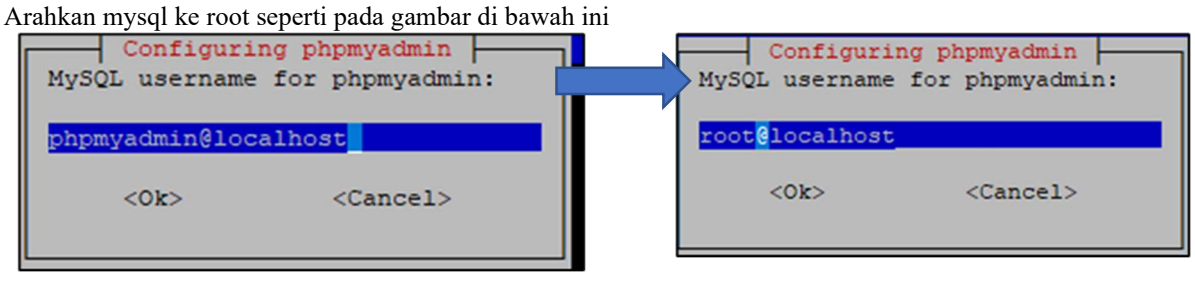
n. Ganti phpmyadmin menjadi root



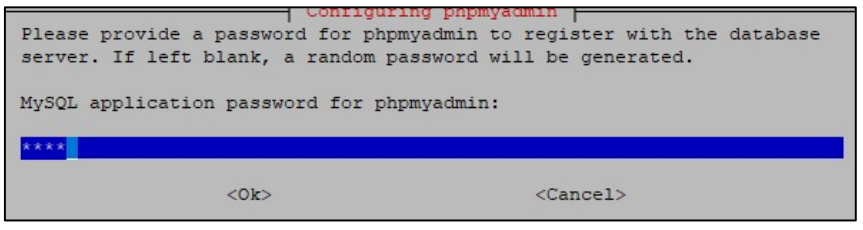
o. Untuk menu berikut polih ok



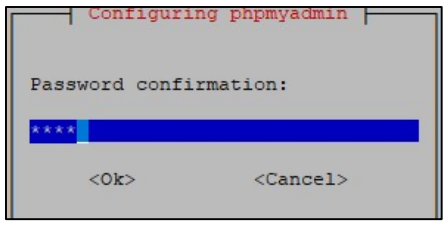
p. Arahkan mysql ke root seperti pada gambar di bawah ini



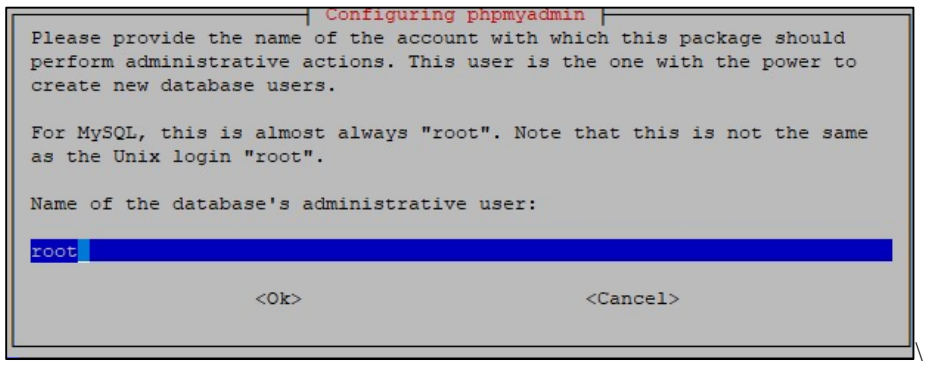
q. Masukkan password root



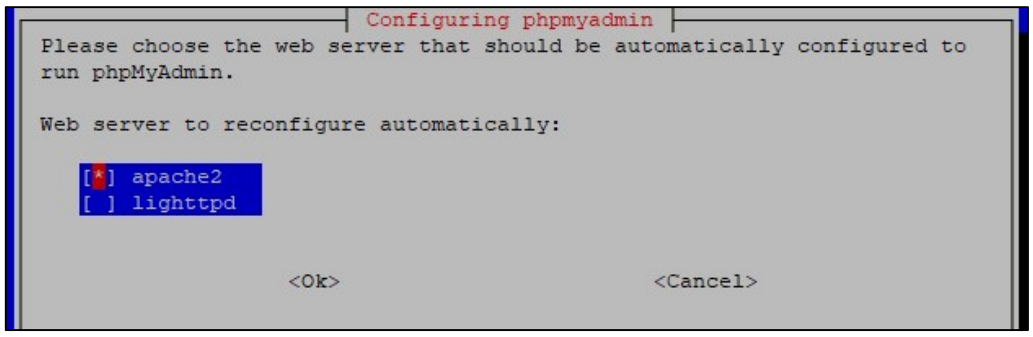
r. Ulangi password pada point "**q**"



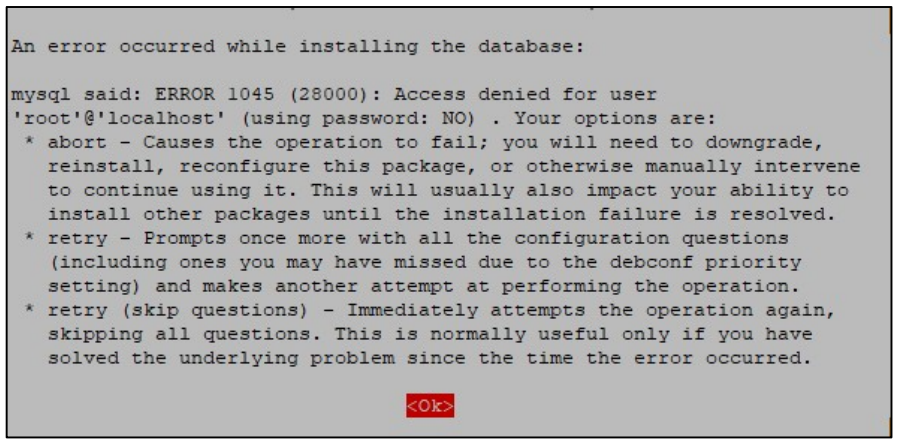
s. Isikan root untuk name of .....



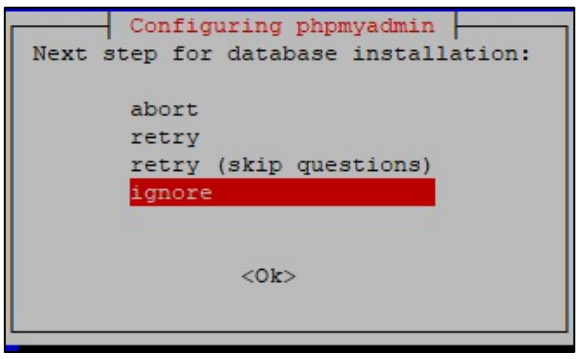
t. Pilih apache 2 untuk web server



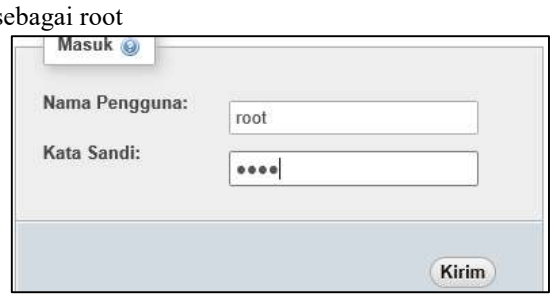
u. Pilih ok jika terjadi eror 1045



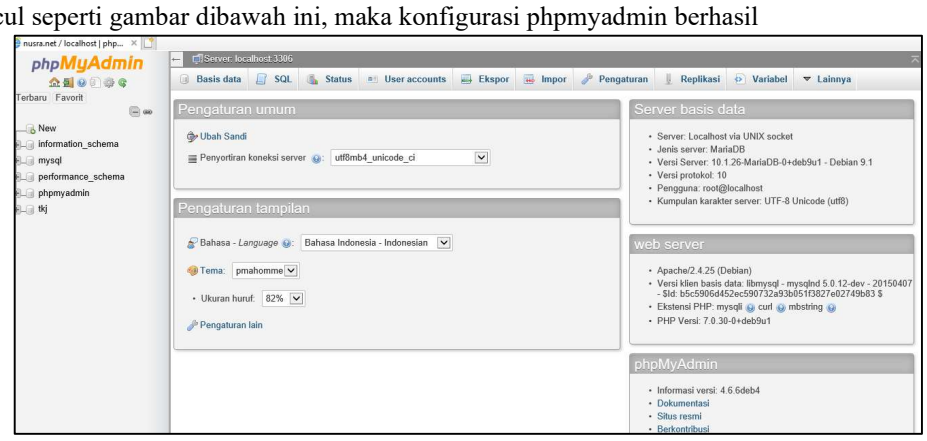
v. Pilih ignore untuk mengalihkan error 1045



w. Lakukan pengujian dengan login sebagai root



x. Jika muncul seperti gambar dibawah ini, maka konfigurasi phpmyadmin berhasil



Konfigurasi Mail server

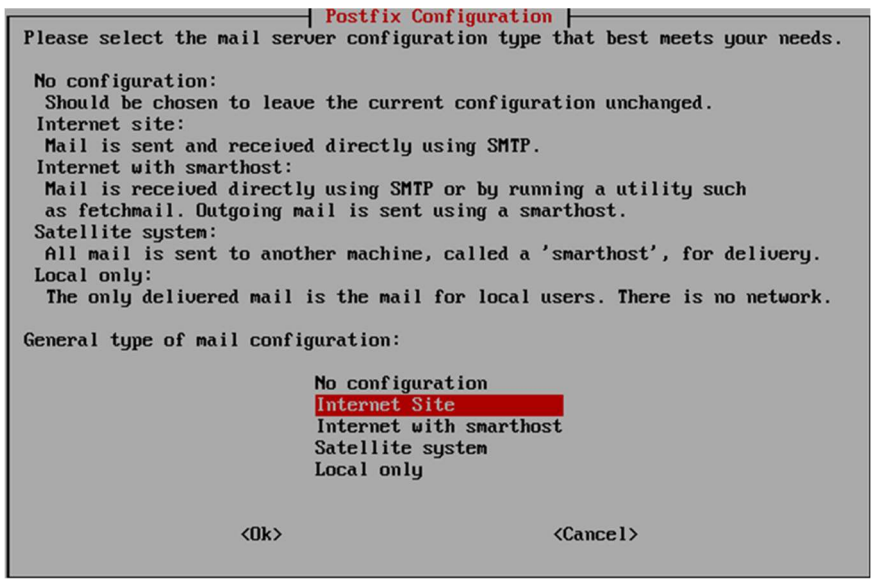
Mail Server adalah aplikasi Server yang digunakan untuk Komunikasi menggunakan Surat Elektronik (Email). Untuk membangun sebuah Mail Server, diperlukan paket-paket Mail Server seperti SMTP Server (Simple Mail Transfer Protocol) disini menggunakan Postfix, kemudian IMAP dan POP3 Server disini menggunakan Courier, dan juga Webmail yang disini menggunakan Roundcube. Asumsinya telah tersedia Komputer Debian 9 yang telah terkonfigurasi IP Address dan DNS Server, dan juga telah tersedia Repositori yang Siap digunakan.

Pastikan sudah terkonfigurasi terlebih dahulu.

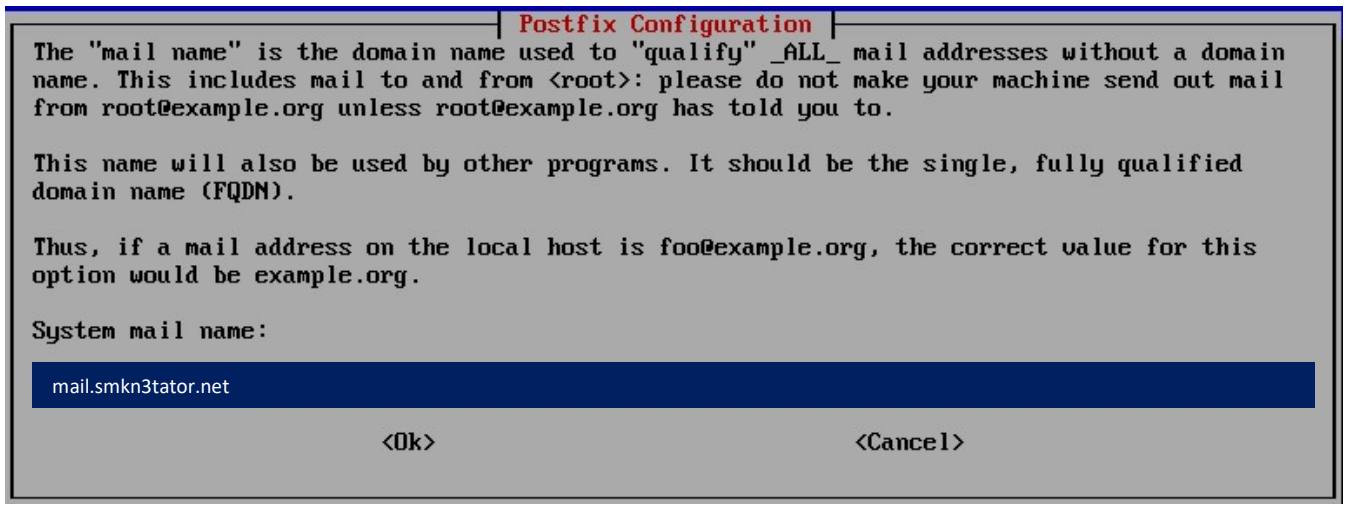
1. pertama install paket aplikasi yang dibutuhkan untuk mail server, perintahnya bisa lihat gambar dibawah ini
   apt install postfix dovecot-imapd dovecot-pop3d mailutils

```
# apt-get install postfix dovecot-imapd dovecot-pop3d mailutils
```

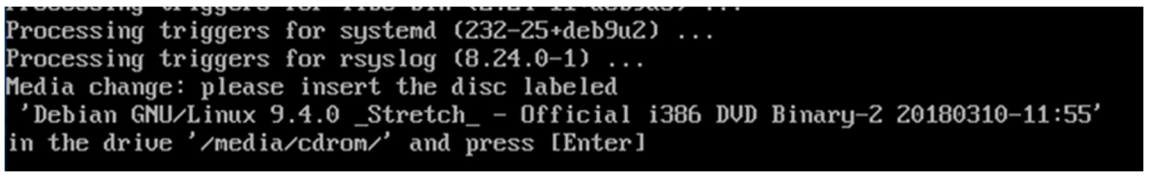
2. pilih Internet Site



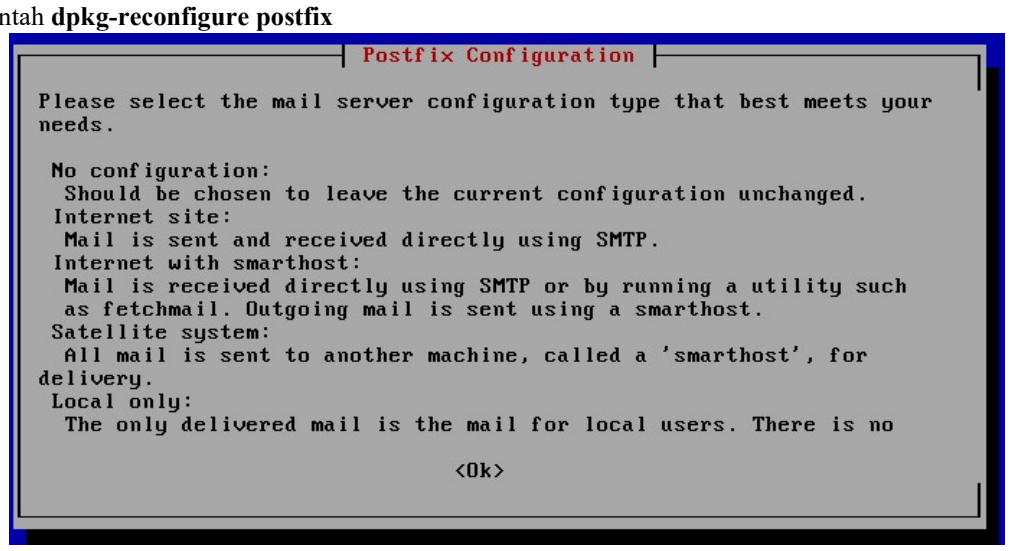
3. Pada system mail name isi dengan nama DNS yang sudah dikonfigurasi misalnya *mail.smkn3.net*



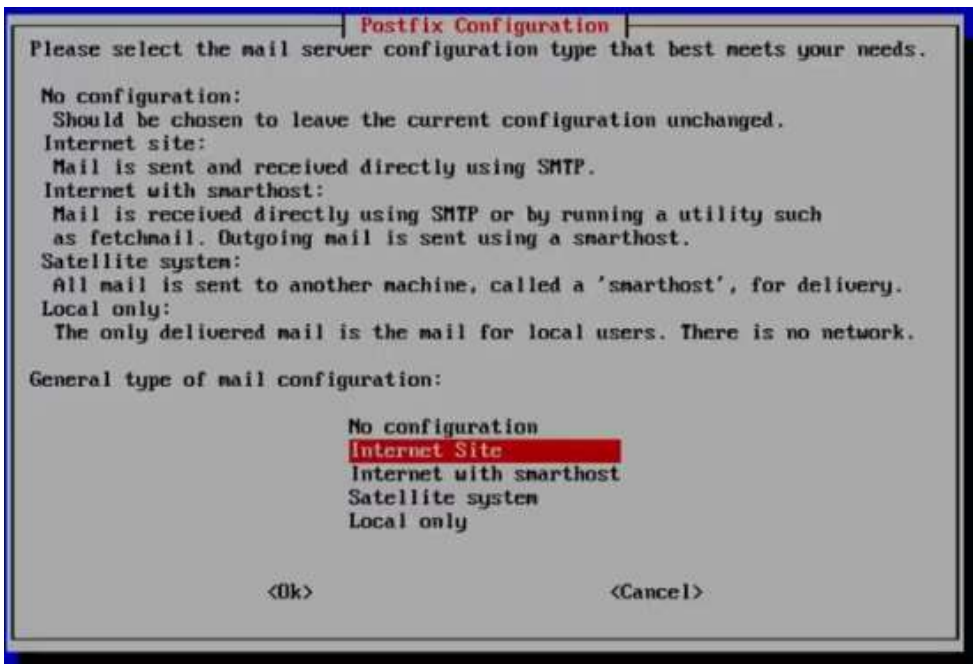
4. Jika diminta untuk memasukan dvd, masukan saja dvd yang dibutuhkannya



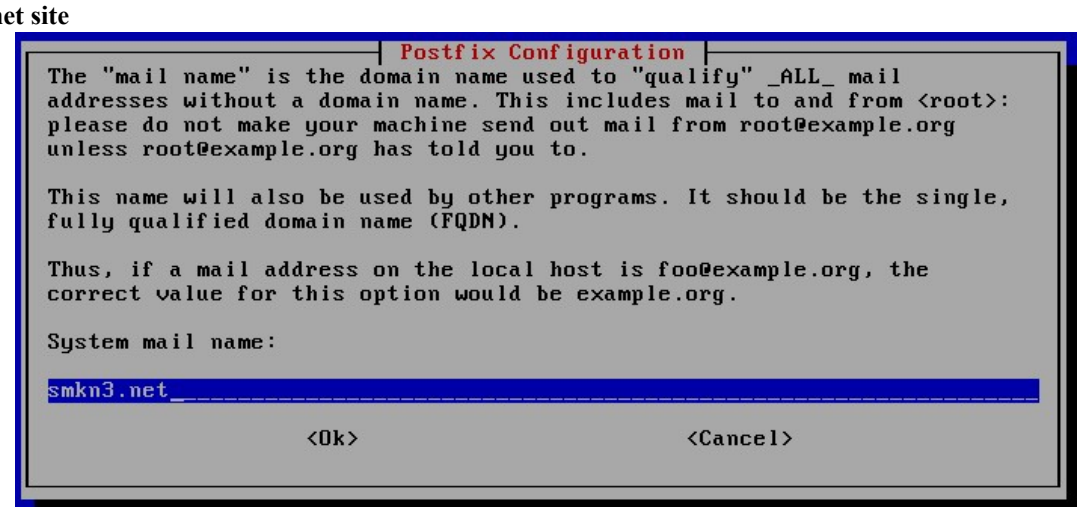
5. Masukkan perintah **dpkg-reconfigure postfix**



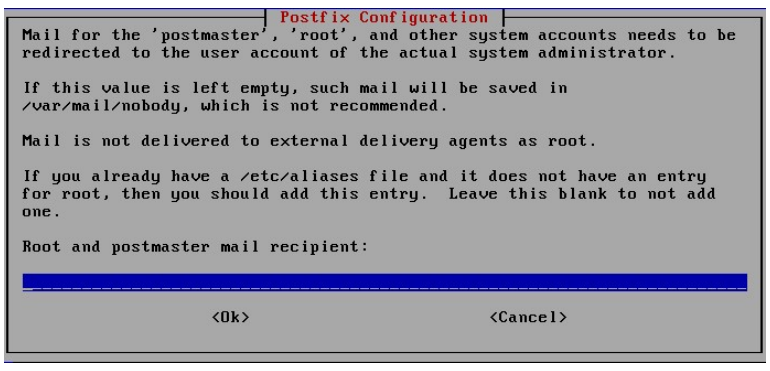
Pilih **Ok**



Pilih **internet site**



Pilih **Ok**



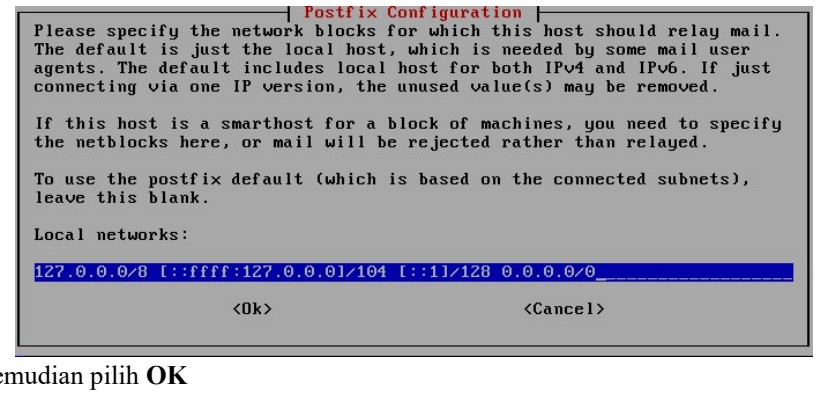
Pilih **OK**



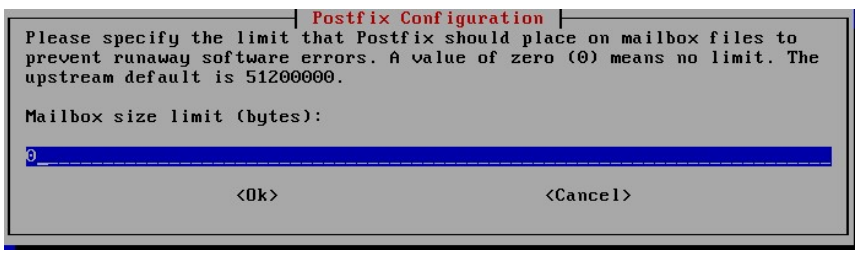
Pilih **Ok**



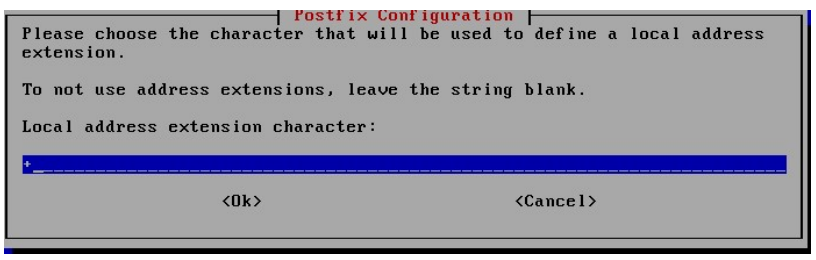
Pilih **No**



Tambahkan **0.0.0.0/0** kemudian pilih **OK**



Polih **Ok**



Pilih **Ok**



Pilih **ipv4** kemudian **Ok**

6. Lakukan konfigurasi pada **/etc/postfix/main.cf**



7. Tambahkan scrib “**home_mailbox =Maildir/**” pada akhir skrib

```
 GNU nano 2.7.4                File: /etc/postfix/main.cf                Modified

readme_directory = no

# See http://www.postfix.org/COMPATIBILITY_README.html -- default to 2 on
# fresh installs.
compatibility_level = 2

# TLS parameters
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
smtpd_use_tls=yes
smtpd_tls_session_cache_database = btree:${data_directory}/smtpd_scache
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache

# See /usr/share/doc/postfix/TLS_README.gz in the postfix-doc package for
# information on enabling SSL in the smtp client.

smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
myhostname = mutu.smkmutu.net
alias_maps = hash:/etc/aliases
alias_database = hash:/etc/aliases
myorigin = /etc/mailname
mydestination = mutu.smkmutu.net, smkmutu.net, mutu.smkmutu.net, localhost.smkmutu.net, localhost
relayhost =
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128 0.0.0.0/0
mailbox_size_limit = 0
recipient_delimiter = +
inet_interfaces = all
inet_protocols = ipv4
home_mailbox =Maildir/
```

8. Restary postfix dengan perintah "**systemctl restart postfix**"

9. Lakukan pengeditan pada file dovecot.conf dengan perintah "**nano /etc/dovecot/dovecot.conf**"

10. Cari kata "**#listen = *, ::** "

```
# Default values are shown for each setting, it's
# those. These are exceptions to this though: No s
# or plugin settings are added by default, they're
# Paths are also just examples with the real defau
# options. The paths listed here are for configure
# --sysconfdir=/etc --localstatedir=/var

# Enable installed protocols
!include_try /usr/share/dovecot/protocols.d/*.prot

# A comma separated list of IPs or hosts where to
# "*" listens in all IPv4 interfaces, "::" listens
# If you want to specify non-default ports or anyt
# edit conf.d/master.conf.
#listen = *, ::

# Base directory where to store runtime data.
```

Ubah menjadi

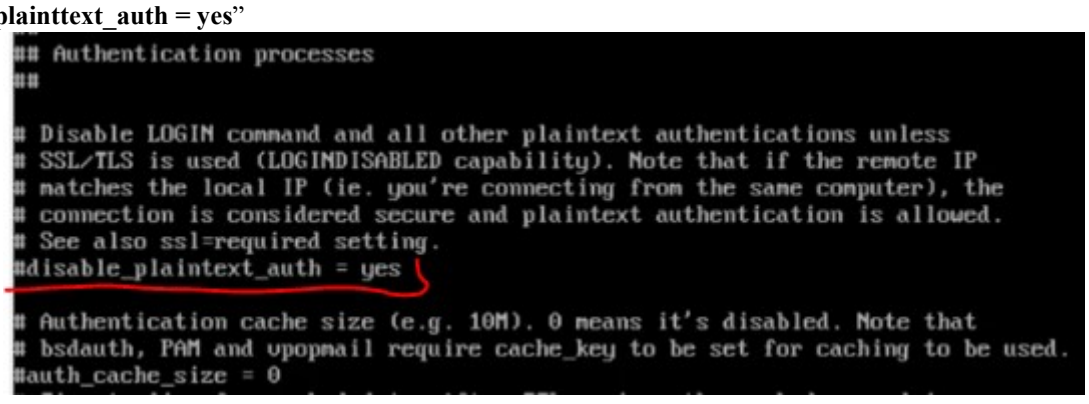
```
# Default values are shown for each setting, it's not required
# those. These are exceptions to this though: No sections (e.g.
# or plugin settings are added by default, they're listed only
# Paths are also just examples with the real defaults being bas
# options. The paths listed here are for configure --prefix=/us
# --sysconfdir=/etc --localstatedir=/var

# Enable installed protocols
!include_try /usr/share/dovecot/protocols.d/*.protocol

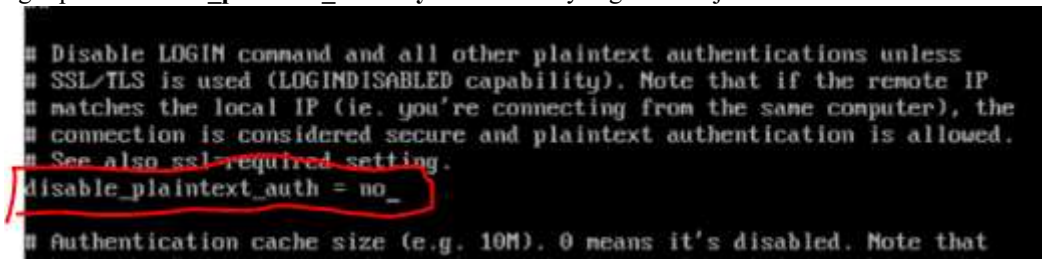
# A comma separated list of IPs or hosts where to listen in for
# "*" listens in all IPv4 interfaces, "::" listens in all IPv6
# If you want to specify non-default ports or anything more com
# edit conf.d/master.conf.
listen = *

# Base directory where to store runtime data.
```

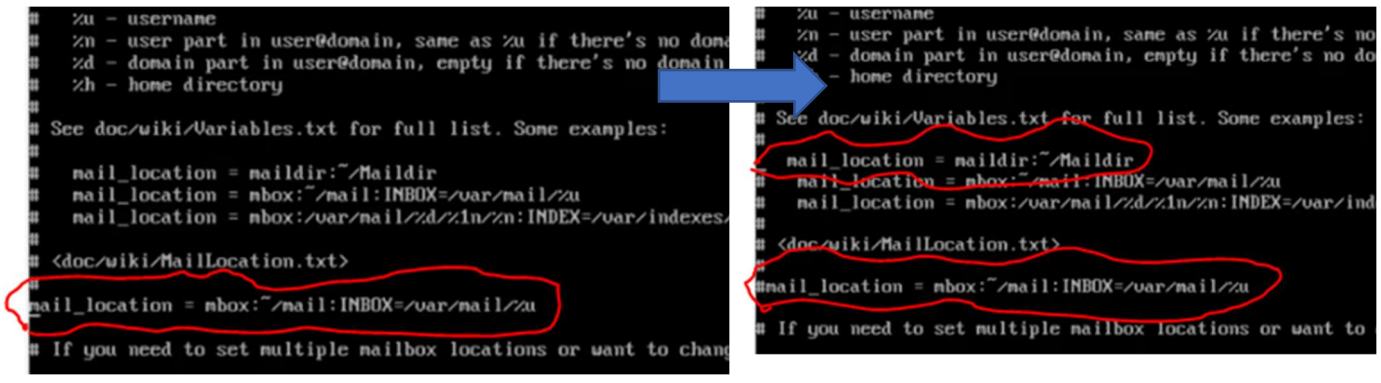
11. Lakukan pengeditan pada **10-auth.conf** dengan perintah "**nano /etc/dovecot/conf.d/10-auth.conf** " cari kalimat "**#disable_plainttext_auth = yes**"



Hapus tanda pagar pada "**disable_plaintext_auth = yes**" dan kata yes ganti menjadi no



12. Lakukan pengeditan pada **10-mail.conf** dengan perintah "**nano /etc/dovecot/conf.d/10-mail.conf** " lakukan pengeditan seperti gambar dibawah ini

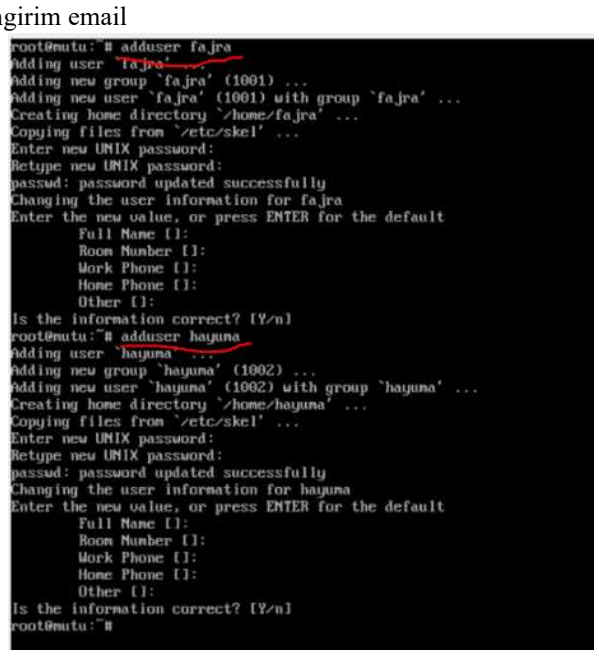


13. Ketikkan perintah "**maildirmake.dovecot /etc/skel/Maildirmake**"

14. Restart postfix dengan perintah "**systemctl restart postfix**"

15. Restart dovecot dengan perintah "**systemctl restart dovecot**"

16. Tambahkan 2 user baru untuk mengirim email



17. Lakukan pengiriman email dengan menggunakan perintah **telnet localhost 25**

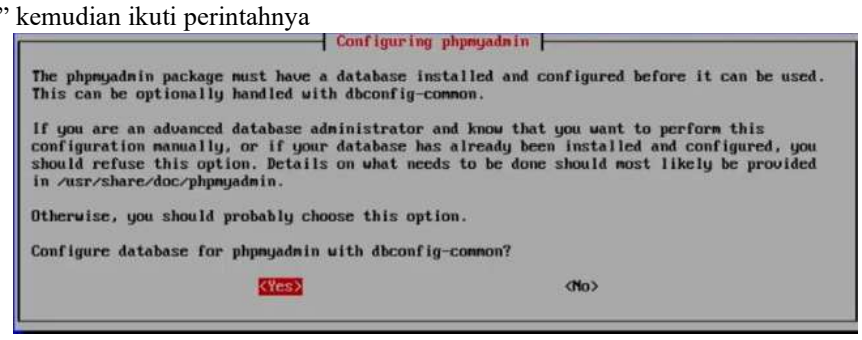
```
root@mutu:~# telnet localhost 25
Trying ::1...
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
220 mutu.smkmutu.net ESMTP Postfix (Debian/GNU)
mail from:fajra
250 2.1.0 Ok
rcpt to:hayuma
250 2.1.5 Ok
data
354 End data with <CR><LF>.<CR><LF>
Halo Hayuma, Apa kabar?
.
250 2.0.0 Ok: queued as 9B9FA6CA
quit
221 2.0.0 Bye
Connection closed by foreign host.
root@mutu:~# _
```

18. Periksa email masuk dengan perintah **telnet localhost 110**

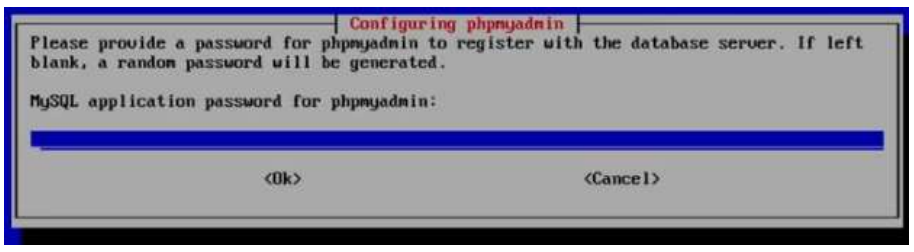
```
root@mutu:~# telnet localhost 110
Trying ::1...
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
+OK Dovecot ready.
user hayuma
+OK
pass 123456
+OK Logged in.
stat
+OK 1 402
retr 1
+OK 402 octets
Return-Path: <fajra@smkmutu.net>
X-Original-To: hayuma
Delivered-To: hayuma@smkmutu.net
Received: from localhost (localhost [127.0.0.1])
        by mutu.smkmutu.net (Postfix) with SMTP id 9B9FA6CA
        for <hayuma>; Thu, 31 Dec 2020 13:14:02 +0700 (WIB)
Message-Id: <20201231061414.9B9FA6CA@mutu.smkmutu.net>
Date: Thu, 31 Dec 2020 13:14:02 +0700 (WIB)
From: fajra@smkmutu.net

Halo Hayuma, Apa kabar?
.
quit
+OK Logging out.
Connection closed by foreign host.
```

19. Lakukan penginstalan paket ***apache2, php, mysql-server, phpMyAdmin*** dengan perintah "**apt install apache2 php mysql-server phpmyadmin**" kemudian ikuti perintahnya



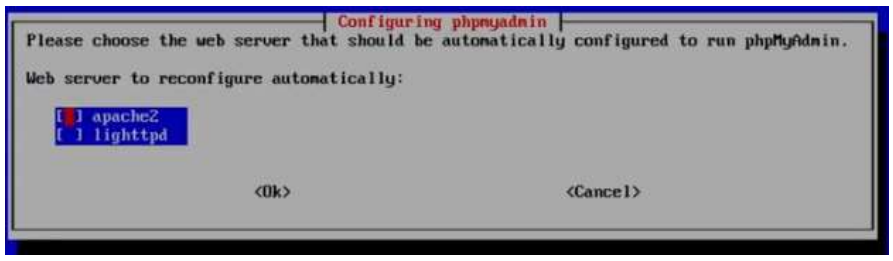
Pilih **yes**



Isikan password



Ulangi password

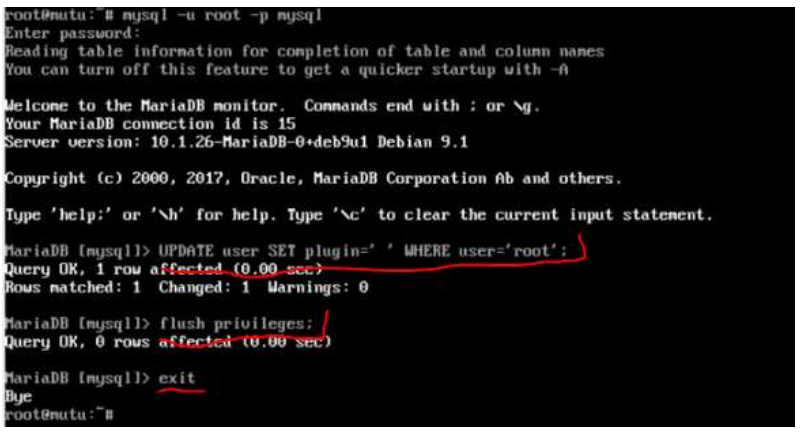


Pilih **apache2**

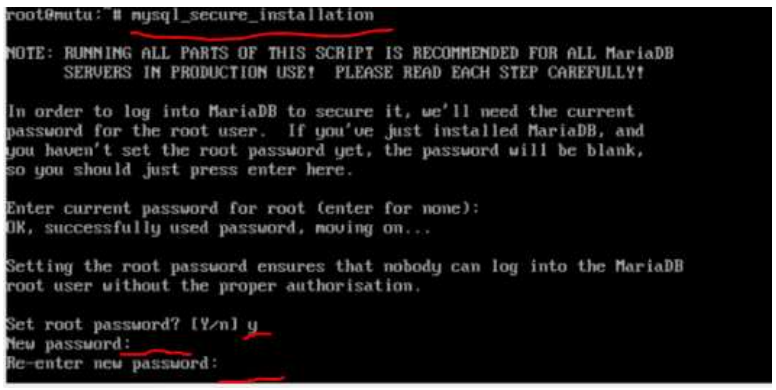
20. Lakukan pengujian pada browser client dengan mengetikkan alamat domain (*smkn3.net/phpmyadmin*)



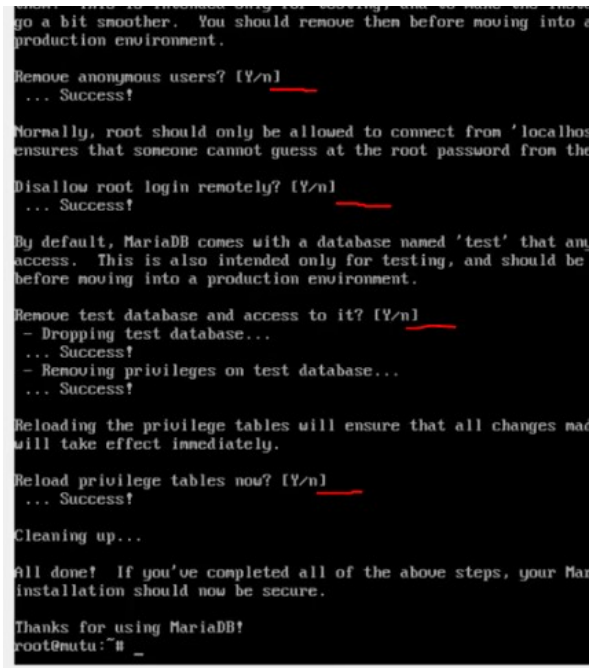
21. Buat password rood untuk dapat mengakses phpMyAdmin pada clien dengan perintah "**mysql -u root -p mysql**" ketikkan perintah "**UPDATE user SET plugin=' ' WHERE user='root';** " dan perintah " **flush privileges** " seperti gambar dibawah ini



22. Ketikkan perintah "**mysql_secure_installation**"

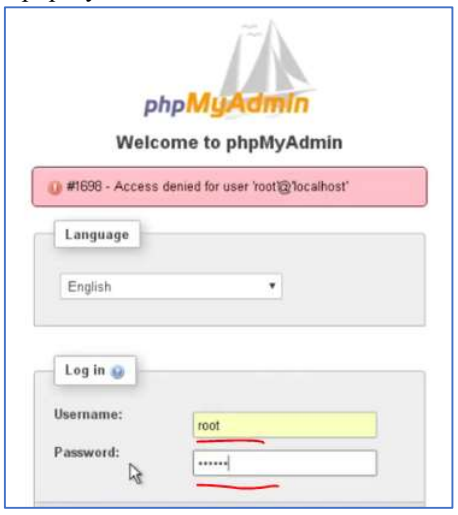


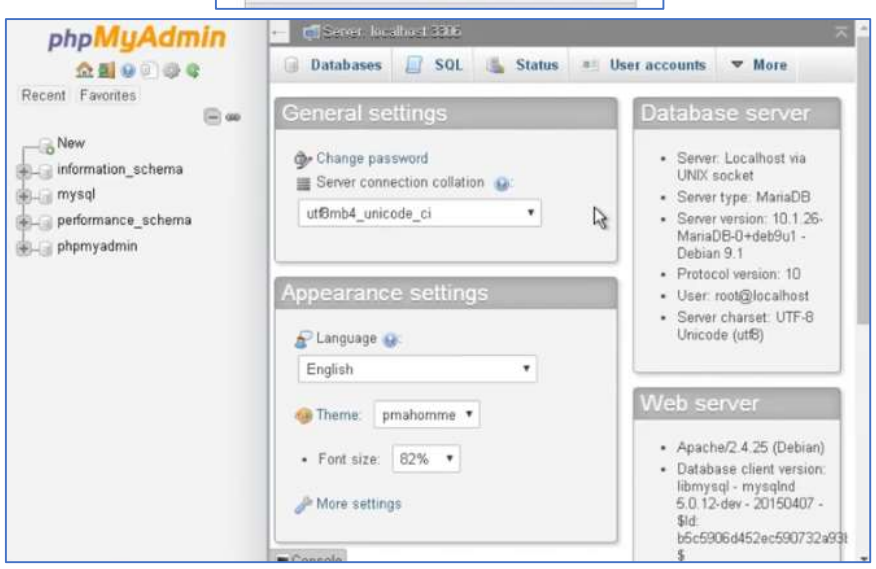
Keterangan:
Pada set password tekan Y
New password: (masukkan password baru)
Re_entry new password : (ulangi password pada new password)



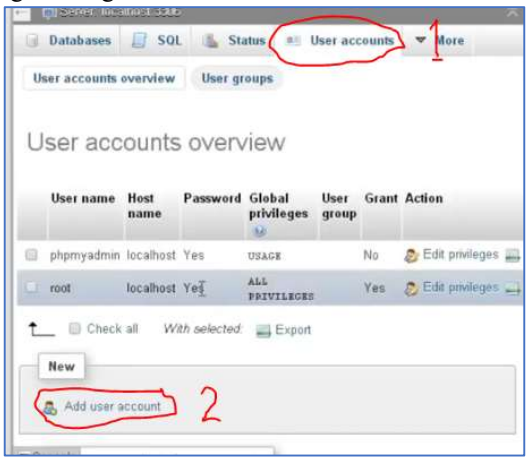
Tekan tombol **enter** pada *"Remove anonymous uer?, Dislow root login remotely?, Remove test databases and accest to it?, Reload privilege tables now?"*

23. Masuk ke client pada php myadmin, masukkan *user : root, password : (password yang telah dibuat pada point 22 di atas)* kemudian klik **Go**, untuk dapat mengakses phpMyAdmin

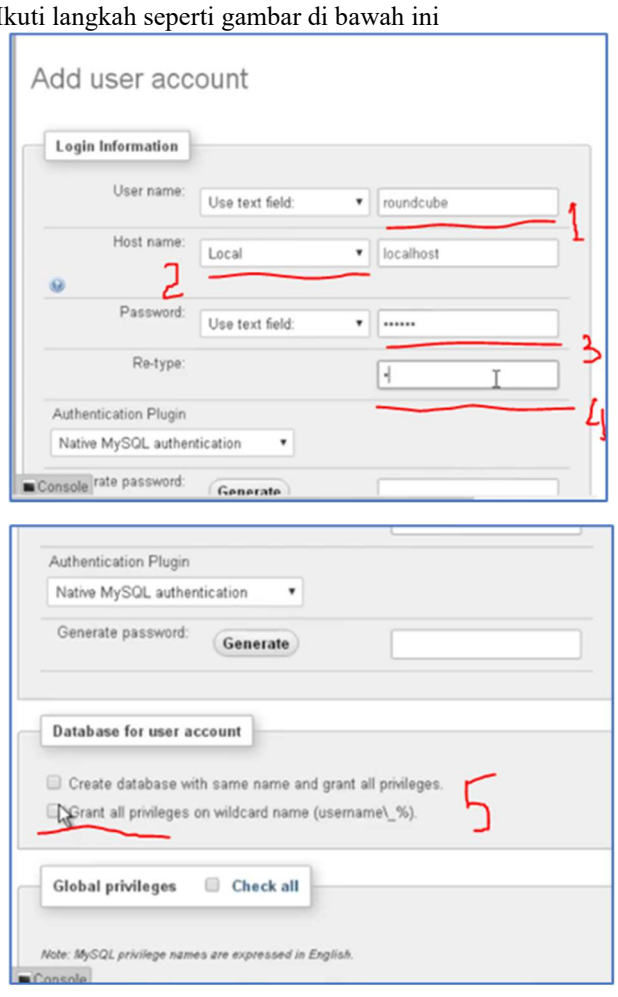


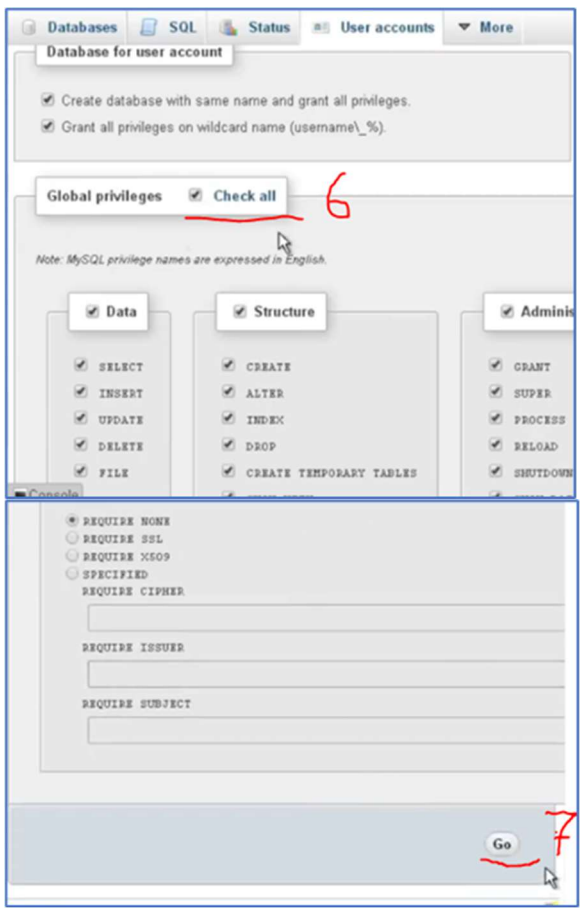


24. Buat user baru pada phpMyAdmin dengan mengkilk **User accounts** kemudian **Add User accounts**



25. Ikuti langkah seperti gambar di bawah ini

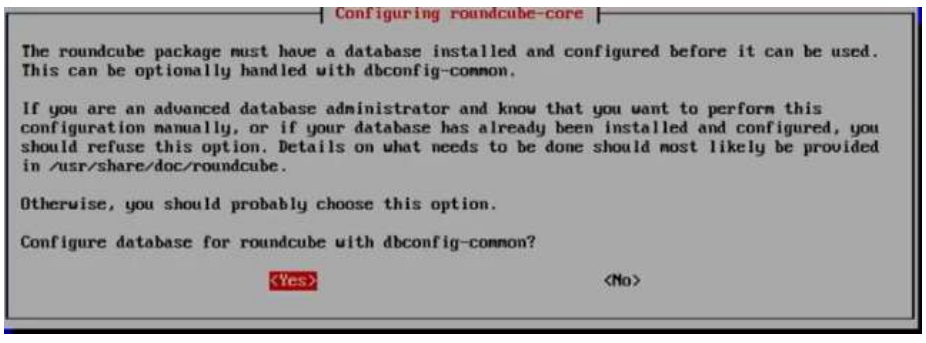




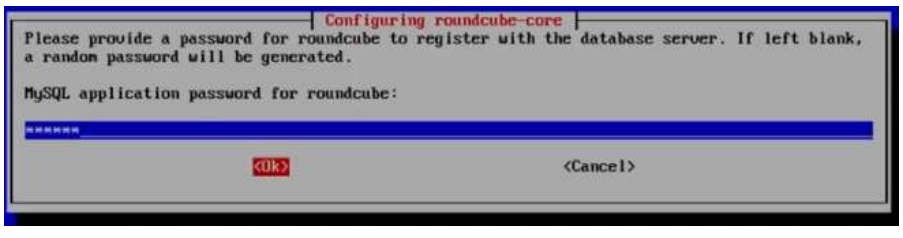
Keterangan:

1. User name : roundcube
2. Host name : local
3. Password : 1234
4. Re-type : 1234
5. Centang pada "create database with same name and grand all privilages", dan "grant all privileges on wildcard name (username_%)"
6. Pada Global privileges centang check all
7. Klik **Go**

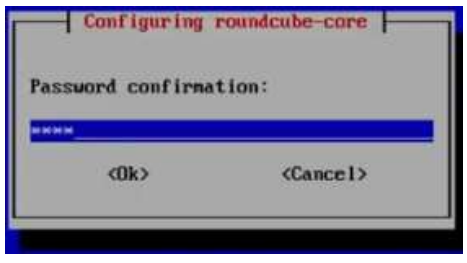
26. Lakukan penginstallan rouncube pada linux Debian dengan perintah "**apt install roundcube** ".
*Jika terjadi error pada saat pengistalan ketikkan perintah "**apt -f install**", ulangi penginstallan rouncube*



Pilih Yes

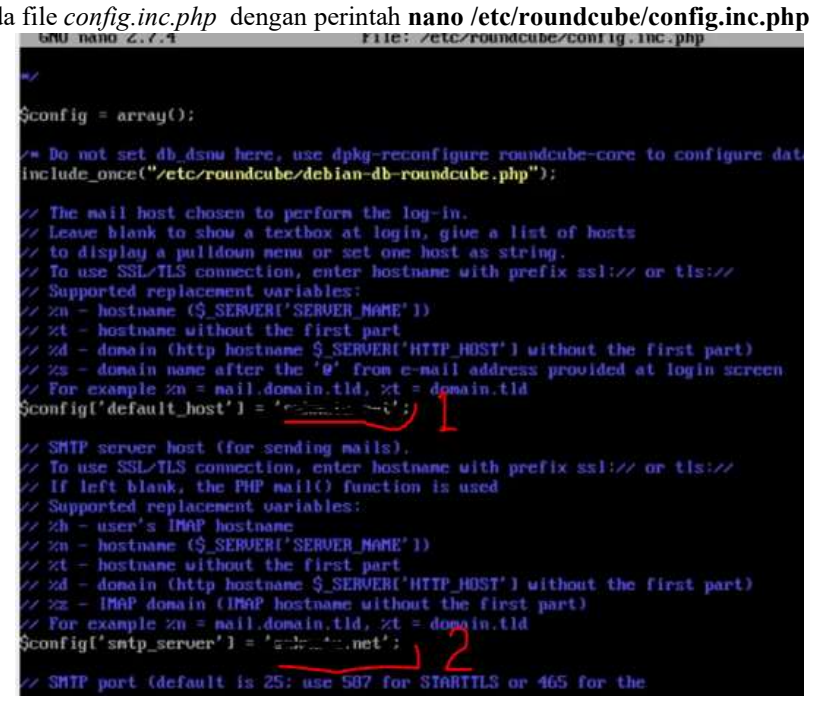


Masukkan password kemudian Ok



Ulangi ketikkan password

27. Lakukan pengeditan pada file *config.inc.php* dengan perintah **nano /etc/roundcube/config.inc.php**

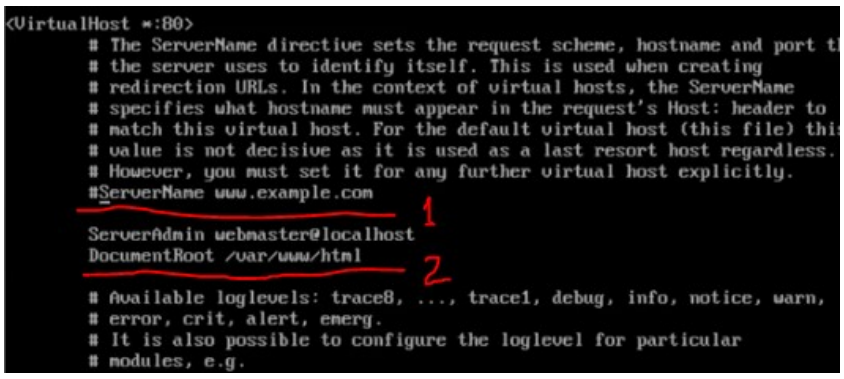


Kettikkan nama domain "**smkn3.net**" pada *$config[‘default_host’] = ‘ ‘* dan *$config[‘smtp_server’] = ‘ ‘*

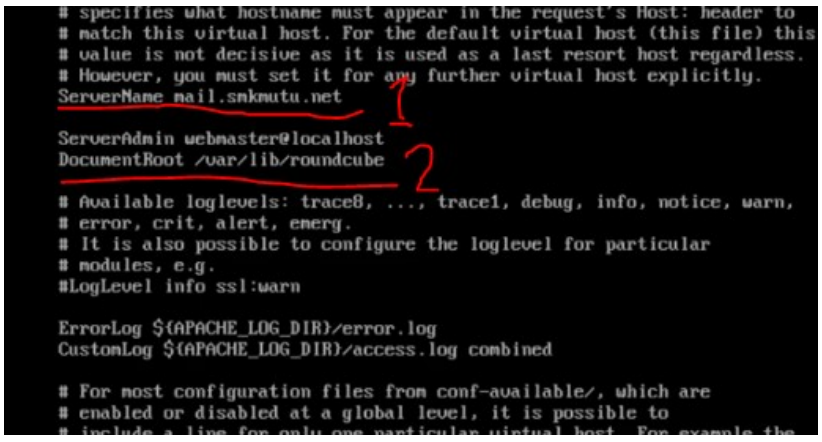
28. Masuk ke directory sites-available dengan perintah **cd /etc/apache2/sites-available**

29. Copy file *000-default.conf* dengan nama mail.conf. dengan perintah **cp 000-default.conf mail.conf**

30. Lakukan pengeditan pada file *mail.conf* dengan perintah **nano mail.conf**



Menjadi



31. Aktifkan *mail.conf* dengan perintah **a2ensite mail.conf**

32. Restart apache2, rouncube, serta postfix dengan perintah
    systemctl restart apache2
    systemctl restart roundcube
    systemctl restart postfix

33. Ketikkan alamat url “mail.smkn3.net” pada browser clien (windows) untuk menampilkan roundcube

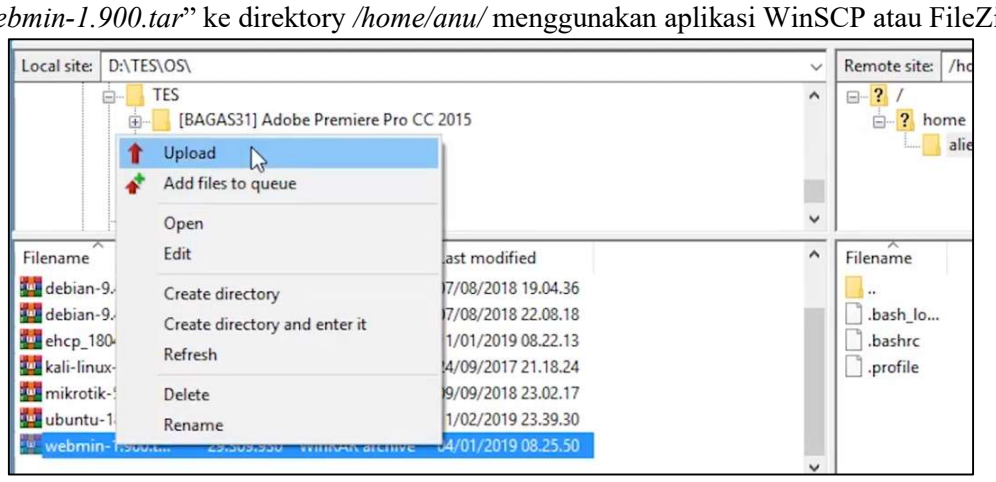


34. Masukkan username dan password yang telah di buat pada Langkah ke **16** di atas untuk dapat mengirim dan menerima email



K. Konfigurasi control panel webmin server

1. Siapkan file "*webmin-1.900.tar*"
2. Upload file "*webmin-1.900.tar*" ke direktory */home/anu/* menggunakan aplikasi WinSCP atau FileZilla



3. Masuk ke direktory *home/anu/* dengan perintah **cd /home/anu**
4. Ekstrak file "webmin-1.900.tar" menggunakan perintah **tar -xzvf webmin-1.900.tar.gz**
5. Setelah pengekstratan selesai, gunakan perintah **cd webmin-1.900** kemudian instal paket dengan perintah **./setup.sh** ikuti petunjuk saat penginstalan
6. Buka web browser pada client, ketikkan alamat ip atau alamat domain pada url. Contoh: 192.168.10.1:10000



7. Ketikkan user dan password (user = admin, password = kosong)

